

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\140-C2-Final 23rd March.pvdb

Date of Analysis : Sep 9, 2026 9:33am

PV Elite 28, January 2026



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Vessel Design Summary:

Step: 40

9:33am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1800.000 x 2800.000 mm ID

Vessel Design Length, Tangent to Tangent 7847.80 mm

Specified Datum Line Distance 0.00 mm

Shell Side Design Temperature 235 °C

Channel Side Design Temperature 275 °C

Shell Side Design Pressure 1250.000 KPa.

Channel Side Design Pressure 4950.000 KPa.

Shop Shell Side Test Pressure 2222.044 KPa.

Field Shell Side Test Pressure 1625.000 KPa.

Shop Channel Side Test Pressure 7370.678 KPa.

Field Channel Side Test Pressure 6435.000 KPa.

Wind Design Code ASCE/SEI 7-16

Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Cone	SA-516 70	...	...	K02700	
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-336 F11	2	...	K11572	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Ch Shell	4967.673	0.00	6523.000	3.0000	N/A	No
Port shell-1	1267.183	103.42	3148.000	3.0000	N/A	No
cone-1	1276.985	103.42	1468.000	3.0000	N/A	No
Main Shell	1277.475	103.42	1438.000	3.0000	N/A	No
Main Shell 2	1277.475	103.42	1438.000	3.0000	N/A	No
Cylinder	1277.475	103.42	1438.000	3.0000	N/A	No
Ellipse	1277.475	103.42	1438.000	3.0000	Yes	No

Liquid Level: 2800.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
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Vessel Design Summary:

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Type	mm	Length mm	Thickness mm	Thickness mm	Internal mm	External mm	Eff	Eff
Cylinder	1000.0	1000.0	44.0	44.0	33.9	...	1.00	1.00
Cylinder	1497.0	230.0	24.0	24.0	11.3	...	1.00	1.00
Conical	3297.0	1800.0	20.0	20.0	17.5	...	1.00	1.00
Cylinder	5297.0	2000.0	18.0	18.0	16.1	...	1.00	1.00
Cylinder	6797.0	1500.0	18.0	18.0	16.1	...	1.00	1.00
Cylinder	7797.0	1000.0	18.0	18.0	16.1	...	1.00	1.00
Ellipse	7847.8	50.8	18.0	18.0	16.0	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	Tsht	44	No Calc	...	3996
Tsht	20	44	No Calc	...	No Calc
20	30	24	13.3163	103.419	596
30	40	20	16.1454	103.419	197
40	50	18	16.267	103.419	141
50	60	18	16.267	103.419	141
60	70	18	16.267	103.419	141
70	80	18	10.4622	103.419	412

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	Tsht	999.97	No Calc	No Calc	No Calc
Tsht	20	No Calc	No Calc	No Calc	No Calc
20	30	6814.16	156084	No Calc	No Calc
30	40	6814.16	12543.6	No Calc	No Calc
40	50	6814.16	9152.61	No Calc	No Calc
50	60	6814.16	9152.61	No Calc	No Calc
60	70	6814.16	9152.61	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N6	WRC-297	0.986	Shell	Passed
Noz k6	WRC-297	0.848	Shell	Passed
N5	WRC-297	0.855	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	400.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	2000.000	mm
Wear Pad Width	780.000	mm
Wear Pad Thickness	24.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	900.000	mm
Baseplate Length	2500.000	mm
Baseplate Thickness	36.000	mm



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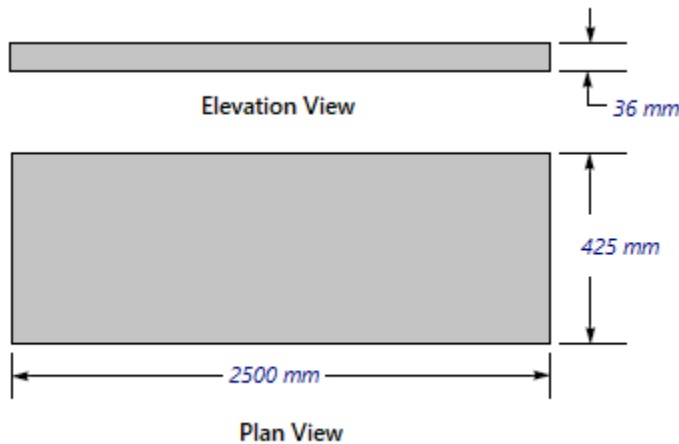
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Vessel Design Summary:

Step: 40 9:33am Sep 9, 2026

Baseplate Width	425.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	14.000	mm
Web Thickness	14.000	mm
Height of Center Web	625.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	91206.67	Kgf
Maximum Transverse Saddle Shear Load	4678.70	Kgf
Maximum Longitudinal Saddle Shear Load	4497.52	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	78626.05	Kgf
Maximum Transverse Saddle Shear Load	531.26	Kgf
Maximum Longitudinal Saddle Shear Load	447.90	Kgf

Weights:

Fabricated - Bare w/o removable internals	37090.6	kg.
Shop Test - Fabricated + water (full)	77349.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	37090.6	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	37090.6	kg.
Empty - Fab. + intls. + details + wghts.	39728.7	kg.
Operating - Empty + operating liquid (no ca)	79987.9	kg.
Field Test - Empty weight + test medium (full)	79987.9	kg.

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Nozzle Summary:

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M2-Manway	1368	OK	...	...		OK	Passed
N6	1438	OK	...	OK	11.33	OK	Passed
D3	1438	...	...	OK	7.80	OK	No Calc[*]
Noz K3	1438	...	...	OK	7.80	OK	No Calc[*]
Noz K5	1438	...	...	OK	7.80	OK	No Calc[*]
Noz k6	1438	...	...	OK	7.80	OK	No Calc[*]
Noz K4	1438	...	...	OK	7.80	OK	No Calc[*]
Noz K8	1438	...	...	OK	7.80	OK	No Calc[*]
D5	1438	...	...	OK	7.80	OK	No Calc[*]
D1	1438	...	...	OK	7.80	OK	No Calc[*]
N5	1438	OK	...	OK	8.73	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1368 Nozzle : M2-Manway [Shellside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
40	M2-Manway	4797.000	90.000	1174.920
40	N6	4797.000	0.000	590.201
40	D3	3832.257	73.313	113.600
50	Noz K3	6653.530	0.000	113.600
50	Noz K5	6172.091	0.000	113.600
50	Noz k6	5897.000	0.000	113.600
50	Noz K4	6419.320	0.000	113.600
50	Noz K8	5578.514	0.000	113.600
50	D5	6066.589	73.313	113.600
50	D1	6474.030	73.313	113.600
60	N5	7172.607	90.000	190.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 37 9:33am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D3	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
Noz K3	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
Noz K5	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
Noz k6	50 mm	Actual	LWN	77.7	13.462	...	...	318	600
Noz K4	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
Noz K8	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
D5	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
D1	50 mm	Actual	LWN	77.7	13.462	...	...	318	150
N5	3 in	LWN	LWN	120.5	15.748	...	...	371	600
N6	324 mm	Actual	WNF	324.0	17.450	...	...	377	300
M2-Manway	581 mm	Actual	WNF	610.0	14.270	...	...	426	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D3	SA-105	18.000	10.000	...	...	...
Noz K3	SA-105	18.000	10.000	...	...	...
Noz K5	SA-105	18.000	10.000	...	...	...
Noz k6	SA-105	18.000	10.000	...	...	...
Noz K4	SA-105	18.000	10.000	...	...	...
Noz K8	SA-105	18.000	10.000	...	...	...
D5	SA-105	18.000	10.000	...	...	...
D1	SA-105	18.000	10.000	...	...	...
N5	SA-105	18.000	10.000	...	...	...
N6	SA-266 2	18.000	10.000	...	...	...
M2-Manway	SA-266 2	18.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
D3	3832.257	73.3	300.00	0.00	Main Shell
Noz K3	6653.530	0.0	300.00	0.00	Main Shell 2
Noz K5	6172.090	0.0	300.00	0.00	Main Shell 2
Noz k6	5897.000	0.0	300.00	0.00	Main Shell 2
Noz K4	6419.319	0.0	300.00	0.00	Main Shell 2
Noz K8	5578.514	0.0	300.00	0.00	Main Shell 2
D5	6066.589	73.3	300.00	0.00	Main Shell 2
D1	6474.030	73.3	300.00	0.00	Main Shell 2
N5	7172.607	90.0	350.00	0.00	Node: 60
N6	4797.000	0.0	350.00	0.00	Main Shell



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Nozzle Schedule:

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M2-Manway	4797.000	90.0	350.00	0.00	Main Shell
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Bill of Material:

Step: 36 9:33am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	CYLINDER: 24.0mm THK X 1800.0mm ID X 230.0mm	SA-516 70
1	ECCENTRIC CONE: 20.0mm THK X 2750.0mm X 1800.0mm ID X 1800.0 mm	SA-516 70
1	CYLINDER: 18.0mm THK X 2800.0mm ID X 2000.0mm	SA-516 70
1	CYLINDER: 18.0mm THK X 2800.0mm ID X 1500.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm THK X 2800.0mm ID X 50.8mm	SA-516 70
2	INSULATION: 1000mm X 100mm THK	H0t Insulation
10	BAFFLES: 1768.0mm DIA X 19.0mm THK	SA-516 70
8	TIE RODS: 19mm DIA X 3000mm	SA-193 B7M
1	TUBESHEET: 1908.0mm X 267.0mm THK	SA-336 F11
2660	TUBES: 3300.0mm X 25.4mm DIA X 1.6mm THK	SA-213 TP304L
1	INSULATION: 230mm X 100mm THK	H0t Insulation
1	INSULATION: 1800mm X 100mm THK	H0t Insulation
1	SADDLE: 400.0mm X 120 DEG	SA-516 70
1	INSULATION: 2000mm X 100mm THK	H0t Insulation
1	INSULATION: 1500mm X 100mm THK	H0t Insulation
1	INSULATION: 750mm X 100mm THK	H0t Insulation
1	CLASS 150 GR 1.1, 581.5mm BLIND FLANGE(S)	SA-105
1	CYLINDRICAL SEGMENT 44.0mm THK X 3110.1mm X 1000.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 44.0mm THK X 2544.6mm X 1000.0mm	SA-387 11
2	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 2700.4mm X 1000.0mm	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 39 9:33am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Port shell-1	[8]	B	-2	-48	-29	0.395	24.000	1.00	No
cone-1	[8]	B	-24	-33	-29	0.842	20.000	1.00	No
Main Shell	[8]	B	-27	-35	-29	0.859	18.000	1.00	No
Main Shell 2	[8]	B	-11	-18	-29	0.859	18.000	1.00	No
	[8]	B	-11	-18	-29	0.859	18.000	1.00	No
	[10]	B	-11	-19	-29	0.853	18.000	1.00	No
	[7]	B	-11	-18	-29	0.859	18.000	1.00	No
Left Saddle	[24]	B	-27	-35	-29	0.859	18.000	1.00	No
M2-Manway	[1]	B	-27	-35	-29	0.861	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-38					
N6	[1]	B	-27	-35	-29	0.852	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
D3	[1]	A	-16	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
Noz K3	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
Noz K5	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
Noz k6	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Noz K4	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
Noz K8	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
D5	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
D1	[1]	A	1	-104		0.025	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-38					
N5	[1]	A	9	2		0.861	18.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Tubesheet: SS	[13]	B	23	-104		0.266	66.750	1.00	No
Warmest MDMT:			23	2					
LHS Ch Shell	[8]	A	33	33		0.997	44.000	1.00	Yes
Tubesheet: CS	[14]	A	42	35		0.870	66.750	1.00	No
Warmest MDMT:			42	35					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	2.0	-12.0	Fail
Channel/Tube	35.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3



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MDMT Summary:

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- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:33am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:33am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1250	KPa.
Channel Side Design Pressure	4950	KPa.
Shell Side Design Temperature	235.0	°C
Channel Side Design Temperature	275.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	25500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

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Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7	
Element From Node	10	
Element To Node	20	
Element Type	Cylinder	
Description	LHS Ch Shell	
Distance "FROM" to "TO"	1000	mm
Inside Diameter	1800	mm
Element Thickness	44	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	44	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4950	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	148	N./mm ²
Allowable Stress, Operating	148	N./mm ²
Allowable Stress, Hydrotest	278.99	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	249	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1800	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	10	
Detail Type	Insulation	



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Input Echo:

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Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Port shell-1
Distance "FROM" to "TO"	230 mm
Inside Diameter	1800 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	235 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	219 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1750 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	230 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Conical
Description	cone-1
Distance "FROM" to "TO"	1800 mm
Inside Diameter	1800 mm
Element Thickness	20 mm
Internal Corrosion Allowance	3 mm



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Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1250	KPa.
Design Temperature Internal Pressure	235	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	2750	mm
Design Length of Cone	1800	mm
Half Apex Angle of Cone	27.824104	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	Yes	

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2750 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1800 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	2000 mm
Inside Diameter	2800 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	235 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	900 mm
Width of Saddle	400 mm
Height of Saddle at Bottom	2000 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	780 mm
Thickness of Wear Plate	24 mm
Contact Angle, Wear Plate (degrees)	132.0



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Input Echo:

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Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2800	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	M2-Manway	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	581.5	mm
Nozzle Schedule	30	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1914.8	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N6	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	324	mm
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	160.17	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	D3	
Dist. from "FROM" Node / Offset dist	535.26	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	73.312599	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.061	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	



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Input Echo:

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Element To Node	60
Element Type	Cylinder
Description	Main Shell 2
Distance "FROM" to "TO"	1500 mm
Inside Diameter	2800 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	235 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2800 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	50
Detail Type	Nozzle
Detail ID	Noz K3
Dist. from "FROM" Node / Offset dist	1356.5 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.061 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	50
Detail Type	Nozzle
Detail ID	Noz K5
Dist. from "FROM" Node / Offset dist	875.09 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.061 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	50
Detail Type	Nozzle
Detail ID	Noz k6
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	50.8 mm



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Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.702	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz K4	
Dist. from "FROM" Node / Offset dist	1122.3	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.061	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz K8	
Dist. from "FROM" Node / Offset dist	281.51	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.061	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	D5	
Dist. from "FROM" Node / Offset dist	769.59	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	73.312599	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.061	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	D1	
Dist. from "FROM" Node / Offset dist	1177	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	73.312599	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.061	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	6/7
Element From Node	60
Element To Node	70



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Input Echo:

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Element Type	Cylinder
Description	
Distance "FROM" to "TO"	1000 mm
Inside Diameter	2800 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	235 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2800 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	60
Detail Type	Nozzle
Detail ID	N5
Dist. from "FROM" Node / Offset dist	375.61 mm
Nozzle Diameter	89 mm
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	44.376 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	2800 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	235 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70



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Input Echo:

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Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2800 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	750.8 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

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XY Coordinate Calculations:

Step: 2 9:33am

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Ch Shell		1000	...	1000	...
Port shell-1		1497	...	230	...
cone-1		3297	...	1800	...
Main Shell		5297	...	2000	...
Main Shell 2		6797	...	1500	...
60	70	7797	...	1000	...
70	80	7847.8	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Ch Shell		4967.7	44	3	1800	148	1
Port shell-1		1267.2	24	3	1800	138	2
cone-1		1277	20	3	2750	138	3
Main Shell		1277.5	18	3	2800	138	4
Main Shell 2		1277.5	18	3	2800	138	5
60	70	1277.5	18	3	2800	138	6
70	80	1277.5	18	3	2800	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Ch Shell		4950	6523	7029	44	33.9349	1
Port shell-1		1250	3148	3622	24	11.3384	2
cone-1		1250	1468	1761	20	17.5041	3
Main Shell		1250	1438	1761	18	16.0612	4
Main Shell 2		1250	1438	1761	18	16.0612	5
60	70	1250	1438	1761	18	16.0612	6
70	80	1250	1438	1772	18	15.9637	7

Warning: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1261 KPa.
 Shell Side MAPnc = 1730 KPa.
 Channel Side MAWP = 5368 KPa.
 Channel Side MAPnc = 5691 KPa.

Elements Suitable for Design Internal Pressure.

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Element: LHS Ch Shell

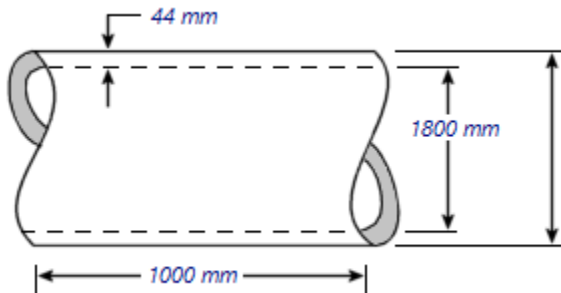
Step: 4

9:33am

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	4967.673 KPa.	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	0 °C
Length	1000.000 mm	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	148.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1800.000 mm	Actual Stress	112.4 N./mm ²
Spec. Min. Thk t,e, or sa	44.0000 mm	Spec. Nominal Thk.	44.000 mm
Surface Area	59313.3 cm ²	SG of Contents	1.000
Element Volume	2545146.2 cm ³	Weight of Contents	2543.6 Kg
Empty Weight	1975.6 Kg	Operating Weight	4519.2 Kg
		Comp. All. Ope. Stress	109.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4968 \cdot 903) / (148 \cdot 1 - 0.6 \cdot 4968) \\
 &= 30.9349 + 3.0000 = 33.9349 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.673 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148 \cdot 1 \cdot 41) / (903 + 0.6 \cdot 41) \\
 &= 6541 - 17.7 = 6523 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148 \cdot 1 \cdot 44) / (900 + 0.6 \cdot 44) \\
 &= 7029 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4968 \cdot (903 + 0.6 \cdot 41)) / (1 \cdot 41) \\
 &= 112.397 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS Ch Shell

Step: 4

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 2.386 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 44$, $t_r = 40.9$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 33 °C

Min Metal Temp. at Required thickness (UCS 66.1) 33 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	6523.441 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	44.000 mm	Basic MDMT	33 °C
Stress/Thk. Ratio	0.997	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	33 °C
		Joint Efficiency E^*	100 %

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Element: Port shell-1

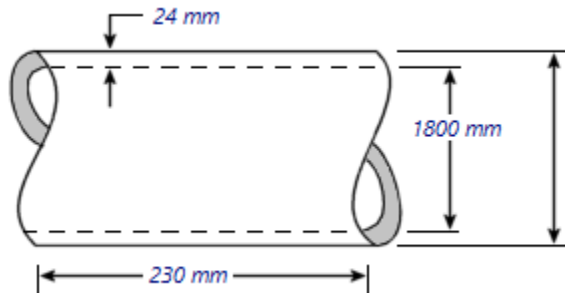
Step: 5

9:33am

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Port shell-1

Cylindrical Shell Design Information:			
Design Pressure P	1267.183 KPa.	Int. Design Temperature	235 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	230.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1800.000 mm	Actual Stress	55.3 N./mm ²
Spec. Min. Thk t,e, or sa	24.0000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	13353.0 cm ²	SG of Contents	1.000
Element Volume	275324.4 cm ³	Weight of Contents	275.2 Kg
Empty Weight	245.2 Kg	Operating Weight	520.3 Kg
		Comp. All. Ope. Stress	108.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1267 \cdot 903) / (138 \cdot 1 - 0.6 \cdot 1267) \\
 &= 8.3384 + 3.0000 = 11.3384 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.183 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 21) / (903 + 0.6 \cdot 21) \\
 &= 3165 - 17.2 = 3148 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 24) / (900 + 0.6 \cdot 24) \\
 &= 3622 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1267 \cdot (903 + 0.6 \cdot 21)) / (1 \cdot 21) \\
 &= 55.252 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Port shell-1 Step: 5 9:33am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.316 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 24$, $t_r = 8.3$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.4$, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1261.040 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.395	Temperature Difference	46 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Port shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	1848.00	6814.16	88.00	3.6873	0.0004064	39.35

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 39.3) / (3 \cdot 88) = 596 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.316	1848.00	6814.16	179.13	3.6873	0.0001435	13.90

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 13.9) / (3 \cdot 179) = 103 \text{ KPa.}$$

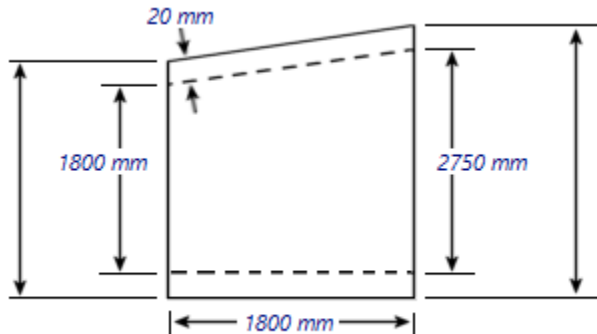
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	1848.00	156084.28	88.00	50.0000	0.0001447	14.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14) / (3 \cdot 88) = 212 \text{ KPa.}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: cone-1

Conical Head/Section Design Information:			
Design Pressure P	1276.985 KPa.	Int. Design Temperature	235 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	1800.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2750.000 mm	Actual Stress	117.8 N./mm ²
Spec. Min. Thk t,e, or sa	20.0000 mm	Spec. Nominal Thk.	20.000 mm
Surface Area	135697.4 cm ²	SG of Contents	1.000
Element Volume	4997971.5 cm ³	Weight of Contents	4994.9 Kgf
Empty Weight	2277.3 Kgf	Operating Weight	7272.2 Kgf
		Comp. All. Ope. Stress	106.6 N./mm ²
		To End Diameter	1800.000 mm
Line of Support	Neither End		

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$ per Appendix 1-4 (e)
 $= (1277 \cdot 2757) / (2 \cdot 0.88 \cdot (138 \cdot 1 - 0.6 \cdot 1277))$
 $= 14.5041 + 3.0000 = 17.5041$ mm

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.985 KPa.

$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$ per App 1-4(e)
 $= (2 \cdot 138 \cdot 1 \cdot 17 \cdot 0.88) / (2757 + 1.2 \cdot 17 \cdot 0.88)$
 $= 1495 - 27 = 1468$ KPa.

Maximum Allowable Pressure, New and Cold [MAPNC]:

$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$ per App 1-4(e)
 $= (2 \cdot 138 \cdot 1 \cdot 20 \cdot 0.88) / (2750 + 1.2 \cdot 20 \cdot 0.88)$
 $= 1761$ KPa.

Actual stress at given pressure and thickness, corroded [Sact]:



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 Element: cone-1 Step: 6 9:33am Sep 9,2026

$$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (1277 \cdot (2757 + 1.2 \cdot 17 \cdot 0.88)) / (2 \cdot 1 \cdot 17 \cdot 0.88)$$

$$= 117.849 \text{ N/mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.097 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 20$, $t_r = 14.3$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1261.040 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.842	Temperature Difference	9 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: cone-1

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	2790.00	6814.16	164.12	2.4424	0.0002511	24.31

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 24.3) / (3 \cdot 164) = 197 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.145	2790.00	6814.16	212.24	2.4424	0.0001701	16.46

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 16.5) / (3 \cdot 212) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	2790.00	12543.57	164.12	4.4959	0.0001316	12.74

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 12.7) / (3 \cdot 164) = 103 \text{ KPa.}$$



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Element: Main Shell

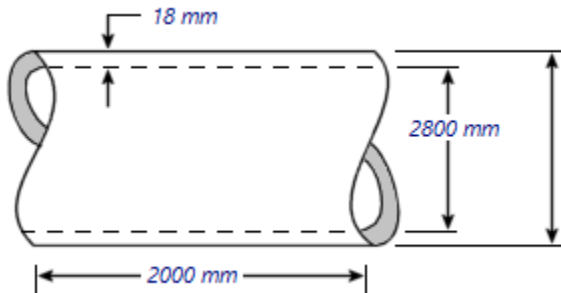
Step: 7

9:33am

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1277.475 KPa.	Int. Design Temperature	235 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	2000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2800.000 mm	Actual Stress	120.3 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	178191.1 cm ²	SG of Contents	1.000
Element Volume	9108719.0 cm ³	Weight of Contents	9103.2 Kg
Empty Weight	2470.1 Kg	Operating Weight	11573.3 Kg
		Comp. All. Ope. Stress	93.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1277 \cdot 1403) / (138 \cdot 1 - 0.6 \cdot 1277) \\
 &= 13.0612 + 3.0000 = 16.0612 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 27.475 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 15) / (1403 + 0.6 \cdot 15) \\
 &= 1466 - 27.5 = 1438 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (1400 + 0.6 \cdot 18) \\
 &= 1761 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1277 \cdot (1403 + 0.6 \cdot 15)) / (1 \cdot 15) \\
 &= 120.260 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Main Shell Step: 7 9:33am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18$, $t_r = 12.9$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-35 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1261.040 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.859	Temperature Difference	8 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-35 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2836.00	6814.16	189.07	2.4027	0.0002065	19.99

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 20) / (3 \cdot 189) = 141 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.267	2836.00	6814.16	213.76	2.4027	0.0001713	16.58

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 16.6) / (3 \cdot 214) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2836.00	9152.61	189.07	3.2273	0.0001516	14.68

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.7) / (3 \cdot 189) = 104 \text{ KPa.}$$



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Element: Main Shell 2

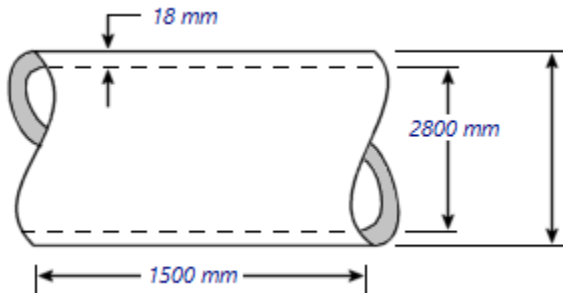
Step: 8

9:33am

Sep 9, 2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell 2

Cylindrical Shell Design Information:			
Design Pressure P	1277.475 KPa.	Int. Design Temperature	235 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2800.000 mm	Actual Stress	120.3 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	133643.4 cm ²	SG of Contents	1.000
Element Volume	9237937.0 cm ³	Weight of Contents	9232.3 Kg
Empty Weight	1852.6 Kg	Operating Weight	11084.9 Kg
		Comp. All. Ope. Stress	93.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1277 \cdot 1403) / (138 \cdot 1 - 0.6 \cdot 1277) \\
 &= 13.0612 + 3.0000 = 16.0612 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 27.475 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 15) / (1403 + 0.6 \cdot 15) \\
 &= 1466 - 27.5 = 1438 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (1400 + 0.6 \cdot 18) \\
 &= 1761 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1277 \cdot (1403 + 0.6 \cdot 15)) / (1 \cdot 15) \\
 &= 120.260 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Main Shell 2 Step: 8 9:33am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18$, $t_r = 12.9$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1261.040 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.859	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell 2

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2836.00	6814.16	189.07	2.4027	0.0002065	19.99

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 20) / (3 \cdot 189) = 141 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.267	2836.00	6814.16	213.76	2.4027	0.0001713	16.58

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 16.6) / (3 \cdot 214) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2836.00	9152.61	189.07	3.2273	0.0001516	14.68

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.7) / (3 \cdot 189) = 104 \text{ KPa.}$$



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Element: # 6

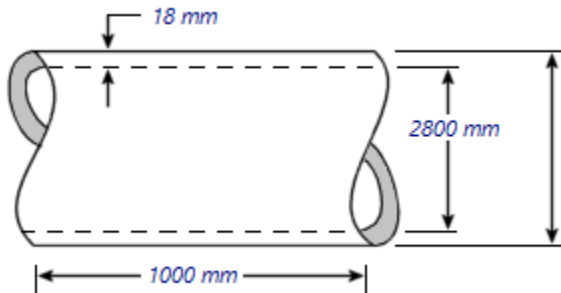
Step: 9

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Element # 6

Cylindrical Shell Design Information:			
Design Pressure P	1277.475 KPa.	Int. Design Temperature	235 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	1000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2800.000 mm	Actual Stress	120.3 N./mm ²
Spec. Min. Thk t, e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	89095.6 cm ²	SG of Contents	1.000
Element Volume	6158624.5 cm ³	Weight of Contents	6154.9 Kg
Empty Weight	1235.1 Kg	Operating Weight	7389.9 Kg
		Comp. All. Ope. Stress	93.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1277 \cdot 1403) / (138 \cdot 1 - 0.6 \cdot 1277) \\
 &= 13.0612 + 3.0000 = 16.0612 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 27.475 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 15) / (1403 + 0.6 \cdot 15) \\
 &= 1466 - 27.5 = 1438 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (1400 + 0.6 \cdot 18) \\
 &= 1761 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1277 \cdot (1403 + 0.6 \cdot 15)) / (1 \cdot 15) \\
 &= 120.260 \text{ N./mm}^2
 \end{aligned}$$



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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18$, $t_r = 12.9$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -18 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1261.040 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.859	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Element # 6

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2836.00	6814.16	189.07	2.4027	0.0002065	19.99

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 20) / (3 \cdot 189) = 141 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.267	2836.00	6814.16	213.76	2.4027	0.0001713	16.58

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 16.6) / (3 \cdot 214) = 103 \text{ KPa.}$$

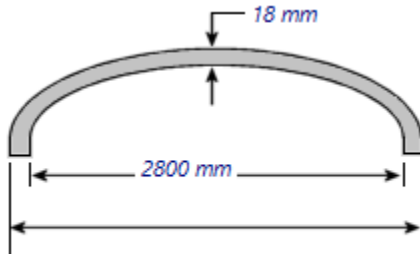
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2836.00	9152.61	189.07	3.2273	0.0001516	14.68

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.7) / (3 \cdot 189) = 104 \text{ KPa.}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Element # 7

Elliptical Head Design Information:			
Design Pressure P	1277.475 KPa.	Int. Design Temperature	235 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	50.800 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2800.000 mm	Actual Stress	119.3 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	91709.9 cm ²	SG of Contents	1.000
Element Volume	3186883.2 cm ³	Weight of Contents	3184.9 Kg
Empty Weight	1388.7 Kg	Operating Weight	4573.6 Kg
		Comp. All. Ope. Stress	93.0 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1277 \cdot 2806 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1277)$$

$$= 12.9637 + 3.0000 = 15.9637 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 27.475 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 15) / (1 \cdot 2806 + 0.2 \cdot 15)$$

$$= 1478 - 27.5 = 1450 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 18) / (1 \cdot 2800 + 0.2 \cdot 18)$$

$$= 1772 \text{ KPa.}$$



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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1277 \cdot (1 \cdot 2806 + 0.2 \cdot 15)) / (2 \cdot 1 \cdot 15)$$

$$= 119.282 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1277 \cdot 1403) / (138 \cdot 1 - 0.6 \cdot 1277) + 3$$

$$= 16.061 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 27.475 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 15) / (1403 + 0.6 \cdot 15)$$

$$= 1466 - 27.5 = 1438 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2806 / (2 \cdot 703)))^2 / 6$$

$$= 0.997158$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.784 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 18, tr = 12.8, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -19 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 18, tr = 12.9, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -18 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1261.040 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.859	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.500 mm

EXTERNAL PRESSURE RESULTS for: Element # 7

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.



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Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
15.000	2836.00	189.07	0.0007346	70.09

$$\text{MAEP} = B / (K0 * \text{Do} / t) = 70.1 / (0.9 * 189) = 412 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.462	2836.00	380.05	0.0003655	35.38

$$\text{MAEP} = B / (K0 * \text{Do} / t) = 35.4 / (0.9 * 380) = 103 \text{ KPa.}$$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1639.352 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1625.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2222.044 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1387.144 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1787.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1787.500 KPa.
App 27-4	= M.A.W.P.	1261.040 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6978.890 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7370.678 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5905.215 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5368.377 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	155.6	279.0	0.558	7388.32
Port shell-1	85.3	234.0	0.365	2239.69
cone-1	176.2	234.0	0.753	2249.00
Main Shell	176.3	234.0	0.754	2249.49
Main Shell 2	176.3	234.0	0.754	2249.49
60 70	176.3	234.0	0.754	2249.49
70 80	175.2	234.0	0.749	2249.49

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M2-Manway	Nozzle	138.00	136.60	1.010
N6	Nozzle	138.00	136.60	1.010
D3	Nozzle	138.00	136.60	1.010
Noz K3	Nozzle	138.00	136.60	1.010
Noz K5	Nozzle	138.00	136.60	1.010
Noz k6	Nozzle	138.00	136.60	1.010
Noz K4	Nozzle	138.00	136.60	1.010
Noz K8	Nozzle	138.00	136.60	1.010
D5	Nozzle	138.00	136.60	1.010
D1	Nozzle	138.00	136.60	1.010
N5	Nozzle	138.00	136.60	1.010

Minimum 1.010

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Ch Shell	148.00	148.00	1.000



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Port shell-1	138.00	138.00	1.000
cone-1	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell 2	138.00	138.00	1.000
	138.00	138.00	1.000
	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	100.35	1.146
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M2-Manway	47.18	179.40	0.263
N6	19.99	179.40	0.111
D3	5.59	179.40	0.031
Noz K3	5.59	179.40	0.031
Noz K5	5.59	179.40	0.031
Noz k6	5.59	179.40	0.031
Noz K4	5.59	179.40	0.031
Noz K8	5.59	179.40	0.031
D5	5.59	179.40	0.031
D1	5.59	179.40	0.031
N5	7.71	179.40	0.043

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1639.352 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1625.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2222.044 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1387.144 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1787.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1787.500 KPa.
App 27-4	= M.A.W.P.	1261.040 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6978.890 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7370.678 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5905.215 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5368.377 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1250 • 1
 = 1625.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 4950 • 1
 = 6435.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	146.0	279.0	0.523	6452.64
Port shell-1	71.6	234.0	0.306	1642.64
cone-1	152.5	234.0	0.652	1651.96
Main Shell	155.6	234.0	0.665	1652.45
Main Shell 2	155.6	234.0	0.665	1652.45
60 70	155.6	234.0	0.665	1652.45
70 80	154.3	234.0	0.659	1652.45

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M2-Manway	Nozzle	138.00	136.60	1.010
N6	Nozzle	138.00	136.60	1.010
D3	Nozzle	138.00	136.60	1.010
Noz K3	Nozzle	138.00	136.60	1.010
Noz K5	Nozzle	138.00	136.60	1.010
Noz k6	Nozzle	138.00	136.60	1.010
Noz K4	Nozzle	138.00	136.60	1.010
Noz K8	Nozzle	138.00	136.60	1.010
D5	Nozzle	138.00	136.60	1.010



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D1	Nozzle	138.00	136.60	1.010
N5	Nozzle	138.00	136.60	1.010

Minimum 1.010

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Ch Shell	148.00	148.00	1.000
Port shell-1	138.00	138.00	1.000
cone-1	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell 2	138.00	138.00	1.000
	138.00	138.00	1.000
	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	100.35	1.146
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M2-Manway	34.66	179.40	0.193
N6	14.68	179.40	0.082
D3	4.11	179.40	0.023
Noz K3	4.11	179.40	0.023
Noz K5	4.11	179.40	0.023
Noz k6	4.11	179.40	0.023
Noz K4	4.11	179.40	0.023
Noz K8	4.11	179.40	0.023
D5	4.11	179.40	0.023
D1	4.11	179.40	0.023
N5	5.66	179.40	0.032

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:33a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Port shell-1

Shell Test Pressure		1625.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1800.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		6435.00 KPa.
Channel Thickness	tc	44.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1800.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2660
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces) L		3033.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	876.669 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	23585.6 cm ²
Perimeter of Tube Layout	Cp	5684.2 mm



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ASME TS Calc: Pressure Test

Case: 1 9:33a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F11

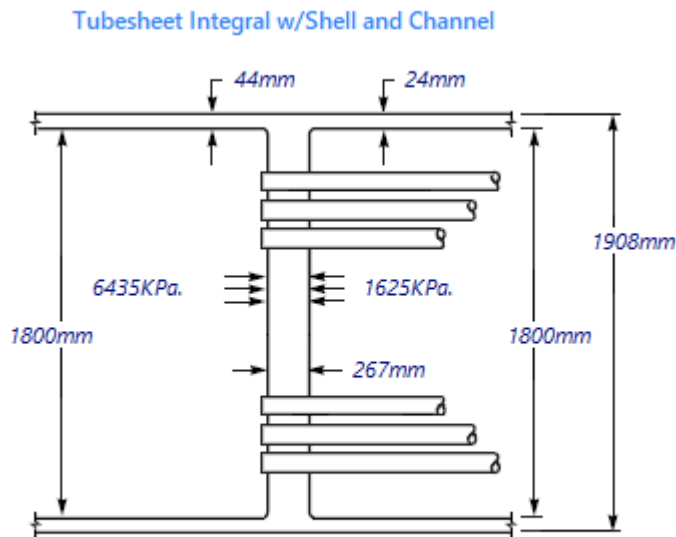
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K11572
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 267.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1908.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Channel Side Min. Design Pressure Ptd,min 103.0000 KPa.

Tubesheet Integral with Both
Tubesheet Extended as Flange No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.



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 ASME TS Calc: Pressure Test Case: 1 9:33a Sep 9,2026

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (6435 \cdot (25.4/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 6435)$
 $= 0.6951 + 0.0000 = 0.6951 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	21.11 °C
Elastic Modulus for Material			193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	3909.67	15.39	50.0000	0.0046419	81.86

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 81.9) / (3 \cdot 15.4) = 7090 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6104	25.40	3909.67	41.61	50.0000	0.0006352	53.94

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 53.9) / (3 \cdot 41.6) = 1728 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6951 mm
Allowable Internal Pressure at Corroded thickness	15758.77 KPa.
Required Internal Design Pressure	6435.00 KPa.
Allowable External Pressure at Corroded thickness	7090.01 KPa.
Required External Design Pressure	1728.00 KPa.
Required Thickness due to Shell Side pressure	0.6104 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 877 + 25.4$
 $= 1778.738 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.8 - 25.4) / 31.8$
 $= 0.200$



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ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1800/1779$$

$$= 1.0120$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1800/1779$$

$$= 1.0120$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/267$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t/E) \cdot (StT/S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 1.65 (195117184/204080160) \cdot$$

$$(115/138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$$h/p = 8.409449 ; \mu^* = 0.200000 ; E^* = 31703676 \text{ KPa.}$$

$$E^*/E = 0.155349 ; \nu^* = 0.458572$$

Note: As h/p (8.409) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1800 + 24) 24 }$$

$$= 0.0087 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0087 \cdot 202701232 \cdot 24^3 / (6 (1 - 0.3^2))$$

$$= 4458976256.0000 \text{ KPa.}\cdot\text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1800 + 24) \cdot 4.46E+09/267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2)$$

$$= 15409498.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 1800^2 / (4 \cdot 202701232 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2 (1800 + 24)))$$

$$= 0.14185E+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



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$$= 1.01 \cdot 4.46\text{E}+09 \cdot 0.0087 \cdot 0.1418464 (1 + 267 \cdot 0.0087)$$

$$= 18462.3730 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1800 + 44)44}$$

$$= 0.0064 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.0064 \cdot 204080160 \cdot 44^3 / (6(1 - 0.3^2))$$

$$= 20319592448.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1800 + 44 / \cos(0)) \cdot 20.32\text{E}+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1)$$

$$= 49083820.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1800^2 / (4 \cdot 204080160 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44)))$$

$$= 0.76993\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [zeta α_c]:

$$zeta\alpha_c = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 20.32\text{E}+09 [0.076992802 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)]$$

$$= 27321.2363 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1908 / 1779$$

$$= 1.0727$$

Coefficient [F]:

$$= (1 - v_s) / E_s \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 31703676 \cdot (15409498 + 49083820 + 204080160 \cdot \ln(1.07))$$

$$= 1.3459$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot Pt]$$

$$= 1779^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 0 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 6435]$$

$$= -30786730.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -30786730 + 27321 \cdot 6435 - 18462 \cdot 0$$

$$= 145025424.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:



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Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((145025424) - 1779^2/32 \cdot 1.35 (0 - 6435)) / (1 + 1.35)$$

$$= 426848256.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= 426848256 + 1779^2/64 \cdot (3 + 0.46) \cdot (0 - 6435)$$

$$= -673396736.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 426848256 | ; | -673396736 |)$$

$$= 673396736.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 673396736 / (0.2 (267 - 8)^2)$$

$$= 301.1743 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 275$$

$$= 522.49 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 625973120 / (0.2 (198 - 8)^2)$$

$$= 520.1150 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 198 + 0 + 0$$

$$= 198.0209 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(198 ; 63.9)$$

$$= 198.0209 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 0 - 6435 |$$

$$= 6435.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 267 / 1779 \cdot 0.8 \cdot 0.95 \cdot 275$$

$$= 25.097 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 209) \cdot (4 \cdot 23586 / 5684) \cdot | 0 - 6435 | + 0$$

$$= 63.8829 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:



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Note: For a given Shell thickness of: 24.000mm
 Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Note: For a given Channel thickness of: 44.000 mm
 Min. Channel len. adjacent to the tubesheet is: 506.565mm

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 1.65) 1.65$
 $= 1.231 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 1.65 (25.4 - 1.65) 161$
 $= 2027.372 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
 $= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$
 $= 1924.4156 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 161/161$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$
 $= 1245.2101 \text{ Kg}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$
 $= 2.141 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
 Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 2027 is >= tube strength 2027 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	301.2 <=	522.5 N./mm ²	Ok
Tubesheet shear stress	50.0 <=	209.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	198.021	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok



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Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 877 + 25.4 \\ &= 1778.738 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1800 / 1779 \\ &= 1.0120 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1800 / 1779 \\ &= 1.0120 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 267 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2tt(E_tT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 1.65 (195117184 / 204080160) \cdot (115 / 138) \cdot (0); 25.4 - 2 \cdot 1.65) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) } \\ &= 31.8 / \sqrt{ 1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2) } \\ &= 31.7500 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.2000 \end{aligned}$$

$h/p = 8.409449$; $\mu^* = 0.200000$; $E^* = 31703676 \text{ KPa}$.

$E^*/E = 0.155349$; $v^* = 0.458572$

Note: As h/p (8.409) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$



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$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{(1800 + 24)24}$$

$$= 0.0087 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0087 \cdot 202701232 \cdot 24^3 / (6(1 - 0.3^2))$$

$$= 4458976256.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1800 + 24) \cdot 4.46E+09 / 267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2)$$

$$= 15409498.0 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1800^2 / (4 \cdot 202701232 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 24)))$$

$$= 0.14185E+00 \text{ mm/N./mm}^2$$

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 4.46E+09 \cdot 0.0087 \cdot 0.1418464 (1 + 267 \cdot 0.0087)$$

$$= 18462.3730 \text{ mm}^2$$

Channel coefficient [βc]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} \right) / \sqrt{(D_c + t_c)t_c}$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{(1800 + 44)44}$$

$$= 0.0064 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0064 \cdot 204080160 \cdot 44^3 / (6(1 - 0.3^2))$$

$$= 20319592448.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1800 + 44 / \cos(0)) \cdot 20.32E+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1)$$

$$= 49083820.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1800^2 / (4 \cdot 204080160 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44)))$$

$$= 0.76993E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta_c (CmTheta + \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 20.32E+09 [0.076992802 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)]$$

$$= 27321.2363 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1908/1779$$

$$= 1.0727$$



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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E * (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.46) / 31703676 * (15409498 + 49083820 + \\
 &\quad 204080160 \cdot \ln(1.07)) \\
 &= 1.3459
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 * [(p_s - 1) * (p_s^2 + 1) * P_s - (p_c - 1) * (p_c^2 + 1) * P_t] \\
 &= 1779^2 / 16 * [(1.01 - 1) * (1.01^2 + 1) * 1625 - \\
 &\quad (1.01 - 1) * (1.01^2 + 1) * 6435] \\
 &= -23012302.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s \\
 &= -23012302 + 27321 \cdot 6435 - 18462 \cdot 1625 \\
 &= 122798504.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= (M^* - D_o^2 / 32 \cdot F(P_s - P_t)) / (1 + F) \\
 &= ((122798504) - 1779^2 / 32 \cdot 1.35 (1625 - 6435)) / (1 + 1.35) \\
 &= 325194784.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 \cdot (3 + \nu) \cdot (P_s - P_t) \\
 &= 325194784 + 1779^2 / 64 \cdot (3 + 0.46) \cdot (1625 - 6435) \\
 &= -497210560.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(|M_p|; |M_o|) \\
 &= \max(|325194784|; |-497210560|) \\
 &= 497210560.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu * (h - h_g')^2) \\
 &= 6 \cdot 497210560 / (0.2 (267 - 8)^2) \\
 &= 222.3756 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot \beta_T \cdot S_y \\
 &= 2 \cdot 0.95 \cdot 275 \\
 &= 522.49 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu * (h - h_g')^2) \\
 &= 6 \cdot 442802048 / (0.2 (168 - 8)^2) \\
 &= 522.1657 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 168 + 0 + 0 \\
 &= 167.5048 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(168; 47.8) \\
 &= 167.5048 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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ASME TS Calc: Pressure Test

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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 1625 - 6435 |$$

$$= 4810.000 \text{ KPa.}$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.2 \cdot 267 / 1779 \cdot 0.8 \cdot 0.95 \cdot 275$$

$$= 25.097 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 209) \cdot (4 \cdot 23586 / 5684) \cdot | 1625 - 6435 | + 0$$

$$= 47.7508 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 24.000mm

Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Note: For a given Channel thickness of: 44.000 mm

Min. Channel len. adjacent to the tubesheet is: 506.565mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 161$$

$$= 2027.372 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$$

$$= 1924.4156 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 161 / 161$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$$

$$= 1245.2101 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) } - 0.75 \cdot d_o)$$

$$= 2 (\sqrt{ ((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1) } - 0.75 \cdot 25.4)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm



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 ASME TS Calc: Pressure Test Case: 1 9:33a Sep 9,2026

Tube-Tubesheet Jt allowable, 2027 is >= tube strength 2027 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	222.4 <=	522.5 N./mm ²	Ok
Tubesheet shear stress	37.4 <=	209.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	167.505	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

	--Reqd. Thk. + CA		----	Tubesheet	Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	198.021	...	301	522	50	209	Ps+Pt D1	Ok
D2uc	102.328	...	84	522	13	209	Ps+Pt D2	Ok
D3uc	167.505	...	222	522	37	209	Ps+Pt D3	Ok
Max:	198.0209	... mm		0.576		0.239	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	370	...	442	Ok
D2uc	...	...	...	370	...	442	Ok
D3uc	...	...	...	370	...	442	Ok
Max	...	...	...	...	...	...	

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.



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ASME TS Calc: Pressure Test

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Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	198.021	267.000	Ok
Tube thickness :	0.695	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.651	Ok
Tube-tubesheet groove weld leg :	1.070	1.651	Ok

Given Shell Thickness: 24.0000 mm

Given Channel Thickness: 44.0000 mm

Min Shell length of thk, (24.000) adj. to tubesheet: 374.123 mm

Min Channel length of thk, (44.000) adj. to tubesheet: 506.565 mm

[Note: This is a full strength Tube to Tubesheet Joint.](#)[PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.](#)



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	Tsht	999.97	1888	41	0.0090075	138.006
Tsht	20	No Calc	...	41	No Calc	No Calc
20	30	6814.16	1848	21	0.00040644	39.3462
30	40	6814.16	2790	17	0.00025107	24.3051
40	50	6814.16	2836	15	0.00020653	19.9933
50	60	6814.16	2836	15	0.00020653	19.9933
60	70	6814.16	2836	15	0.00020653	19.9933
70	80	No Calc	2836	15	0.0007346	70.0899

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	Tsht	44	No Calc	...	3996
Tsht	20	44	No Calc	...	No Calc
20	30	24	13.3163	103.419	596
30	40	20	16.1454	103.419	197
40	50	18	16.267	103.419	141
50	60	18	16.267	103.419	141
60	70	18	16.267	103.419	141
70	80	18	10.4622	103.419	412

Minimum

141

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	Tsht	999.97	No Calc	No Calc	No Calc
Tsht	20	No Calc	No Calc	No Calc	No Calc
20	30	6814.16	156084	No Calc	No Calc
30	40	6814.16	12543.6	No Calc	No Calc
40	50	6814.16	9152.61	No Calc	No Calc
50	60	6814.16	9152.61	No Calc	No Calc
60	70	6814.16	9152.61	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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 Element and Detail Weights: Step: 14 9:33am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	1975.58	2545146	1843.87	2562141	395.115
20	30	245.157	275324	214.865	279233	49.0313
30	40	2277.29	4997971	1938.55	5041685	455.459
40	50	2470.15	9108719	2060.65	9161563	494.029
50	60	1852.61	9237937	1545.49	9277570	370.522
60	70	1235.07	6158624	1030.32	6185047	247.015
70	80	1388.69	3186883	1157.24	3206740	277.737
Total		11444	35510604.00	9790	35713984.00	2288



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	2543.59	500	...	...	Liquid: 30
10	Insl	220.098	500	...	...	Ins: 10
20	Liqd	275.156	115	25	...	Liquid: 40
20	Insl	49.6041	115	...	...	Ins: 10
30	Liqd	4994.92	605.854	-237.5	...	Liquid: 50
30	Insl	540.167	900	...	...	Ins: 10
40	Sadl	1515.19	900	1679	...	Left Saddle
40	Liqd	9103.16	1000	...	...	Liquid: 60
40	Insl	650.11	1000	...	...	Ins: 10
40	Nozl	2297.81	1500	...	1690.73	M2-Manway
40	Nozl	192.205	1500	1562	...	N6
40	Nozl	12.0731	535.257	409.303	1365.37	D3
50	Liqd	9232.3	750	...	...	Liquid: 70
50	Insl	487.582	750	...	...	Ins: 10
50	Nozl	12.0731	1356.53	1425.4	...	Noz K3
50	Nozl	12.0731	875.091	1425.4	...	Noz K5
50	Nozl	14.0426	600	1425.4	...	Noz k6
50	Nozl	12.0731	1122.32	1425.4	...	Noz K4
50	Nozl	12.0731	281.514	1425.4	...	Noz K8
50	Nozl	12.0731	769.589	409.303	1365.37	D5
50	Nozl	12.0731	1177.03	409.303	1365.37	D1
60	Liqd	6154.87	500	...	...	Liquid: 60
60	Insl	325.055	500	...	...	Ins: 60
60	Nozl	53.2514	375.607	...	1444.5	N5
70	Liqd	3184.94	284.133	...	...	Liquid: 70
70	Insl	365.483	375.4	...	...	Ins: 70
10	FTsh	3753.13	1133.5	...	...	
10	Tube	13424.1	2650	...	...	Tubes
20	Baf1	202.254	330	...	...	Baffles
20	Baf1	202.254	660	...	...	Baffles
20	Baf1	202.254	990	...	...	Baffles
20	Baf1	202.254	1320	...	...	Baffles
20	Baf1	202.254	1650	...	...	Baffles
20	Baf1	202.254	1980	...	...	Baffles
20	Baf1	202.254	2310	...	...	Baffles
20	Baf1	202.254	2640	...	...	Baffles
20	Baf1	202.254	2970	...	...	Baffles



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Element and Detail Weights:

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20 Baf1	202.679	3300	...	...	Baffles
---------	---------	------	-----	-----	---------

Total Weight of Each Detail Type:

Saddles	1515.2
Liquid	35488.9
Insulation	2638.1
Nozzles	2641.8
Exchanger Components	19200.2
Liquid in Tubes	4770.3

Sum of the Detail Weights 66254.5 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	13733.5	13733.5	13733.5	13733.5	13733.5	13733.5
Saddles	1515.2	1515.2	1515.2	1515.2	1515.2	1515.2
Nozzles	2641.8	2641.8	2641.8	2641.8	2641.8	2641.8
Exchanger	19200.2	19200.2	19200.2	19200.2	19200.2	19200.2
Insulation	...	...	...	...	2638.1	2638.1 *
Ope. Liquid	...	...	...	...	...	35488.9
Tube Ope Lqd	...	...	...	...	...	4770.3
Test Liquid	...	35488.9	...	...	...	...
Tube Tst Lqd	...	4770.3	...	...	...	...
Totals	37090.6	77349.8	37090.6	37090.6	39728.7	79987.9

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 20.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	37090.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	77349.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	37090.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	37090.6 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	39728.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	79987.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	78003.6 kg.
Field Test Wt.	- Empty weight + test medium (full)	79987.9 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	4773178.0 cm ³
Weight of Ope Liq in tubes :	4770.3 kg.
Weight of Water in tubes :	4770.3 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	59313.3	56548.7



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Element and Detail Weights:

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20	30	13353	13006.2
30	40	135697	133052
40	50	178191	175929
50	60	133643	131947
60	70	89095.6	87964.6
70	80	91709.9	89453.1

Total 701003.6 687900.9 cm²

Total 754.6 740.5 ft²

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Nozzle Flange MAWP:

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
M2-Manway	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
N6	4254.0	5100.0	235	300	GR 1.1	2214.1	4167.0	4254.0	4167.0
D3	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
Noz K3	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
Noz K5	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
Noz k6	8501.1	10200.0	235	600	GR 1.1	6128.5	6623.1	8501.1	6623.1
Noz K4	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
Noz K8	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
D5	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
D1	1261.0	1960.0	235	150	GR 1.1	...	...	...	...
N5	8501.1	10200.0	235	600	GR 1.1	5319.7	7431.9	8501.1	7431.9

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1261.040 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N6	357.251	0.50	1816	2295	1261.0	4167.0	Pass
Noz k6	77.851	0.50	50	382	1261.0	6623.1	Pass
N5	111.125	0.50	129	612	1261.0	7431.9	Pass

Evaluating Flange on Nozzle [N6]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) 357.251 \leq \pi \cdot 357.251^3 [(4254.041 - 1261.040) + .500 \cdot 4254.041]$$

$$= 0.32342E+05 \leq 0.74791E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4254.041, 0.500 \cdot 4254.041 + 4254.041 - 32342.3 / (\pi \cdot 357.251^3))$$

$$= \min(4254.041, 4166.974)$$

$$= 4166.974 \text{ KPa.}$$

Evaluating Flange on Nozzle [Noz k6]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(50) + 4 \cdot \text{abs}(382) 77.851 \leq \pi \cdot 77.851^3 [(8501.077 - 1261.040) + .500 \cdot 8501.077]$$

$$= 0.92641E+03 \leq 0.17370E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz k6 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8501.077, 0.500 \cdot 8501.077 + 8501.077 - 926.4 / (\pi \cdot 77.851^3))$$



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Nozzle Flange MAWP:

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= min(8501.077, 6623.125)
= 6623.125 KPa.

Evaluating Flange on Nozzle [N5]:

= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
= $16(129) + 4 \cdot \text{abs}(612)111.125 \leq \pi \cdot 111.125^3 [(8501.077 - 1261.040) + .500 \cdot 8501.077]$
= $0.23387E+04 \leq 0.50516E+04$ Check Passes

Maximum Pressure for Flange on Nozzle N5 [Pmax]:

= min(PR, $FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)$)
= min(8501.077, $0.500 \cdot 8501.077 + 8501.077 - 2338.7 / (\pi \cdot 111.125^3)$)
= min(8501.077, 7431.919)
= 7431.919 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Natural Frequency Calculation: Step: 16 9:33am Sep 9, 2026

The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 29.3868 Hz.

The Natural Frequency for the Vessel (Ope...) is 30.9137 Hz.

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Wind Load Calculation:

Step: 17 9:33am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	25500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.91)/(1 + 1.7 \cdot 3.4 \cdot 0.19)))$$

$$= \min(0.85, 0.88)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	30.914 Hz
Natural Frequency of Vessel (Empty)	29.387 Hz
Natural Frequency of Vessel (Test)	31.983 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.538
Structure Height to Diameter ratio	3.286

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 2000 + 25500 = 27500 \text{ mm}$$

$$= 90.2 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (90.2 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (90.2/900)^{2/9.5}$$

$$= 1.24$$

Type of Hill: No Hill



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 Wind Load Calculation: Step: 17 9:33am Sep 9,2026

Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0 \cdot 0 \cdot 0)^2$
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 1.24 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16 \cdot 0.85 \cdot 0.54 \cdot 27$
 = 197 lbs. [89.5] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Ch Shell	27500.0	0.000	0.000	0.000	1.239	1.000	78.120
Port shell-1	27500.0	0.000	0.000	0.000	1.239	1.000	78.120
cone-1	27500.0	0.000	0.000	0.000	1.239	1.000	78.120
Main Shell	27500.0	0.000	0.000	0.000	1.239	1.000	78.120
Main Shell 2	27500.0	0.000	0.000	0.000	1.239	1.000	78.120
Node 60 to 70	27500.0	0.000	0.000	0.000	1.239	1.000	78.120
Node 70 to 80	27500.0	0.000	0.000	0.000	1.239	1.000	78.120

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	27500	2505.6	25056	78.12	98.4806
20	30	27500	2457.6	5652.48	78.12	22.2166
30	40	27500	3018	54324	78.12	213.516
40	50	27500	3643.2	72864	78.12	286.386
50	60	27500	3643.2	54648	78.12	214.79
60	70	27500	3643.2	36432	78.12	143.193
70	80	27500	3643.2	21355.7	78.12	83.9368

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(11.20.75)$
 = 0.122 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/30.9)
 = 0.032

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.12$, 0.032) per 12.8.2
 = 0.032
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 76488 \cdot 1.25$
 = 4497.522 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 76488$
 = 2398.7 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	900	9561.06	417.296



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Earthquake Load Calculation:

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20	30	900	9561.06	417.296
30	40	900	9561.06	417.296
40	50	1400	9561.06	649.127
40	50	1400	9561.06	649.127
50	60	1400	9561.06	649.127
60	70	1400	9561.06	649.127
70	80	1400	9561.06	649.127

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 19 9:33am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4197.000 mm
Center of Gravity of the Liquid	5008.593 mm
Center of Gravity of the Insulation	4766.354 mm
Center of Gravity of the Nozzles	4885.543 mm
Center of Gravity of the Tubesheet(s)	1133.500 mm
Center of Gravity of the Tubes	2650.000 mm
Center of Gravity of the Baffles	3082.312 mm
Center of Gravity of Bare Shell New and Cold	4294.055 mm
Center of Gravity of Bare Shell Corroded	4207.343 mm
Vessel CG in the Operating Condition	4073.234 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3351.293 mm
Vessel CG in the Test Condition	4096.337 mm

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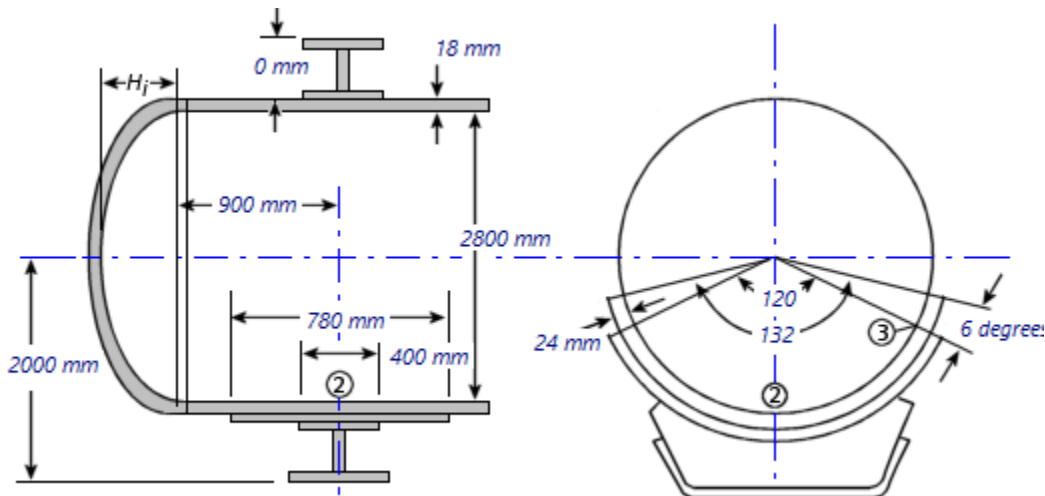
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1409.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6580.80	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		43870.75	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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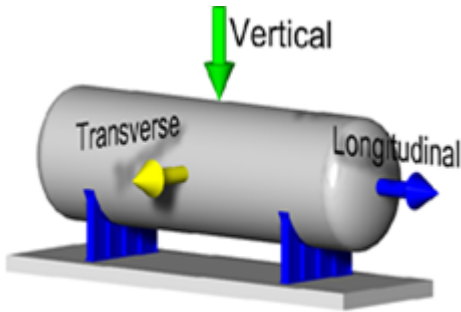
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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.43	-234.00
Long. Stress at Top of Midspan	3.43	138.00
Long. Stress at Bottom of Midspan	3.43	138.00
Long. Stress at Top of Saddles	-1.00	-234.00
Long. Stress at Top of Saddles	1.00	138.00
Long. Stress at Bottom of Saddles	0.55	138.00
Tangential Shear in Shell	14.43	110.40
Circ. Stress at Horn of Saddle	19.12	207.00
Circ. Compressive Stress in Shell	1.20	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 291 \cdot 0.00256 \cdot 57.6^2 \cdot 1$$

$$= 2471.7 \text{ lbf}$$

$$= 1121.156 \text{ Kg}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 9.96^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2$$

$$= 794.3 \text{ lbf}$$

$$= 360.3 \text{ Kg}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw}$$

$$= 0.1 \cdot 35575 \cdot 1 + 1121$$

$$= 4678.7 \text{ Kg}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$$

$$= 0.1 \cdot 35575 \cdot 1 + 360$$

$$= 3917.8 \text{ Kg}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$= F_t / \text{Num of Saddles} \cdot B / E$$

$$= 4679/1 \cdot 2000/2498$$

$$= 3746.5 \text{ Kg}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$= F_l \cdot B / L_s$$

$$= 3918 \cdot 2000/7848$$

$$= 998.5 \text{ Kg}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$= \text{Saddle Load} \cdot g_y$$

$$= 38214 \cdot 0.05$$



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= 1910.677 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 38214 + 1911 + max(998, 3747)
 = 43870.7 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	45385.94	Kgf
Transverse Shear Load	4678.70	Kgf
Longitudinal Shear Load	3917.83	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(2800/2 + 18 + 24)sin(120/2), 2•1409)
 = 2497.617 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0242	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0199			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -43871•900[1-(1-900/6581+(1409²-0²)/
 (2•900•6581))/(1+(4•0)/(3•6581))]
 = 1217.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 43871•6581/4((1+2(1409²-0²)/(6581²))/(1+(4•0)/
 (3•6581))-4•900/6581)
 = 39310.7 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 1409/(2•18) - 39311/(π•1409²•18)
 = -3.43 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 1409/(2 • 18) + 39311/(π • 1409² • 18)
 = 3.43 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•1409/(2•18)-1218/(0.11•π•1409²•18)
 = -1.00 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•1409/(2•18)+1218/(0.19•π•1409²•18)
 = 0.55 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 43871 (6581 - 2 • 900)/(6581 + (4 • 0/3))
 = 31871.1 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 31871 / (1409 \cdot 18)$$

$$= 14.43 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1409 \cdot 18)}$$

$$= 124.219 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1409 \cdot 18)}, 2 \cdot 900)$$

$$= 648.44 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 43871 \cdot 0.1 / (648(18 + 1 \cdot 24))$$

$$= -1.20 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -43871 / (4(18 + 1 \cdot 24)648) -$$

$$12 \cdot 0.024 \cdot 43871 \cdot 1409 / (6581(18 + 1 \cdot 24)^2)$$

$$= -19.12 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1400 \cdot 18)} + 780)18^3) / 12$$

$$= 49.943 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 49.9 + 185(198 - 9)^2$$

$$= 66143.000 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1027.6	18.0	9.0	185.0	166478.0	0.661E+05
Wearplate	780.0	24.0	30.0	187.2	561600.0	0.529E+05
Web	14.0	522.0	303.0	73.1	2214322.8	0.246E+05
BasePlate	425.0	36.0	582.0	153.0	8904598.0	0.226E+06
Totals	...	...	...	598.3	11846999.0	0.369E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 11846999/598$$

$$= 198.026 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 43871$$

$$= 8928.6484 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 2.1187 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2000 - 1400 \cdot \sin(120/2) / 1.05$$

$$= 842.209 \text{ mm}$$

Bending Moment, M = Fh * d = 7519.9385 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.9525 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(43871 + 1515)425) / (4 \cdot 2500 \cdot 173)}$$

$$= 18.094 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1400 + 18 + 24)$$

$$= 2497.617 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2498 / (5 - 1)$$

$$= 624.404 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 624 \cdot 425 / 2$$

$$= 1326.859 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 45386 / (2500 \cdot 425)$$

$$= 4.272 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1327 \cdot 4.27$$

$$= 5667.835 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 54 + 43.7$$

$$= 97.748 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 5668/97.7$$

$$= 5.686 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 14$$

$$= 386.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/97.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7467/97.7)}$$

$$= 87.399 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2000 - \cos(120/2) (1400 + 18 + 24) - 36$$

$$= 1243.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1243/87.4$$

$$= 14.222$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 3918 / (2 \cdot 2500) \cdot 624 \cdot 1243/2$$

$$= 304.082 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($14.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 123) - (14.2^3) / (8 \cdot 123^3))$$

$$= 151.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.69/151 + (304 \cdot 200/74666672) / 173$$



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= 0.084

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	624.4	14.0	0.0	87.4	0.0	14.3
Totals	...	...	...	141.5	...	0.748E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/141

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 400/2

= 200.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 2000 - cos(1442/2)(36 + 0 + 0) - 0

= 522.000 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(7481/141)}$

= 72.721 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 522/72.7

= 7.178

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3918/(2 • 2500)

= 0.784 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.78 • 624 • 522

= 255.400 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 255/374031

= 6.696 N./mm²

Compressive Allowable, KL/r < Cc (7.18 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.18)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (7.18) / (8 \cdot 123) - (7.18^3) / (8 \cdot 123^3))$

= 153.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 7.86/154 + (255 • 200/7481)/173

= 0.090



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Saddle Calcs: Operating Case:

Step: 21

9:33am

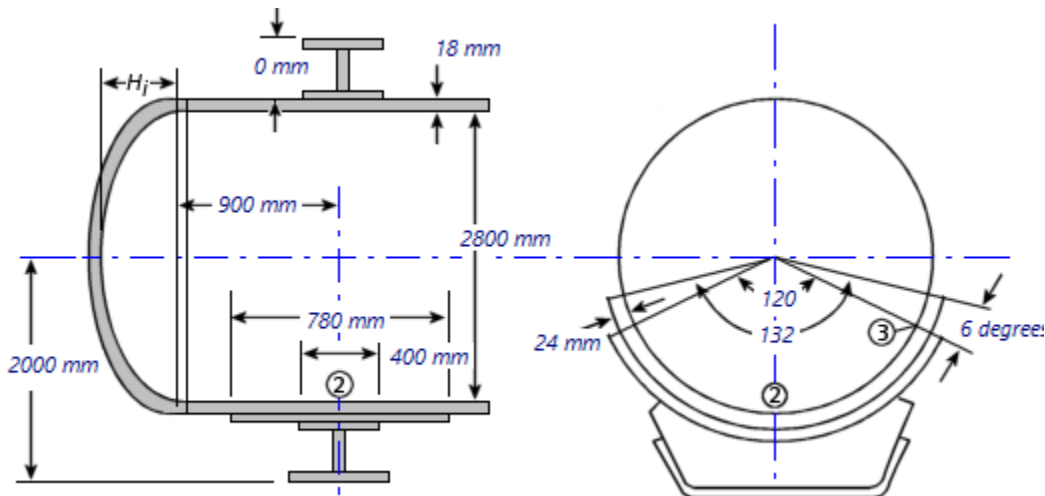
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1410.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6580.80	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	89691.48	Kgf
Pressure used in Analysis	P	1263.723 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	51.01	138.00
Long. Stress at Bottom of Midspan	67.83	138.00
Long. Stress at Top of Saddles	56.95	138.00
Long. Stress at Bottom of Saddles	60.79	138.00
Tangential Shear in Shell	35.36	110.40
Circ. Stress at Horn of Saddle	44.93	172.50
Circ. Compressive Stress in Shell	2.74	138.00



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Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1063/1 + 0) \cdot 2000/2498$$

$$= 2552.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(896, 0) \cdot 2000/7848$$

$$= 228.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4498, 0) \cdot 2000/7848$$

$$= 1146.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (4498/1 + 0) \cdot 2000/2498$$

$$= 10804.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 78887 + \max(228, 2552, 1146, 10804)$$

$$= 89691.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(3.04 \cdot 1.2)^2)$$

$$= 8784.681 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	91206.67	Kgf
Transverse Shear Load	4497.52	Kgf
Longitudinal Shear Load	4497.52	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2800/2 + 18 + 24)\sin(120/2), 2 \cdot 1410)$$

$$= 2497.617 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0242	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0199			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -89691 \cdot 900 [1 - (1 - 900/6581 + (1410^2 - 0^2)/(2 \cdot 900 \cdot 6581))/(1 + (4 \cdot 0)/(3 \cdot 6581))]$$

$$= 2518.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 89691 \cdot 6581/4((1 + 2(1410^2 - 0^2)/(6581^2))/(1 + (4 \cdot 0)/$$



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Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9, 2026

$$(3 \cdot 6581) - 4 \cdot 900 / 6581 \\ = 80397.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1264 \cdot 1410 / (2 \cdot 15) - 80397 / (\pi \cdot 1410^2 \cdot 15) \\ = 51.01 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1264 \cdot 1410 / (2 \cdot 15) + 80397 / (\pi \cdot 1410^2 \cdot 15) \\ = 67.83 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1264 \cdot 1410 / (2 \cdot 15) - 2518 / (0.11 \cdot \pi \cdot 1410^2 \cdot 15) \\ = 56.95 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1264 \cdot 1410 / (2 \cdot 15) + 2518 / (0.19 \cdot \pi \cdot 1410^2 \cdot 15) \\ = 60.79 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 89691 (6581 - 2 \cdot 900) / (6581 + (4 \cdot 0 / 3)) \\ = 65158.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 65159 / (1410 \cdot 15) \\ = 35.36 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{1410 \cdot 15} \\ = 113.456 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(400 + 1.56 \cdot \sqrt{1410 \cdot 15}, 2 \cdot 900) \\ = 626.91 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(138 / 138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 89691 \cdot 0.1 / (627(15 + 1 \cdot 24)) \\ = -2.74 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2) \\ = -89691 / (4(15 + 1 \cdot 24)627) - 12 \cdot 0.024 \cdot 89691 \cdot 1410 / (6581(15 + 1 \cdot 24)^2) \\ = -44.93 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm



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Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9,2026

Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	219.0	N./mm ²
Height of Web at Center	Hw, c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1403 \cdot 15)} + 780)15^3)/12$$

$$= 28.302 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 28.3 + 151(207 - 7.5)^2$$

$$= 60001.324 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1006.3	15.0	7.5	150.9	113209.6	0.600E+05
Wearplate	780.0	24.0	27.0	187.2	505440.0	0.606E+05
Web	14.0	522.0	300.0	73.1	2192398.8	0.229E+05
BasePlate	425.0	36.0	579.0	153.0	8858699.0	0.212E+06
Totals	...	...	...	564.2	11669747.0	0.356E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 11669747/564$$

$$= 206.827 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2)/(\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 89691$$

$$= 18254.1602 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.3316 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 131.4001 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$$

$$= 2000 - 1403 \cdot \sin(120/2)/1.05$$

$$= 839.728 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 15328.8379 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 8.7420 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 146.0001 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(89691 + 1515)425 / (4 \cdot 2500 \cdot 146))}$$

$$= 27.948 \text{ mm}$$



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 Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9,2026

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (1400 + 18 + 24)$
 $= 2497.617 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 2498 / (5 - 1)$
 $= 624.404 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 624 \cdot 425 / 2$
 $= 1326.859 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 91207 / (2500 \cdot 425)$
 $= 8.584 \text{ Kg/cm}^2$

Axial Load [P]:
 $= \text{Ap} \cdot \text{Bp}$
 $= 1327 \cdot 8.58$
 $= 11389.968 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 54 + 43.7$
 $= 97.748 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / \text{Ar}$
 $= 11390 / 97.7$
 $= 11.427 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 400 - 14$
 $= 386.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= \text{AY} / \text{A}$
 $= 0 / 97.7$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 400 / 2$
 $= 200.000 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(7467 / 97.7)}$
 $= 87.399 \text{ mm}$



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Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9,2026

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2000 - \cos(120/2) (1400 + 18 + 24) - 36$$

$$= 1243.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (1400 + 18 + 24)$$

$$= 2497.617 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2500 - 50.8$$

$$= 2449.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2498 - 2449) / 2$$

$$= 24.209 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1243^2 + 24.2^2}$$

$$= 1243.236 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 219)}$$

$$= 134.248$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1243 / 87.4$$

$$= 14.225$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 4498 / (2 \cdot 2500) \cdot 624 \cdot 1243 / 2$$

$$= 349.141 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.2 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 134^2)) 219 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 134) - (14.2)^3 / (8 \cdot 134^3))$$

$$= 127.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.4 / 128 + (349 \cdot 200 / 74666672) / 146$$

$$= 0.152$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	624.4	14.0	0.0	87.4	0.0	14.3
Totals	...	...	...	141.5	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 141$$

$$= 0.000 \text{ mm}$$



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Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9, 2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2000 - \cos(1442/2)(36 + 0 + 0) - 0$$

$$= 522.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7481/141}$$

$$= 72.721 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 522/72.7$$

$$= 7.178$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4498 / (2 \cdot 2500)$$

$$= 0.900 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.9 \cdot 624 \cdot 522$$

$$= 293.189 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 293/374031$$

$$= 7.687 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.18 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.18)^2 / (2 \cdot 134^2)) 219 /$$

$$(5/3 + 3 \cdot (7.18) / (8 \cdot 134) - (7.18^3) / (8 \cdot 134^3))$$

$$= 129.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 15.8/130 + (293 \cdot 200/7481) / 146$$

$$= 0.175$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	80402.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1146.2 Kg
Maximum Transverse Force	Ft	4497.5 Kg
Maximum Longitudinal Force	F1	4497.5 Kg
Saddle Bolted to Steel Foundation		No



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Saddle Calcs: Operating Case: Step: 21 9:33am Sep 9, 2026

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4498 / (4.87 \cdot 8)$$

$$= 11.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4498 / (4.87 \cdot 8)$$

$$= 11.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 4498 / (82.8 \cdot 8)$$

$$= 0.6659 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2000 \cdot 4498 + 0$$

$$= 8995.22 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 8995/80402$$

$$= 111.88 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.67, 0]$$

$$= 0.6659 \text{ cm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 18, tr = 12.9, c = 3 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.86, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-11 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit

-27 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-35 °C

Min Metal Temp. w/o impact per UG-20(f)

-29 °C

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Saddle Calcs: Test Case:

Step: 22

9:33am

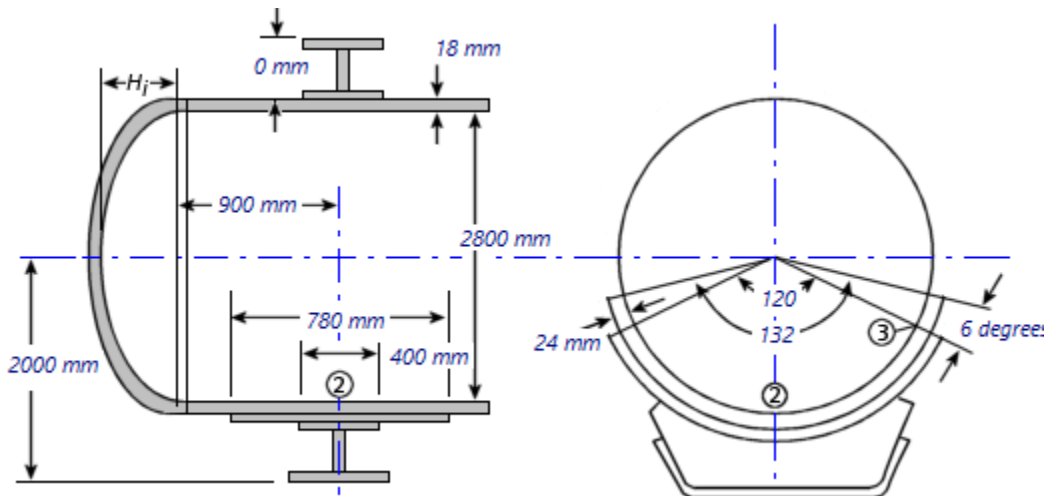
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1409.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6580.80	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	77110.87	Kgf
Pressure used in Analysis	P	1638.811 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	58.11	234.00
Long. Stress at Bottom of Midspan	70.18	234.00
Long. Stress at Top of Saddles	62.39	234.00
Long. Stress at Bottom of Saddles	65.12	234.00
Tangential Shear in Shell	25.36	138.58
Circ. Stress at Horn of Saddle	33.61	350.99
Circ. Compressive Stress in Shell	2.11	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (531/1 + 0) \cdot 2000/2498$$

$$= 1276.2 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(448, 0) \cdot 2000/7848$$

$$= 114.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 75835 + \max(114, 1276, 0, 0)$$

$$= 77110.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(3.04 \cdot 1.2)^2)$$

$$= 8784.681 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 78626.05 Kgf

Transverse Shear Load 531.26 Kgf

Longitudinal Shear Load 447.90 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2800/2 + 18 + 24)\sin(120/2), 2 \cdot 1409)$$

$$= 2497.617 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0242 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0199

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -77111 \cdot 900 [1 - (1 - 900/6581 + (1409^2 - 0^2)/(2 \cdot 900 \cdot 6581))/(1 + (4 \cdot 0)/(3 \cdot 6581))]$$

$$= 2140.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 77111 \cdot 6581/4((1 + 2(1409^2 - 0^2)/(6581^2))/(1 + (4 \cdot 0)/(3 \cdot 6581)) - 4 \cdot 900/6581)$$

$$= 69095.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 1639 \cdot 1409/(2 \cdot 18) - 69096/(\pi \cdot 1409^2 \cdot 18)$$

$$= 58.11 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 1639 \cdot 1409 / (2 \cdot 18) + 69096 / (\pi \cdot 1409^2 \cdot 18)$$

$$= 70.18 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1639 \cdot 1409 / (2 \cdot 18) - 2140 / (0.11 \cdot \pi \cdot 1409^2 \cdot 18)$$

$$= 62.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1639 \cdot 1409 / (2 \cdot 18) + 2140 / (0.19 \cdot \pi \cdot 1409^2 \cdot 18)$$

$$= 65.12 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 77111 (6581 - 2 \cdot 900) / (6581 + (4 \cdot 0/3))$$

$$= 56019.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 56019 / (1409 \cdot 18)$$

$$= 25.36 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1409 \cdot 18)}$$

$$= 124.219 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1409 \cdot 18)}, 2 \cdot 900)$$

$$= 648.44 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 77111 \cdot 0.1 / (648(18 + 1 \cdot 24))$$

$$= -2.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -77111 / (4(18 + 1 \cdot 24)648) -$$

$$12 \cdot 0.024 \cdot 77111 \cdot 1409 / (6581(18 + 1 \cdot 24)^2)$$

$$= -33.61 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1400 \cdot 18)} + 780)^3) / 12$$

$$= 49.943 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 49.9 + 185 (198 - 9)^2$$

$$= 66143.000 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1027.6	18.0	9.0	185.0	166478.0	0.661E+05
Wearplate	780.0	24.0	30.0	187.2	561600.0	0.529E+05
Web	14.0	522.0	303.0	73.1	2214322.8	0.246E+05
BasePlate	425.0	36.0	582.0	153.0	8904598.0	0.226E+06
Totals	...	...	...	598.3	11846999.0	0.369E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 11846999/598$$

$$= 198.026 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 77111$$

$$= 15693.7324 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.7240 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2000 - 1400 \cdot \sin(120/2) / 1.05$$

$$= 842.209 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 13217.6660 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.9472 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(77111 + 1515) \cdot 425) / (4 \cdot 2500 \cdot 173)}$$

$$= 23.816 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1400 + 18 + 24)$$

$$= 2497.617 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2498 / (5 - 1)$$

$$= 624.404 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 624 \cdot 425 / 2$$

$$= 1326.859 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 78626 / (2500 \cdot 425)$$

$$= 7.400 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1327 \cdot 7.4$$

$$= 9818.890 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 54 + 43.7$$

$$= 97.748 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 9819 / 97.7$$

$$= 9.851 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 400 - 14$$

$$= 386.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 97.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (7467 / 97.7) }$$

$$= 87.399 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 2000 - \cos(120 / 2) (1400 + 18 + 24) - 36$$

$$= 1243.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1243/87.4$$

$$= 14.222$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 448 / (2 \cdot 2500) \cdot 624 \cdot 1243 / 2$$

$$= 34.764 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 123) - (14.2^3) / (8 \cdot 123^3))$$

$$= 151.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 9.85/151 + (34.8 \cdot 200/74666672) / 173$$

$$= 0.070$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	624.4	14.0	0.0	87.4	0.0	14.3
Totals	...	...	...	141.5	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/141$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 2000 - \cos(1442/2) (36 + 0 + 0) - 0$$

$$= 522.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (7481/141) }$$

$$= 72.721 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 522/72.7$$

$$= 7.178$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 448 / (2 \cdot 2500)$$

$$= 0.090 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.09 \cdot 624 \cdot 522$$

$$= 29.198 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 29.2/374031
= 0.766 N./mm²

Compressive Allowable, $KL/r < C_c$ ($7.18 < 123$) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (7.18)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (7.18) / (8 \cdot 123) - (7.18^3) / (8 \cdot 123^3))$
= 153.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
= $13.6/154 + (29.2 \cdot 200/7481) / 173$
= 0.093

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load QO (Weight)	QO	77349.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	114.1 Kg
Maximum Transverse Force	Ft	531.3 Kg
Maximum Longitudinal Force	F1	447.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $448 / (4.87 \cdot 8)$
= 1.1 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $531 / (4.87 \cdot 8)$
= 1.3 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (BoltShearAllowable \cdot Nbolts)$
= $448 / (29 \cdot 8)$
= 0.1895 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
= $2000 \cdot 531 + 0$
= 1062.54 Kg-m.

Eccentricity (e):

= $Rmom / QO$
= $1063/77350$
= 13.74 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.19, 0]

= 0.1895 cm²

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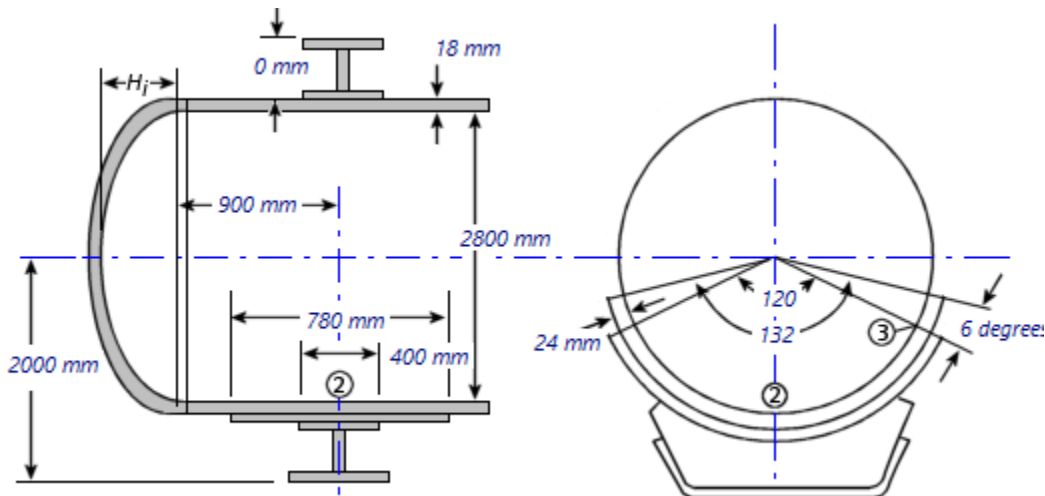
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1410.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	6580.80	mm
Distance from Saddle to Vessel tangent	a1 or a	900.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	89691.48	Kgf
Pressure used in Analysis	P	-103.419 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-13.27	-153.61
Long. Stress at Top of Midspan		13.27	138.00
Long. Stress at Bottom of Midspan		3.55	138.00
Long. Stress at Top of Saddles		-7.33	-153.61
Long. Stress at Top of Saddles		7.33	138.00
Long. Stress at Bottom of Saddles		-3.49	-153.61
Long. Stress at Bottom of Saddles		3.49	138.00



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Tangential Shear in Shell	35.36	110.40
Circ. Stress at Horn of Saddle	44.93	172.50
Circ. Compressive Stress in Shell	2.74	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1063/1 + 0) \cdot 2000/2498$$

$$= 2552.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(896, 0) \cdot 2000/7848$$

$$= 228.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4498, 0) \cdot 2000/7848$$

$$= 1146.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (4498/1 + 0) \cdot 2000/2498$$

$$= 10804.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 78887 + \max(228, 2552, 1146, 10804)$$

$$= 89691.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (3.04 \cdot 1.2)^2)$$

$$= 8784.681 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	91206.67	Kgf
Transverse Shear Load	4497.52	Kgf
Longitudinal Shear Load	4497.52	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2800/2 + 18 + 24)\sin(120/2), 2 \cdot 1410)$$

$$= 2497.617 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0242	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0199			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -89691 \cdot 900 [1 - (1 - 900/6581 + (1410^2 - 0^2)/$$

$$(2 \cdot 900 \cdot 6581)/(1 + (4 \cdot 0)/(3 \cdot 6581))]$$

$$= 2518.1 \text{ Kg-m.}$$



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Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 89691 \cdot 6581 / 4 \cdot ((1 + 2(1410^2 - 0^2) / (6581^2)) / (1 + (4 \cdot 0) / (3 \cdot 6581)) - 4 \cdot 900 / 6581)$$

$$= 80397.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 1410 / (2 \cdot 15) - 80397 / (\pi \cdot 1410^2 \cdot 15)$$

$$= -13.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 1410 / (2 \cdot 15) + 80397 / (\pi \cdot 1410^2 \cdot 15)$$

$$= 3.55 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 1410 / (2 \cdot 15) - 2518 / (0.11 \cdot \pi \cdot 1410^2 \cdot 15)$$

$$= -7.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -103 \cdot 1410 / (2 \cdot 15) + 2518 / (0.19 \cdot \pi \cdot 1410^2 \cdot 15)$$

$$= -3.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 89691 (6581 - 2 \cdot 900) / (6581 + (4 \cdot 0 / 3))$$

$$= 65158.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 65159 / (1410 \cdot 15)$$

$$= 35.36 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{1410 \cdot 15}$$

$$= 113.456 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1410 \cdot 15}, 2 \cdot 900)$$

$$= 626.91 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 89691 \cdot 0.1 / (627(15 + 1 \cdot 24))$$

$$= -2.74 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -89691 / (4(15 + 1 \cdot 24)627) - 12 \cdot 0.024 \cdot 89691 \cdot 1410 / (6581(15 + 1 \cdot 24)^2)$$

$$= -44.93 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm



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Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	219.0	N./mm ²
Height of Web at Center	Hw,c	625.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1403 \cdot 15)} + 780)15^3)/12$
 $= 28.302 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 28.3 + 151(207 - 7.5)^2$
 $= 60001.324 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1006.3	15.0	7.5	150.9	113209.6	0.600E+05
Wearplate	780.0	24.0	27.0	187.2	505440.0	0.606E+05
Web	14.0	522.0	300.0	73.1	2192398.8	0.229E+05
BasePlate	425.0	36.0	579.0	153.0	8858699.0	0.212E+06
Totals	...	...	...	564.2	11669747.0	0.356E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 11669747/564$
 $= 206.827 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 89691$
 $= 18254.1602 \text{ Kg}$

Tension Stress, St = (Fh/As) = 4.3316 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 131.4001 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 2000 - 1403 \cdot \sin(120/2) / 1.05$
 $= 839.728 \text{ mm}$

Bending Moment, M = Fh * d = 15328.8379 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 8.7420 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 146.0001 N./mm²

Minimum Thickness of Baseplate per Moss:



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$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(89691 + 1515) \cdot 425}{4 \cdot 2500 \cdot 146}}$$

$$= 27.948 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1400 + 18 + 24)$$

$$= 2497.617 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2498 / (5 - 1)$$

$$= 624.404 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 624 \cdot 425 / 2$$

$$= 1326.859 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 91207 / (2500 \cdot 425)$$

$$= 8.584 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1327 \cdot 8.58$$

$$= 11389.968 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 54 + 43.7$$

$$= 97.748 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 11390 / 97.7$$

$$= 11.427 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 14$$

$$= 386.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 97.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7467/97.7}$$

$$= 87.399 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2000 - \cos(120/2) (1400 + 18 + 24) - 36$$

$$= 1243.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (1400 + 18 + 24)$$

$$= 2497.617 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2500 - 50.8$$

$$= 2449.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2498 - 2449) / 2$$

$$= 24.209 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1243^2 + 24.2^2}$$

$$= 1243.236 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 219)}$$

$$= 134.248$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1243 / 87.4$$

$$= 14.225$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 4498 / (2 \cdot 2500) \cdot 624 \cdot 1243 / 2$$

$$= 349.141 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.2 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 134^2)) 219 / (5/3 + 3 \cdot (14.2) / (8 \cdot 134) - (14.2)^3 / (8 \cdot 134^3))$$

$$= 127.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.4/128 + (349 \cdot 200/74666672) / 146$$

$$= 0.152$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	624.4	14.0	0.0	87.4	0.0	14.3
Totals	...	...	...	141.5	...	0.748E+04

Distance to Centroid from Datum [ytot]:



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= AY / A
= 0/141
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 400/2
= 200.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
= 2000 - cos(1442/2)(36 + 0 + 0) - 0
= 522.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7481/141}$
= 72.721 mm

Slenderness ratio [KL/r]:

= KL/r
= 1 • 522/72.7
= 7.178

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
= 4498/(2 • 2500)
= 0.900 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
= 0.9 • 624 • 522
= 293.189 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
= 293/374031
= 7.687 N./mm²

Compressive Allowable, $KL/r < C_c$ (7.18 < 134) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (7.18)^2 / (2 \cdot 134^2)) 219 /$
 $(5/3 + 3 \cdot (7.18) / (8 \cdot 134) - (7.18^3) / (8 \cdot 134^3))$
= 129.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
= 15.8/130 + (293 • 200/7481)/146
= 0.175

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Conical Section: cone-1

Cone: 5 9:33am Sep 9,2026

Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

cone-1

Design Internal Pressure, Small End	1276.985	KPa.
Design Internal Pressure, Large End	1276.985	KPa.
Temperature for Internal Pressure	235.0	°C
Design External Pressure	103.419	KPa.
Temperature for External Pressure	200.0	°C

CONE Properties:

Cone Material		SA-516 70	
Cone Allowable Stress at Temperature	Ss	138.00	N./mm ²
Cone Allowable Stress At Ambient		138.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	20.0000	mm
Corrosion Allowance for Cone	Cac	3.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	1800.000	mm
Diameter of Large End of Cone	Dl	2800.000	mm
Half Apex Angle for Cone		27.82	degrees
Axial Length of Cone	Lc	1800.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	
Operating Allowable Stress	Ss	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	24.0000	mm
Corrosion Allowance	Cas	3.0000	mm
Axial Length of Attached Element	Ls	6814.164	mm

LARGE-END Properties:

Attached Large End Material Specification		SA-516 70	
Operating Allowable Stress	Sl	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	18.0000	mm
Corrosion Allowance	Cal	3.0000	mm
Axial Length of Attached Element	Ll	6814.164	mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



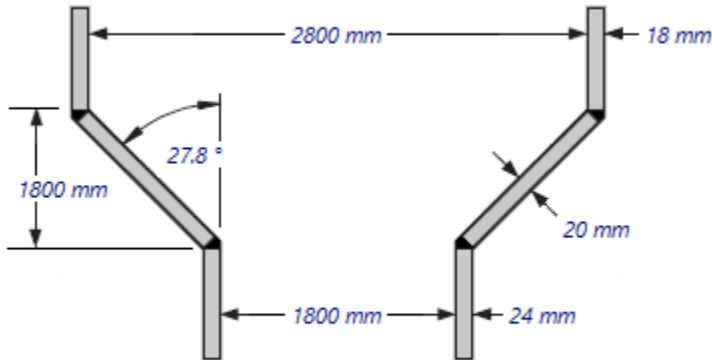
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Cone Section: cone-1

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Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 200 °C

Elastic Modulus of Cone Material	0.191E+09 KPa. at 200 °C
Elastic Modulus of Small Cylinder Material	0.193E+09 KPa. at 200 °C
Elastic Modulus of Large Cylinder Material	0.193E+09 KPa. at 200 °C
Elastic Modulus of Large End Reinforcement	0.193E+09 KPa. at 200 °C
Elastic Modulus of Small End Reinforcement	0.193E+09 KPa. at 200 °C

Elastic Modulus Data from ASME Section II Part D at 235 °C

Elastic Modulus of Cone Material	0.189E+09 KPa. at 235 °C
Elastic Modulus of Small Cylinder Material	0.190E+09 KPa. at 235 °C
Elastic Modulus of Large Cylinder Material	0.190E+09 KPa. at 235 °C
Elastic Modulus of Large End Reinforcement	0.190E+09 KPa. at 235 °C
Elastic Modulus of Small End Reinforcement	0.190E+09 KPa. at 235 °C

Axial forces and moments are not computed for Horizontal geometries.
Compute them manually and enter the loads using the Force/Moment dialog.
Place the loads at the ends of the cone as needed.

Note: Neither end of Cone is a Line of Support

Maximum Centroid Reinforcement Distance Large End	36.4606 mm
Maximum Centroid Reinforcement Distance Small End	34.8246 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00925

Large End Maximum half apex angle w/o reinforcement [δ]:

$$= 326.6 \cdot \sqrt{\left(\frac{P}{Ss \cdot E1} \right)}$$
 per 1-5 equation 1

$$= 326.6 \cdot \sqrt{\left(\frac{1277}{138 \cdot 1} \right)}$$



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= 31.419 degrees

Large end actual half apex angle [alpha]: 27.824 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the large end under internal pressure.

Required Area of Reinforcement for Large End Under External Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00075

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 104 \cdot \sqrt{\frac{P}{(Ss \cdot E1)}}$ per 1-8 equation 1
 $= 104 \cdot \sqrt{\frac{103}{(138 \cdot 1)}}$
 = 2.847 degrees

Large end actual half apex angle [alpha]: 27.824 degrees

Intermediate Value [k]:
 $= \max(Y / (Srl \cdot Erl), 1)$
 $= \max(26.58E+09 / (138 \cdot 192580832), 1)$
 = 1.0000

where [Y] is:
 = Large End All. Stress • Large End Elastic Modulus (Ext. temp.)
 $= 138 \cdot 192580832$
 $= 26575673344.0 \text{ (N./mm}^2\text{)}$

Allowable Stress of Large End Material (Ext. Temp)	138.0	N./mm ²
Allowable Stress of Cone Material (Ext. Temp)	138.0	N./mm ²

Required Area of Reinforcement, Large End, External [Arl]:
 $= (k \cdot QL \cdot RI \cdot \tan(\text{angle}) / (Ss \cdot E1)) \cdot (1 - 1/4 \cdot ((P \cdot RI - QL) / QL) \cdot (\delta / \alpha))$
 $= (1 \cdot 7.48 \cdot 1418 \cdot 0.53 / (138 \cdot 1)) \cdot$
 $(1 - 1/4 \cdot ((103 \cdot 1418 - 7.48) / 7.48) \cdot (2.85 / 27.8))$
 = 3.875 cm²

Force per Length, Cone Large End External Pressure, compressive side [QL-]:
 $= P_{ext}(RI/2) + f1$
 $= 103 (1418/2) + 0$
 = 7.477 Kgf/mm

Axial Load per Unit Circumference excluding Pressure [f1]:
 $= F_{axial} / (\pi(DI + TI)) + \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +0 / (\pi(2836 - 18)) + 0 / (\pi(1418 - 18/2)^2)$
 = 0.0 Kgf/mm

Force per Length, Cone Large End External Pressure, tensile side [QL+]:
 $= P_{ext}(RI/2) + f1$
 $= 103 (1418/2) + 0$
 = 7.477 Kgf/mm

Axial Load per Unit Circumference excluding Pressure [f1]:
 $= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +0 / (\pi(2836 - 18)) - 0 / (\pi(1418 - 18/2)^2)$
 = 0.0 Kgf/mm

Available Area of Reinforcement, Large End, External [Ael]:
 $= 0.55 [\sqrt{(DI \cdot ts)(ts - t)} + \sqrt{(DI \cdot tc / \cos(\alpha))(tc - tr)}]$ per 1-8 eqn. 3
 $= 0.55 [\sqrt{(2836 \cdot 15)(15 - 13.3)} +$
 $\sqrt{(2836 \cdot 17/0.88)(17 - 13.1)}]$
 = 6.9157 cm²

Summary of Reinforcement Area, Large End, External Pressure:

Area of reinforcement required per App. 1-8(1)	3.8750	cm ²
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Conical Section: cone-1

Cone: 5

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Area of reinforcement in shell per App. 1-8(2)	6.9157	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.00925

Small End Maximum half apex angle w/o reinforcement [δ]:

$$= 89 \cdot \sqrt{\left(\frac{P}{(Ss \cdot E1)} \right)} \text{ per 1-5 equation 6}$$

$$= 89 \cdot \sqrt{\left(\frac{1277}{(138 \cdot 1)} \right)}$$

$$= 8.562 \text{ degrees}$$

Small end actual half apex angle [alpha]: 27.824 degrees

Intermediate Value [k]:

$$= \max\left(\frac{Y}{(Sr \cdot Ers)}, 1 \right)$$

$$= \max\left(\frac{26.23E+09}{(138 \cdot 190085616)}, 1 \right)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$$

$$= 138 \cdot 190085616$$

$$= 26231341056.0 \text{ (N./mm}^2\text{)}^2$$

Decay Length, Cone Small End:

$$= 1.4 \cdot \sqrt{\left(\frac{Rs}{(ts - ca)} \right)}$$

$$= 1.4 \cdot \sqrt{\left(\frac{903}{(24 - 3)} \right)}$$

$$= 192.789 \text{ mm}$$

Required Area of Reinforcement, Small End, Internal [Ars]:

$$= k \cdot QS \cdot Rs / (Ss \cdot E1) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha) \text{ per 1-5 equation 7}$$

$$= 1 \cdot 58.8 \cdot 903 / (138 \cdot 1) \cdot (1 - 8.56/27.8) 0.53$$

$$= 13.79 \text{ cm}^2$$

Force per Length, Cone Small End, tensile side [QS+]:

$$= P \left(\frac{Rs}{2} \right) + f_2$$

$$= 1277 \left(\frac{903}{2} \right) + 0$$

$$= 58.795 \text{ Kg/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial} / (\pi(Ds + Ts)) + \text{Moment} / (\pi(Rs + Ts/2)^2)$$

$$= -0 / (\pi(1806 + 24)) + 0 / (\pi(903 + 24/2)^2)$$

$$= 0.0 \text{ Kg/mm}$$

Force per Length, Cone Small End, compressive side [QS-]:

$$= P \left(\frac{Rs}{2} \right) + f_2$$

$$= 1277 \left(\frac{903}{2} \right) + -0$$

$$= 58.795 \text{ Kg/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial} / (\pi(Ds + Ts)) - \text{Moment} / (\pi(Rs + Ts/2)^2)$$

$$= -0 / (\pi(1806 + 24)) - 0 / (\pi(903 + 24/2)^2)$$

$$= -0.0 \text{ Kg/mm}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.78 \left[\sqrt{(Rs \cdot ts)} \cdot (ts - t) + \sqrt{(Rs \cdot tc / \cos(\alpha))} \cdot (tc - tr) \right] \text{ per 1-5 eqn. 8}$$

$$= 0.78 \left[\sqrt{(903 \cdot 21)} \cdot (21 - 8.4) + \right.$$

$$\left. \sqrt{(903 \cdot 17 / 0.88)} \cdot (17 - 9.5) \right]$$

$$= 21.2358 \text{ cm}^2$$

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	13.7856	cm ²
--	---------	-----------------

Area of reinforcement in shell per App. 1-5(8)	21.2358	cm ²
--	---------	-----------------



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 Conical Section: cone-1 Cone: 5 9:33am Sep 9, 2026

Area of reinforcement in stiffening ring 0.0000 cm²

Required Area of Reinforcement for Small End Under External Pressure

Note: Small end is in compression, 1.0 will be used for E1 and E2.

Allowable Stress of Small End Material (Ext. Temp) 138.0 N./mm²
 Allowable Stress of Cone Material (Ext. Temp) 138.0 N./mm²

Intermediate Value [k]:
 = max(Y / (Srs • Ers), 1)
 = max(26.58E+09 / (138 • 192580832), 1)
 = 1.0000

where [Y] is:
 = Small End All. Stress • Small End Elastic Modulus (Ext. temp.)
 = 138 • 192580832
 = 26575673344.0 (N./mm²)²

Area of Reinforcement Required in Small End Shell [Ars]:
 = k • QS • Rs • tan(α) / (Ss • E1)
 = (1 • 4.87 • 924 • 0.53 / (138 • 1))
 = 1.689 cm²

Force per Length, Cone Small End External Pressure, compressive side [QS-]:
 = Pext(Rs/2) + f2
 = 103 (924/2) + 0
 = 4.872 Kg/mm

Axial Load per Unit Circumference excluding Pressure [f2]:
 = Faxial / (π(Ds + Ts)) + Moment / (π(Rs - Ts/2)²)
 = +0 / (π(1848 - 24)) + 0 / (π(924 - 24/2)²)
 = 0.0 Kg/mm

Force per Length, Cone Small End External Pressure, tensile side [QS+]:
 = Pext(Rs/2) + f2
 = 103 (924/2) + 0
 = 4.872 Kg/mm

Axial Load per Unit Circumference excluding Pressure [f2]:
 = Faxial / (π(Ds + Ts)) - Moment / (π(Rs - Ts/2)²)
 = +0 / (π(1848 - 24)) - 0 / (π(924 - 24/2)²)
 = 0.0 Kg/mm

Area of Reinforcement Available in Small End Shell [Aes]:
 = 0.55 [√(Ds • ts) (ts - t) + √(Ds • tc / cos(α)) (tc - tr)]
 = 0.55 [√(1848 • 21) (21 - 10.3) +
 √(1848 • 17 / 0.88) (17 - 6.17)]
 = 22.7977 cm²

Summary of Reinforcement Area, Small End, External Pressure:

Area of reinforcement required per App. 1-8(6)	1.6886	cm ²
Area of reinforcement in shell per App. 1-8(7)	22.7977	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:



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Conical Section: cone-1

Cone: 5

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Stress Type	Stress	Allowable	Location
Tensile Stress	128.95	438.00	Small Cyl. Long.
Compres. Stress	-73.40	-438.00	Small Cyl. Long.
Membrane Stress	127.11	-207.00	Small End Tang.
Tensile Stress	193.18	438.00	Cone Longitudinal
Compres. Stress	-115.59	-438.00	Cone Longitudinal
Compres Stress	149.15	-207.00	Cone Tangential
Tensile Stress	416.14	438.00	Large Cyl. Long.
Compres. Stress	-296.05	-438.00	Large Cyl. Long.
Membrane Stress	-54.23	-207.00	Large End Tang.
Tensile Stress	337.14	438.00	Cone Longitudinal
Compres. Stress	-217.33	-438.00	Cone Longitudinal
Compres Stress	-54.51	-207.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure KPa.	Reason for Failure at this Pressure
MAWP	1469.632	Thickness due to internal pressure, Cone Large End
MAPnc	1730.376	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Nozzle Calcs.: M2-Manway

Nozl: 23

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Input, Nozzle Desc: M2-Manway

From: 40

Pressure for Reinforcement Calculations	P	1263.723	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1636	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4797.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		581.4600	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2700	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	65.0000	mm
Height of Beveled Transition	L'	56.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



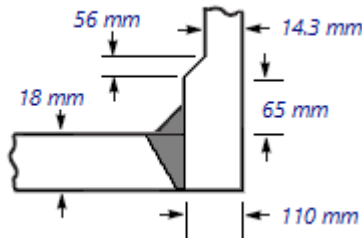
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Nozzle Calcs.: M2-Manway

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Reinforcement CALCULATION, Description: M2-Manway

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	581.460 mm.
Actual Thickness Used in Calculation	14.270 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1264 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1264)$
 $= 12.9198 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1264 \cdot 294 / (137 \cdot 1 - 0.6 \cdot 1264)$
 $= 2.7327 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5737 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 95.7 / 0.5773$
 $= 165.8093 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1174.9202 mm
Parallel to Vessel Wall, opening length	d	587.4601 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	37.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.990$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.990$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.99, 1)$
 $= 0.990$



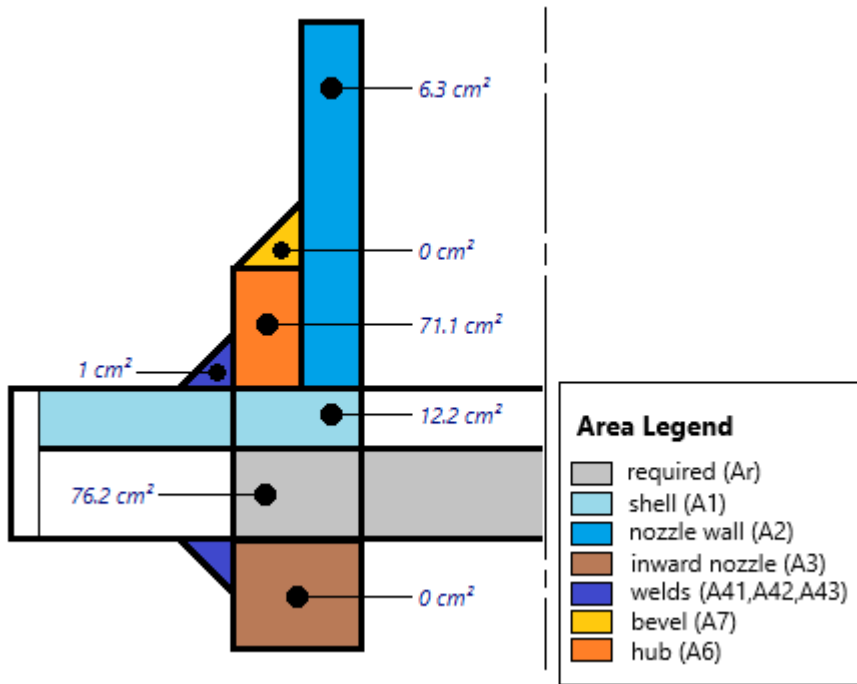
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Nozzle Calcs.: M2-Manway

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	76.179	39.113	NA
Area in Shell	A1	12.175	10.143	NA
Area in Nozzle Wall	A2	6.338	7.198	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.990	0.990	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	71.070	71.070	NA
TOTAL AREA AVAILABLE	Atot	90.573	89.401	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (587 \cdot 12.9 \cdot 1 + 2 \cdot 107 \cdot 12.9 \cdot 1 \cdot (1 - 0.99))$$

$$= 76.179 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 587(1 \cdot 15 - 1 \cdot 12.9) - 2 \cdot 107$$

$$(1 \cdot 15 - 1 \cdot 12.9) \cdot (1 - 0.99)$$

$$= 12.175 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(11.3 - 2.73)0.99$$

$$= 6.338 \text{ cm}^2$$



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Nozzle Calcs.: M2-Manway

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 0.990 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(37.5, 350, 65)) \cdot (110 - 14.3) \cdot 0.99$$

$$= 71.070 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.7327 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.7327 \text{ mm}$$

Available Nozzle Neck Thickness = 14.2700 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 14.3$, $t_r = 2.7$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.24$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 18$, $t_r = 12.9$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-35 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -35 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 1263.72 / 1960.01 = 0.645$$

Weld Size Calculations, Description: M2-Manway

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (76.2 - 12.2 + 2 \cdot 107 \cdot 0.99 \cdot (1 \cdot 15 - 12.9)) 138)$$

$$= 96264.27 \text{ Kg}$$



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Nozzle Calcs.: M2-Manway

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For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (77.4 + 0.99 - 0 \cdot 0.99) \cdot 138$$

$$= 110317.48 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (77.4 + 0 + 0.99 + (31.8)) \cdot 138$$

$$= 155029.25 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (77.4 + 0 + 0.99 + 0 + (31.8)) \cdot 138$$

$$= 155029.25 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 801 \cdot 10 \cdot 0.49 \cdot 137$$

$$= 85924. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 347) \cdot (110 - 3) \cdot 0.7 \cdot 137$$

$$= 1138061. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 801 \cdot (18 - 3) \cdot 0.74 \cdot 138$$

$$= 196638. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (85924 + 1138061) = 1223985 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (85924 + 0 + 196638 + 0) = 282562 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (85924 + 196638 + 0) = 282562 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1223985 Kgf, must exceed W = 96264 Kgf	or W1 = 110317 Kgf
Path 2-2 = 282561 Kgf, must exceed W = 96264 Kgf	or W2 = 155029 Kgf
Path 3-3 = 282561 Kgf, must exceed W = 96264 Kgf	or W3 = 155029 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1368 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 58.5770 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.5770 mm



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Nozzle Calcs.: M2-Manway

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Nozzle Calcs.: N6

Nozl: 24

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Input, Nozzle Desc: N6

From: 40

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1636	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4797.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		324.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



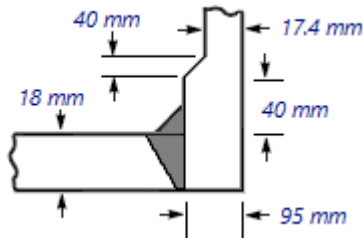
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Nozzle Calcs.: N6

Noz1: 24 9:33am Sep 9, 2026



Reinforcement CALCULATION, Description: N6

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	324.000 mm.
Actual Thickness Used in Calculation	17.450 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1250)$
 $= 12.7788 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1250 \cdot 162 / (137 \cdot 1 + 0.4 \cdot 1250)$
 $= 1.4771 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0900 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 77.6 / 0.5773$
 $= 134.3209 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	590.2008 mm
Parallel to Vessel Wall, opening length	d	295.1004 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	37.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.990$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.990$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.99, 1)$
 $= 0.990$



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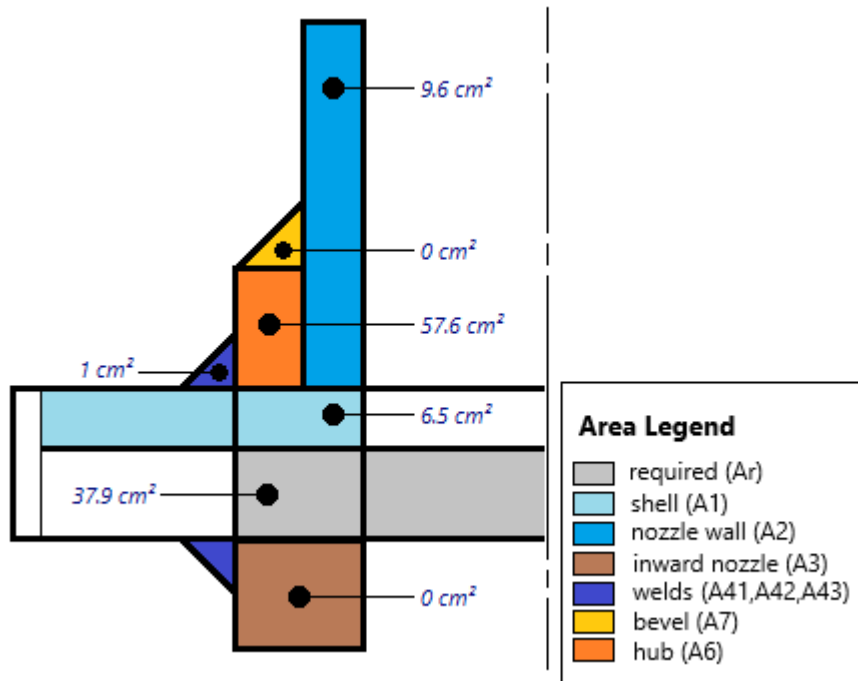
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Nozzle Calcs.: N6

Nozl: 24

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	37.949	19.699	NA
Area in Shell	A1	6.513	5.082	NA
Area in Nozzle Wall	A2	9.631	9.918	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.990	0.990	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	57.573	57.573	NA
TOTAL AREA AVAILABLE	Atot	74.707	73.563	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (295 \cdot 12.8 \cdot 1 + 2 \cdot 92 \cdot 12.8 \cdot 1 \cdot (1 - 0.99))$$

$$= 37.949 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 295(1 \cdot 15 - 1 \cdot 12.8) - 2 \cdot 92$$

$$(1 \cdot 15 - 1 \cdot 12.8) \cdot (1 - 0.99)$$

$$= 6.513 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(14.4 - 1.48)0.99$$

$$= 9.631 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 0.990 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(37.5, 350, 40)) \cdot (95 - 17.4) \cdot 0.99$$

$$= 57.573 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.4771 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 15.7788 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 15.7788 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 4.0520 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(15.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.48, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.6,	Allowable	:	136.6 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	321.7 N./mm ²	Passed
Occasional	:	6.1,	Allowable	:	181.7 N./mm ²	Passed
Shear	:	12.3,	Allowable	:	95.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 17.4$, $t_r = 1.46$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 8 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 18$, $t_r = 12.8$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -27 °C

Min Metal Temp. at Required thickness (UCS 66.1) -35 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -35 °C



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Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 1250.00/5100.00 = 0.245

Weld Size Calculations, Description: N6

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (37.9 - 6.51 + 2 \cdot 92 \cdot 0.99 \cdot (1 \cdot 15 - 12.8)) \cdot 138)$$

$$= 49926.98 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (67.2 + 0.99 - 0 \cdot 0.99) \cdot 138$$

$$= 95959.18 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (67.2 + 0 + 0.99 + (27.3)) \cdot 138$$

$$= 134402.94 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (67.2 + 0 + 0.99 + 0 + (27.3)) \cdot 138$$

$$= 134402.94 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 137$$

$$= 34736. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 155) \cdot (95 - 3) \cdot 0.7 \cdot 137$$

$$= 436168. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 324 \cdot (18 - 3) \cdot 0.74 \cdot 138$$

$$= 79493. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



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$$= (\text{SONW} + \text{SNW}) = (34736 + 436168) = 470904 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (34736 + 0 + 79493 + 0) = 114229 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (34736 + 79493 + 0) = 114229 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 470903 Kgf, must exceed W = 49926 Kgf or W1 = 95959 Kgf
 Path 2-2 = 114229 Kgf, must exceed W = 49926 Kgf or W2 = 134402 Kgf
 Path 3-3 = 114229 Kgf, must exceed W = 49926 Kgf or W3 = 134402 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1438 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 9.4045 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 377.4044 mm

Input Echo, WRC297 Item 1, Description: N6 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2800.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	479.100	mm
Nozzle Thickness	95.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	1250.00	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust No



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Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tn (5.043) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$
 $= 479.1 + 2 \cdot 1.65 \cdot \sqrt{1410.5(18.0 - 3.0)}$
 $= 959.106 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2821.000 mm
 Nozzle Outside Diameter (d) = 479.100 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 92.000 mm
 T / t = 0.163
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.329$$

Nr/P = 0.069
 Mr/P = 0.130
 M0/P = 0.039
 N0/P = 0.063
 MrD/Mc = 0.282
 NrDL/Mc = 0.099
 M0d/Mc = 0.086
 N0DL/Mc = 0.075
 MrD/Ml = 0.099
 NrDL/Ml = 0.089
 M0D/Ml = 0.030
 N0DL/Ml = 0.083

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	116	117	116	117
Outplane Stress Summary	115	111	176	89

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	78	-78	78	-78
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-69	69	69	-69
Inplane Pressure Stress	58	58	58	58



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Inplane Stress Summary	63	46	223	-72
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	115	112	223	161
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	18	92	156	-10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	78	-78	78	-78
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-197	197	197	-197
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	117	116	117

Inplane Stress Summary	-8	232	410	-140
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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Two * Max Shear Stress	27	232	410	140
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3

Hoop Stress Summary	-1	0	17	19
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	0	5	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	2	1	17	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	3	1	3

Hoop Stress Summary	0	0	16	18
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Nozzle Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-3	2	8	-3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	3	2	17	21

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	117	116	117	116	117	116	117
Circ. Pl (SUS)	-3	-3	15	15	-5	-5	17	17
Circ. Q (SUS)	3	-3	43	-43	-119	119	275	-275
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-4	-4	16	16	-2	-2	14	14
Long. Q (SUS)	9	-9	147	-147	-37	37	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	116.0	117.0	116.0	117.0	116.0	117.0	116.0	117.0
Pm+Pl (SUS)	113.0	114.0	131.0	132.0	111.0	112.0	133.0	134.0
Pm+Pl+Q (Total)	116.3	111.2	221.1	162.0	27.3	231.0	408.1	141.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+Pl (SUS)	134.02	207.00	Passed
Pm+Pl+Q (TOTAL)	408.06	413.99	Passed



*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	3	1	3	1	3	1	3
Circ. Pl (SUS)	-3	-3	15	15	-2	-2	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	3.0	1.0	3.0	1.0	3.0	1.0	3.0
Pm+Pl (SUS)	2.0	1.0	16.0	18.0	1.0	2.0	15.0	17.0
Pm+Pl+Q (Total)	2.0	1.0	16.0	19.0	4.0	2.0	15.0	21.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	136.60	Passed
Pm+Pl (SUS)	18.00	204.90	Passed
Pm+Pl+Q (TOTAL)	21.00	409.80	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: D3

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Input, Nozzle Desc: D3

From: 40

Pressure for Reinforcement Calculations	P	1259.782	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1636	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3832.26	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		73.31	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 140-C2-Final 23rd March

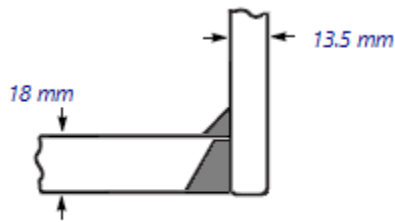
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Nozzle Calcs.: D3

Noz1: 25

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: D3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1260 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1260)$
 $= 12.8793 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1260 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1260)$
 $= 0.2634 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: D3.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4423 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.8793 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.8793 mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.9, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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 FileName : 140-C2-Final 23rd March Page 123 of 223
 Nozzle Calcs.: D3 Nozl: 25 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 13.5, tr = 0.26, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1259.78 / 1960.01 = 0.643$

Weld Size Calculations, Description: D3

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * Wo$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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Nozzle Calcs.: Noz K3

Nozl: 26

9:33am

Sep 9, 2026

Input, Nozzle Desc: Noz K3

From: 50

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6653.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 140-C2-Final 23rd March

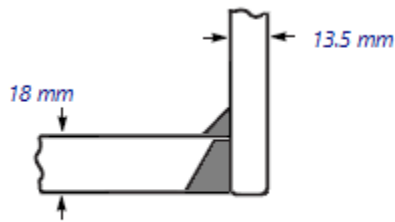
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Nozzle Calcs.: Noz K3

Noz1: 26

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: Noz K3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1250)$
 $= 12.7788 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1250)$
 $= 0.2613 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz K3.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4423 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.7788 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.7788 mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.8, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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 FileName : 140-C2-Final 23rd March Page 126 of 223
 Nozzle Calcs.: Noz K3 Nozl: 26 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 13.5, tr = 0.26, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1250.00 / 1960.01 = 0.638$

Weld Size Calculations, Description: Noz K3

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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Nozzle Calcs.: Noz K5

Nozl: 27

9:33am

Sep 9, 2026

Input, Nozzle Desc: Noz K5

From: 50

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6172.09	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 140-C2-Final 23rd March

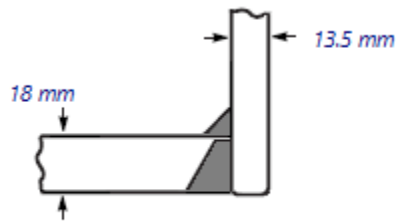
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Nozzle Calcs.: Noz K5

Noz1: 27

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: Noz K5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1250)$
 $= 12.7788 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1250)$
 $= 0.2613 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz K5.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4423 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.7788 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.7788 mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.8, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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 FileName : 140-C2-Final 23rd March Page 129 of 223
 Nozzle Calcs.: Noz K5 Nozl: 27 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 13.5$, $t_r = 0.26$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1250.00 / 1960.01 = 0.638$

Weld Size Calculations, Description: Noz K5

Intermediate Calc. for nozzle/shell Welds T_{min} 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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Nozzle Calcs.: Noz k6

Nozl: 28

9:33am

Sep 9, 2026

Input, Nozzle Desc: Noz k6

From: 50

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5897.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 140-C2-Final 23rd March

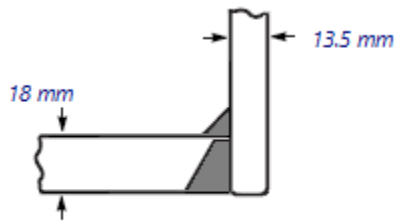
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Nozzle Calcs.: Noz k6

Noz1: 28

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: Noz k6

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1250)$
 $= 12.7788 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1250)$
 $= 0.2613 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz k6.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4423 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.7788 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.7788 mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.8, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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Nozzle Calcs.: Noz k6

Nozl: 28 9:33am Sep 9, 2026

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 14.8,	Allowable	: 136.6 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 328.5 N./mm ²	Passed
Occasional	: 1.4,	Allowable	: 181.7 N./mm ²	Passed
Shear	: 13.2,	Allowable	: 95.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 13.5, tr = 0.26, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1250.00 / 10200.00 = 0.123$

Weld Size Calculations, Description: Noz k6

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

Input Echo, WRC297 Item 1, Description: Noz k6 :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	2800.000 mm
Shell Thickness	18.0000 mm
Shell Stress Concentration Factor	1.000
Vessel Material	SA-516 70
Vessel Cold S.I. Allowable	Smc 138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID
Nozzle Corrosion Allowance	3.0000 mm
Nozzle Diameter	50.800 mm
Nozzle Thickness	13.4620 mm



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Nozzle Calcs.: Noz k6

Nozl: 28 9:33am Sep 9, 2026

Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	136.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1250.00	KPa.
Radial Load	P	-382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_v , (4.318) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (5.774) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{1410.5(18.0 - 3.0)}$$

$$= 557.729 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2821.000 mm
Nozzle Outside Diameter	(d) =	77.724 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	10.462 mm
T / t	=	1.434
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.378$$

Nr/P	=	0.210
Mr/P	=	0.277
M0/P	=	0.122
N0/P	=	0.181
MrD/Mc	=	0.265
NrDL/Mc	=	0.167
M0d/Mc	=	0.145
N0DL/Mc	=	-0.114
MrD/Ml	=	0.251
NrDL/Ml	=	0.165
M0D/Ml	=	0.135
N0DL/Ml	=	-0.047

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: Noz k6

Nozl: 28 9:33am Sep 9,2026

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	116	117	116	117

Outplane Stress Summary	116	125	146	91
-------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-30	30	30	-30
Inplane Pressure Stress	58	58	58	58

Inplane Stress Summary	55	60	122	7
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	116	125	147	91
------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58

Outplane Stress Summary	58	68	88	29
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	27	-27	27	-27



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Nozzle Calcs.: Noz k6

Nozl: 28 9:33am

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Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-31	31	31	-31
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	116	117	116	117

Inplane Stress Summary	112	121	182	64
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	112	121	182	65
------------------------	-----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	7	8	5	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	35	-35	35	-35
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-41	41	41	-41
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-13	0	90	-62
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5

Shear Stress Summary	6	6	3	3
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Nozzle Calcs.: Noz k6

Nozl: 28

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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	25	17	91	70

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4
Hoop Stress Summary	8	10	4	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	35	-35	35	-35
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-44	44	44	-44
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-16	2	94	-66

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5
Shear Stress Summary	3	3	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	26	11	94	73

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	116	117	116	117	116	117	116	117
Circ. Pl (SUS)	3	3	3	3	0	0	6	6
Circ. Q (SUS)	-4	4	28	-28	-4	4	58	-58



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Nozzle Calcs.: Noz k6

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Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	0	0	6	6	5	5	1	1
Long. Q (SUS)	-3	3	57	-57	-5	5	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	116.0	117.0	116.0	117.0	116.0	117.0	116.0	117.0
Pm+Pl (SUS)	119.0	120.0	119.0	120.0	116.0	117.0	122.0	123.0
Pm+Pl+Q (Total)	115.3	124.3	147.2	92.0	112.1	121.1	180.2	65.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	117.00	138.00	Passed
Pm+Pl (SUS)	123.02	207.00	Passed
Pm+Pl+Q (TOTAL)	180.17	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	3	3	3	3	5	5	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-6	6	76	-76	-9	9	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	14.1	15.1	12.2	12.2	16.1	17.1	12.1	12.1
Pm+Pl+Q (Total)	23.3	15.0	88.2	71.4	26.2	11.3	91.4	73.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	136.60	Passed
Pm+Pl (SUS)	17.12	204.90	Passed
Pm+Pl+Q (TOTAL)	91.41	409.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: Noz K4

Nozl: 29

9:33am

Sep 9, 2026

Input, Nozzle Desc: Noz K4

From: 50

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6419.32	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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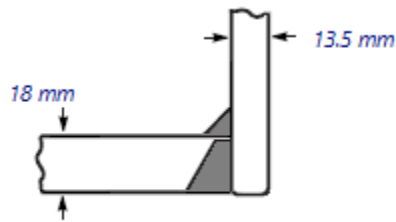
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Nozzle Calcs.: Noz K4

Noz1: 29

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: Noz K4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1250)$
 $= 12.7788 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1250)$
 $= 0.2613 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: Noz K4.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4423 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.7788 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.7788 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.8, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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 FileName : 140-C2-Final 23rd March Page 140 of 223
 Nozzle Calcs.: Noz K4 Nozl: 29 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 13.5$, $t_r = 0.26$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1250.00 / 1960.01 = 0.638$

Weld Size Calculations, Description: Noz K4

Intermediate Calc. for nozzle/shell Welds T_{min} 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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FileName : 140-C2-Final 23rd March

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Nozzle Calcs.: Noz K8

Nozl: 30

9:33am

Sep 9, 2026

Input, Nozzle Desc: Noz K8

From: 50

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5578.51	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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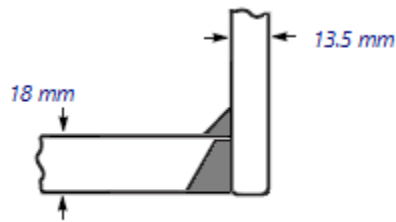
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Nozzle Calcs.: Noz K8

Noz1: 30

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: Noz K8

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1250)$
 $= 12.7788 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1250 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1250)$
 $= 0.2613 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz K8.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4423 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.7788 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.7788 mm
Wall Thickness, shell/head, external pressure	trb2	= 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.8, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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 Nozzle Calcs.: Noz K8 Nozl: 30 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 13.5, tr = 0.26, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1250.00 / 1960.01 = 0.638$

Weld Size Calculations, Description: Noz K8

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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Nozzle Calcs.: D5

Nozl: 31

9:33am

Sep 9, 2026

Input, Nozzle Desc: D5

From: 50

Pressure for Reinforcement Calculations	P	1259.782	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6066.59	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		73.31	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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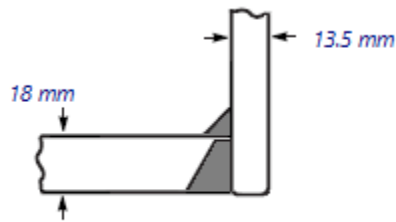
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Nozzle Calcs.: D5

Noz1: 31

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: D5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1260 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1260)$
 $= 12.8793 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1260 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1260)$
 $= 0.2634 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D5.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4423 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.8793 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.8793 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.9, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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FileName : 140-C2-Final 23rd March Page 146 of 223
Nozzle Calcs.: D5 Nozl: 31 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 13.5, tr = 0.26, c = 3 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1259.78 / 1960.01 = 0.643$

Weld Size Calculations, Description: D5

Intermediate Calc. for nozzle/shell Welds Tmin 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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Nozzle Calcs.: D1

Nozl: 32

9:33am

Sep 9, 2026

Input, Nozzle Desc: D1

From: 50

Pressure for Reinforcement Calculations	P	1259.782	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1631	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6474.03	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		73.31	deg
Diameter		50.8000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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FileName : 140-C2-Final 23rd March

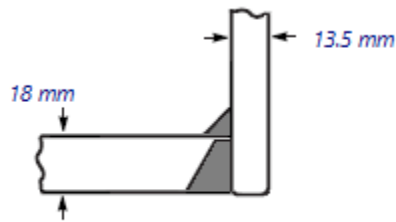
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Nozzle Calcs.: D1

Noz1: 32

9:33am

Sep 9, 2026



Reinforcement CALCULATION, Description: D1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.800 mm.
Actual Thickness Used in Calculation	13.462 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1260 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1260)$
 $= 12.8793 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1260 \cdot 28.4 / (137 \cdot 1 - 0.6 \cdot 1260)$
 $= 0.2634 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4423 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: D1.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4423 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.8793 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.8793 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(15.9, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.44, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



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 FileName : 140-C2-Final 23rd March Page 149 of 223
 Nozzle Calcs.: D1 Nozl: 32 9:33am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 13.5$, $t_r = 0.26$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1259.78 / 1960.01 = 0.643$

Weld Size Calculations, Description: D1

Intermediate Calc. for nozzle/shell Welds T_{min} 10.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5395 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.5395 mm

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Nozzle Calcs.: N5

Nozl: 33

9:33am

Sep 9, 2026

Input, Nozzle Desc: N5

From: 60

Pressure for Reinforcement Calculations	P	1263.723	KPa.
Temperature for Internal Pressure	Temp	235	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2800.00	mm
Design Length of Section	L	6814.1636	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7172.61	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		89.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.7480	mm
Flange Material		SA-105	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	101.0000	mm
Height of Beveled Transition	L'	25.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



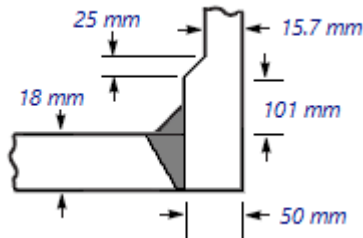
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Nozzle Calcs.: N5

Noz1: 33 9:33am Sep 9,2026



Reinforcement CALCULATION, Description: N5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	89.000 mm.
Actual Thickness Used in Calculation	15.748 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1264 \cdot 1403 / (138 \cdot 1 - 0.6 \cdot 1264)$
 $= 12.9198 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1264 \cdot 47.5 / (137 \cdot 1 - 0.6 \cdot 1264)$
 $= 0.4419 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6116 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 34.3 / 0.5773$
 $= 59.3262 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	190.0000 mm
Parallel to Vessel Wall, opening length	d	95.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	37.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.990$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 137 / 138)$
 $= 0.990$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.99, 1)$
 $= 0.990$



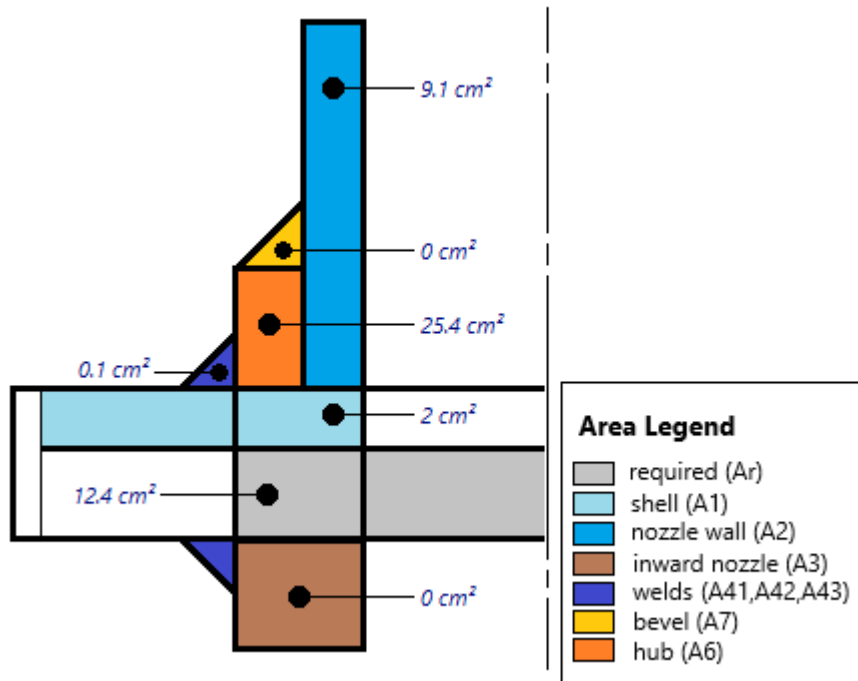
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Nozzle Calcs.: N5

Nozl: 33 9:33am Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.397	6.365	NA
Area in Shell	A1	1.956	1.630	NA
Area in Nozzle Wall	A2	9.136	9.010	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.097	0.097	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	25.429	25.429	NA
TOTAL AREA AVAILABLE	Atot	36.617	36.165	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (95 \cdot 12.9 \cdot 1 + 2 \cdot 47 \cdot 12.9 \cdot 1 \cdot (1 - 0.99))$$

$$= 12.397 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 95(1 \cdot 15 - 1 \cdot 12.9) - 2 \cdot 47$$

$$(1 \cdot 15 - 1 \cdot 12.9) \cdot (1 - 0.99)$$

$$= 1.956 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 37.5)(12.7 - 0.44)0.99$$

$$= 9.136 \text{ cm}^2$$



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Nozzle Calcs.: N5

Nozl: 33 9:33am

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot fr_2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr_2$$

$$= (10^2 - 0.9) \cdot 0.99 + (0^2 - 0) \cdot 0.99$$

$$= 0.097 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{inp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(37.5, 350, 101)) \cdot (50 - 15.7) \cdot 0.99$$

$$= 25.429 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6116 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 15.9198 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 15.9198 mm
Wall Thickness, shell/head, external pressure	trb2 = 4.0520 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.7300 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.73, \max(15.9, 4.5)]$$

$$= 8.7300 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.61, 8.73)$$

$$= 8.7300 \text{ mm}$$

Available Nozzle Neck Thickness = 15.7480 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 12.7,	Allowable	: 136.6 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 330.5 N./mm ²	Passed
Occasional	: 2.1,	Allowable	: 181.7 N./mm ²	Passed
Shear	: 73.8,	Allowable	: 95.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 18, tr = 12.9, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.86, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	2 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 1263.72/10200.00 = 0.124$$

Weld Size Calculations, Description: N5

Intermediate Calc. for nozzle/shell welds Tmin 15.0000 mm



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 Nozzle Calcs.: N5 Nozl: 33 9:33am Sep 9, 2026

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12.4 - 1.96 + 2 \cdot 0.47 \cdot 0.99 \cdot (1 \cdot 15 - 12.9)) \cdot 138)$
 $= 17415.20 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (34.6 + 0.097 - 0 \cdot 0.99) \cdot 138$
 $= 48773.51 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (34.6 + 0 + 0.097 + (14)) \cdot 138$
 $= 68413.26 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (34.6 + 0 + 0.097 + 0 + (14)) \cdot 138$
 $= 68413.26 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 189 \cdot 10 \cdot 0.49 \cdot 137$
 $= 20263. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 71) \cdot (50 - 3) \cdot 0.7 \cdot 137$
 $= 102216. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 189 \cdot (18 - 3) \cdot 0.74 \cdot 138$
 $= 46371. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (20263 + 102216) = 122479 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (20263 + 0 + 46371 + 0) = 66634 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (20263 + 46371 + 0) = 66634 \text{ Kgf}$

Summary of Failure Path Calculations:



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Path 1-1 = 122479 Kgf, must exceed W = 17415 Kgf or W1 = 48773 Kgf
 Path 2-2 = 66633 Kgf, must exceed W = 17415 Kgf or W2 = 68413 Kgf
 Path 3-3 = 66633 Kgf, must exceed W = 17415 Kgf or W3 = 68413 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1438 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 3.1930 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 371.1930 mm

Input Echo, WRC297 Item 1, Description: N5 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2800.000	mm	
Shell Thickness	18.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	89.000	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 136.60	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1263.72	KPa.
Radial Load	P	-611.87	Kgf
Circumferential Shear	Vc	407.91	Kgf
Longitudinal Shear	Vl	407.91	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	1304.82	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (3.780) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 189.0 + 2 \cdot 1.65 \cdot \sqrt{1410.5(18.0 - 3.0)}$$



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Nozzle Calcs.: N5

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= 669.005 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2821.000 mm
 Nozzle Outside Diameter (d) = 189.000 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 47.000 mm
 T / t = 0.319
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.919$$

Nr/P = 0.110
 Mr/P = 0.238
 M0/P = 0.069
 N0/P = 0.128
 MrD/Mc = 0.311
 NrDL/Mc = 0.075
 M0d/Mc = 0.096
 N0DL/Mc = 0.039
 MrD/Ml = 0.206
 NrDL/Ml = 0.091
 M0D/Ml = 0.065
 N0DL/Ml = 0.107

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	117	118	117	118
Outplane Stress Summary	121	117	142	105

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-26	26	26	-26
Inplane Pressure Stress	58	58	58	58
Inplane Stress Summary	71	47	127	0

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	15	15	15	15



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Shear Stress Summary	16	16	14	14
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	126	120	151	109

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-12	12	12	-12
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	58	58	58	58
Outplane Stress Summary	60	62	86	39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-39	39	39	-39
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	117	118	117	118
Inplane Stress Summary	117	121	199	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	15	15	15	15
Shear Stress Summary	14	14	16	16

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	121	124	201	59

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



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Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	2	1	2

Hoop Stress Summary	2	3	6	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	0	-1	7	-2
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4

Shear Stress Summary	5	5	4	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	10	11	11	14
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	2	1	2

Hoop Stress Summary	3	5	5	6
---------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0



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Axial Stress Summary -1 0 8 -4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	4	4	5	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	10	10	12	15

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	117	118	117	118	117	118	117	118
Circ. Pl (SUS)	1	1	5	5	1	1	3	3
Circ. Q (SUS)	3	-3	19	-19	-1	1	77	-77
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	1	1	3	3	3	3	3	3
Long. Q (SUS)	12	-12	64	-64	-1	1	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	15	15	15	15	15	15	15	15
Pm (SUS)	117.0	118.0	117.0	118.0	117.0	118.0	117.0	118.0
Pm+Pl (SUS)	118.0	119.0	122.0	123.0	118.0	119.0	120.0	121.0
Pm+Pl+Q (Total)	125.2	119.1	150.0	111.1	120.7	123.6	199.0	56.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	118.00	138.00	Passed
Pm+Pl (SUS)	123.00	207.00	Passed
Pm+Pl+Q (TOTAL)	198.96	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



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Circ. Pm (SUS)	1	2	1	2	1	2	1	2
Circ. Pl (SUS)	1	1	5	5	3	3	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	4	-4	-1	1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
Pm+Pl (SUS)	3.0	4.0	6.0	7.0	5.0	6.0	4.0	5.0
Pm+Pl+Q (Total)	8.5	8.9	9.5	12.8	10.0	9.4	9.1	12.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	136.60	Passed
Pm+Pl (SUS)	7.00	204.90	Passed
Pm+Pl+Q (TOTAL)	12.81	409.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 ASME TS Calc: DESIGN

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Port shell-1

Shell Maximum Design Pressure	Psd,max	1250.00 KPa.
Shell Maximum Operating Pressure	Psox,max	510.00 KPa.
Shell Minimum Operating Pressure	Psox,min	15.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1800.000 mm
Shell Temperature for Internal Pressure	Ts	235.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4316.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	44.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1800.000 mm
Channel Design Temperature	TEMPC	275.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2660
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	3033.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		100.35 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	111.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	876.669 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc: DESIGN

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	23585.6 cm ²
Perimeter of Tube Layout	Cp	5684.2 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

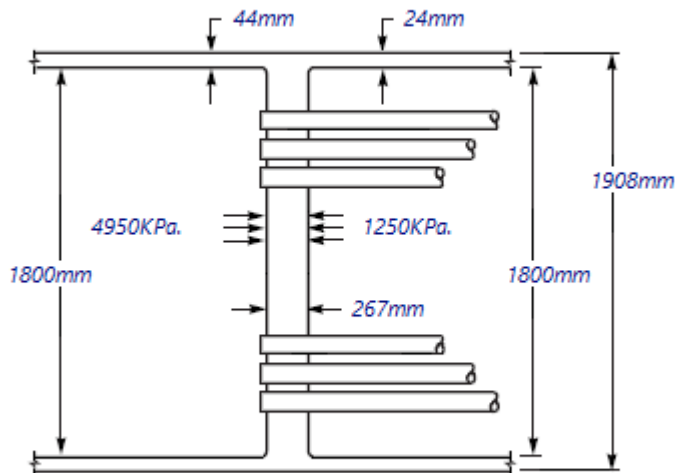
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	267.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1908.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
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Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Channel Side Min. Design Pressure	Ptd,min 103.0000 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: DESIGN

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Elastic Mod. at Design Temperature	235.0 °C	0.19009E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.17745E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(4950 \cdot (25.4/2 - 0)) / (100 \cdot 1 + 0.4 \cdot 4950)$

= $0.6144 + 0.0000 = 0.6144$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	275.00 °C
Elastic Modulus for Material			172442880.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	3909.67	15.39	50.0000	0.0046419	45.13

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 45.1) / (3 \cdot 15.4) = 3909$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7528	25.40	3909.67	33.74	50.0000	0.0009662	34.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 34.2) / (3 \cdot 33.7) = 1353$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7528 mm
Allowable Internal Pressure at Corroded thickness	13751.58 KPa.
Required Internal Design Pressure	4950.00 KPa.
Allowable External Pressure at Corroded thickness	3908.62 KPa.
Required External Design Pressure	1353.00 KPa.
Required Thickness due to Shell Side pressure	0.7528 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 877 + 25.4$

= 1778.738 mm



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Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 267$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944 / 187982736) \cdot$$

$$(100 / 138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 8.409449$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (8.409) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1800 + 24) 24 }$$

$$= 0.0087 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0087 \cdot 190085616 \cdot 24^3 / (6 (1 - 0.3^2))$$

$$= 4181460480.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1800 + 24) \cdot 4.18E+09 / 267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2)$$

$$= 14450448.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 1800^2 / (4 \cdot 190085616 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2 (1800 + 24)))$$



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$$= 0.15126\text{E}+00 \text{ mm/N./mm}^2$$

Parameter [ws]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.01 \cdot 4.18\text{E}+09 \cdot 0.0087 \cdot 0.15126048 (1 + 267 \cdot 0.0087) \\ &= 18462.3730 \text{ mm}^2 \end{aligned}$$

Channel coefficient [βc]:

$$\begin{aligned} &= ((12(1 - v_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)} \\ &= ((12(1 - 0.3^2))^{1/4}) / ((1800 + 44)44) \\ &= 0.0064 \text{ 1/mm} \end{aligned}$$

Channel coefficient [kc]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2)) \\ &= 0.0064 \cdot 187982736 \cdot 44^3 / (6(1 - 0.3^2)) \\ &= 18716825600.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$\begin{aligned} &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ &= 6(1800 + 44 / \cos(0)) \cdot 18.72\text{E}+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1) \\ &= 45212188.0 \text{ KPa} \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned} &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\ &= 1800^2 / (4 \cdot 187982736 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44))) \\ &= 0.83586\text{E}-01 \text{ mm/N./mm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ &= 1.01 \cdot 18.72\text{E}+09 [0.083585896 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)] \\ &= 27321.2422 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1908 / 1779 \\ &= 1.0727 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 29202954 \cdot (14450448 + 45212188 + 187982736 \cdot \ln(1.07)) \\ &= 1.3507 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ &= 1779^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 0 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 4950] \\ &= -23682100.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + w_c \cdot P_t - w_s \cdot P_s$$



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$$= -23682100 + 27321 \cdot 4950 - 18462 \cdot 0$$

$$= 111558056.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((111558056) - 1779^2/32 \cdot 1.35 (0 - 4950)) / (1 + 1.35)$$

$$= 328670592.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 328670592 + 1779^2/64 \cdot (3 + 0.46) \cdot (0 - 4950)$$

$$= -517671776.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 328670592 | ; | -517671776 |)$$

$$= 517671776.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 517671776 / (0.2 (267 - 8)^2)$$

$$= 231.5268 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 507224224 / (0.2 (243 - 8)^2)$$

$$= 275.7003 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 243 + 0 + 0$$

$$= 242.9387 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB} ; H_{reqS})$$

$$= \max(243 ; 93)$$

$$= 242.9387 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 0 - 4950 |$$

$$= 4950.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 267 / 1779 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 222)$$

$$= 13.257 \text{ N./mm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot | 0 - 4950 | + 0$$



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= 93.0290 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 24.000mm

Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1800^2 / (4 \cdot 24 (1800 + 24)) \cdot 0$$

$$= 0.0000 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 940051 / 24^2 \cdot [0.0087 \cdot 0 + 6(1 - 0.46) / 29202954 \cdot$$

$$1779 / 267^3 \cdot (1 + 267 \cdot 0.0087 / 2) \cdot (328670592 + 1779^2 / 32 \cdot$$

$$(0 - 4950))]$$

$$= -157.217 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s, m} | + | \sigma_{s, b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 157.217 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 44.000 mm

Min. Channel len. adjacent to the tubesheet is: 506.565mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1800^2 / (4 \cdot 44 (1800 + 44) \cdot \cos(0)) \cdot 4950$$

$$= 49.420 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4207806 / 44^2 \cdot [(0.083585896 \cdot 1 \cdot 0.0064 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.46) / 29202954) \cdot 1779 / 267^3 \cdot (1 +$$

$$0.0064 \cdot 267 / 2 \cdot 1) \cdot (328670592 + 1779^2 / 32 \cdot (0 - 4950))]$$

$$= 332.710 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c, m} | + | \sigma_{c, b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 382.130 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 187982736 \cdot \sqrt{(1.5 \cdot 148 / 382)}$$

$$= 143279824.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 532566272 / (0.2 (267 - 8)^2)$$

$$= 238.2 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.



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Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + \text{Cats} + \text{Catc} = 247 + 0 + 0 = 247 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS}) = \max(247; 93) = 247 \text{ mm}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 1.65) 1.65 \\ &= 1.231 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 1.65 (25.4 - 1.65) 100 \\ &= 1259.763 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100 \\ &= 1195.7887 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 100/100 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100 \\ &= 773.7455 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right) \\ &= 2.141 \text{ mm} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is $\geq$ tube strength 1260 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	238.2 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	38.5 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	157.2 $\leq$	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	382.1 $\leq$	498.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
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Tubesheet Thickness :	246.847	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 877 + 25.4$$

$$= 1778.738 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 267$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t T / E) \cdot (St T / S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944 / 187982736) \cdot (100 / 138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 8.409449$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $v^* = 0.458572$

Note: As h/p (8.409) is > 2 , data values for $h/p = 2$ were used.



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ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\frac{12(1 - \nu_s^2)}{E_s} \right)^{1/4} / \sqrt{\frac{(D_s + t_s)t_s}{1800 + 24}} \\ = \left(\frac{12(1 - 0.3^2)}{190085616} \right)^{1/4} / \sqrt{\frac{(1800 + 24)24}{4181460480.0000}} \\ = 0.0087 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\ = 0.0087 \cdot 190085616 \cdot 24^3 / (6(1 - 0.3^2)) \\ = 4181460480.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\ = 6(1800 + 24) \cdot 4.18E+09 / 267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2) \\ = 14450448.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\ = 1800^2 / (4 \cdot 190085616 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 24))) \\ = 0.15126E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ = 1.01 \cdot 4.18E+09 \cdot 0.0087 \cdot 0.15126048 (1 + 267 \cdot 0.0087) \\ = 18462.3730 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\frac{12(1 - \nu_c^2)}{E_c} \right)^{1/4} / \sqrt{\frac{(D_c + t_c)t_c}{1800 + 44}} \\ = \left(\frac{12(1 - 0.3^2)}{187982736} \right)^{1/4} / \sqrt{\frac{(1800 + 44)44}{18716825600.0000}} \\ = 0.0064 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\ = 0.0064 \cdot 187982736 \cdot 44^3 / (6(1 - 0.3^2)) \\ = 18716825600.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ = 6(1800 + 44 / \cos(0)) \cdot 18.72E+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1) \\ = 45212188.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ = 1800^2 / (4 \cdot 187982736 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44))) \\ = 0.83586E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ = 1.01 \cdot 18.72E+09 [0.083585896 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)] \\ = 27321.2422 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:



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$$= A / Do$$

$$= 1908/1779$$

$$= 1.0727$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 29202954 \cdot (14450448 + 45212188 + 187982736 \cdot \ln(1.07))$$

$$= 1.3507$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 1779^2/16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 1250 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 4950]$$

$$= -17701772.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -17701772 + 27321 \cdot 4950 - 18462 \cdot 1250$$

$$= 94460408.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - Do^2/32 \cdot F (Ps - Pt) / (1 + F)$$

$$= (94460408) - 1779^2/32 \cdot 1.35 (1250 - 4950) / (1 + 1.35)$$

$$= 250383840.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 250383840 + 1779^2/64 \cdot (3 + 0.46) \cdot (1250 - 4950)$$

$$= -382235648.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|250383840|; |-382235648|)$$

$$= 382235648.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 382235648 / (0.2 (267 - 8)^2)$$

$$= 170.9535 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 360561312 / (0.2 (206 - 8)^2)$$

$$= 275.0895 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 206 + 0 + 0$$

$$= 206.3014 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(206; 69.5)$$



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= 206.3014 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

= $|P_s - P_t|$
 = $|1250 - 4950|$
 = 3700.000 KPa.

Shear Stress check [τ_{limit}]:

= $4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$
 = $4 \cdot 0.2 \cdot 267 / 1779 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222)$
 = 13.257 N./mm²

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

= $1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$
 = $1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot |1250 - 4950| + 0$
 = 69.5368 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 24.000mm

Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

= $D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$
 = $1800^2 / (4 \cdot 24 (1800 + 24)) \cdot 1250$
 = 23.1305 N./mm²

Axial Shell Bending Stress [$\sigma_{s,b}$]:

= $6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$
 = $6 \cdot 940051 / 24^2 \cdot [0.0087 \cdot 0.19 + 6(1 - 0.46) / 29202954 \cdot 1779 / 267^3 \cdot (1 + 267 \cdot 0.0087 / 2) \cdot (250383840 + 1779^2 / 32 \cdot (1250 - 4950))]$
 = -41.347 N./mm²

Shell Membrane + Bending Stress [σ_s]:

= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq 1.5 \cdot S_s$)
 = 64.477 N./mm² (Should be ≤ 207.00)

Note: For a given Channel thickness of: 44.000 mm

Min. Channel len. adjacent to the tubesheet is: 506.565mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
 = $1800^2 / (4 \cdot 44 (1800 + 44) \cdot \cos(0)) \cdot 4950$
 = 49.420 N./mm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:

= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$
 = $6 \cdot 4207806 / 44^2 \cdot [(0.083585896 \cdot 1 \cdot 0.0064 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.46) / 29202954) \cdot 1779 / 267^3 \cdot (1 + 0.0064 \cdot 267 / 2 \cdot 1) \cdot (250383840 + 1779^2 / 32 \cdot (1250 - 4950))]$
 = 282.114 N./mm²

Channel Membrane + Bending Stress [σ_c]:

= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq 1.5 \cdot S_c$)
 = 331.534 N./mm² (Should be ≤ 222.00)



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Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 187982736 \cdot \sqrt{(1.5 \cdot 148/332)}$$

$$= 153824832.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 390234336 / (0.2 \cdot (267 - 8)^2)$$

$$= 174.5 \text{ N./mm}^2 \text{ (Should be } \leq 276.0 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 209 + 0 + 0 = 209 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) = \max(209; 69.5) = 209 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 100/100$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 773.7455 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm



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Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is >= tube strength 1260 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	174.5 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	28.8 <=	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	64.5 <=	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	331.5 <=	498.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	208.879	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 877 + 25.4$$

$$= 1778.738 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1806 / 1779$$

$$= 1.0153$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1806 / 1779$$

$$= 1.0153$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 264$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:



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$$\begin{aligned}
 &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\
 &= \max(25.4 - 2 \cdot 1.65 (177454944/187982736) \cdot \\
 &\quad (100/138) \cdot (0); 25.4 - 2 \cdot 1.65) \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\
 &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) } \\
 &= 31.7500 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.2000
 \end{aligned}$$

$$h/p = 8.314960 ; \mu^* = 0.200000 ; E^* = 29202954 \text{ KPa.}$$

$$E^*/E = 0.155349 ; \nu^* = 0.458572$$

Note: As h/p (8.315) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1806 + 21) 21) } \\
 &= 0.0093 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\
 &= 0.0093 \cdot 190085616 \cdot 21^3 / (6(1 - 0.3^2)) \\
 &= 2992204288.0000 \text{ KPa.} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(1806 + 21) \cdot 2.99E+09 / 264^3 \cdot (1 + 264 \cdot 0.0093 + (264 \cdot 0.0093)^2 / 2) \\
 &= 11500910.0 \text{ KPa.}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\
 &= 1806^2 / (4 \cdot 190085616 \cdot 21) \cdot (1 - 1806 \cdot 0.3 / (2(1806 + 21))) \\
 &= 0.17397E+00 \text{ mm/N.} \cdot \text{mm}^2
 \end{aligned}$$

Parameter [w_s]:

$$\begin{aligned}
 &= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\
 &= 1.02 \cdot 2.99E+09 \cdot 0.0093 \cdot 0.17397252 (1 + 264 \cdot 0.0093) \\
 &= 16924.2324 \text{ mm}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned}
 &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1806 + 41) 41) } \\
 &= 0.0066 \text{ 1/mm}
 \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\
 &= 0.0066 \cdot 187982736 \cdot 41^3 / (6(1 - 0.3^2)) \\
 &= 15674986496.0000 \text{ KPa.} \cdot \text{mm}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\
 &\quad h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\
 &= 6(1806 + 41 / \cos(0)) \cdot 15.67E+09 / 264^3 \cdot (1 + \\
 &\quad 264 \cdot 0.0066 \cdot (1 + 1) / 2 + (264 \cdot 0.0066)^2 / 2 \cdot 1)
 \end{aligned}$$



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= 40261988.0 KPa.

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1806^2 / (4 \cdot 187982736 \cdot 41) \cdot (1 - 1806 \cdot 0.3 / (2(1806 + 41)))$$

$$= 0.90275E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zeta ϵ]:

$$\text{zeta}\epsilon = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta_c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 15.67E+09 [0.090274557 (1 + 0.0066 \cdot 1 \cdot 264) - 0.000 (1/264 + 1 \cdot 0.0066/2)]$$

$$= 26044.1445 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1908/1779$$

$$= 1.0727$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 29202954 \cdot (11500910 + 40261988 + 187982736 \cdot \ln(1.07))$$

$$= 1.2042$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1779^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 0 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -30467594.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -30467594 + 26044 \cdot 4950 - 16924 \cdot 0$$

$$= 98450936.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F(P_s - P_t) / (1 + F)$$

$$= (98450936) - 1779^2 / 32 \cdot 1.2 (0 - 4950) / (1 + 1.2)$$

$$= 312042944.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 312042944 + 1779^2 / 64 \cdot (3 + 0.46) \cdot (0 - 4950)$$

$$= -534299392.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|312042944|; |-534299392|)$$

$$= 534299392.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 534299392 / (0.2 (264 - 8)^2)$$



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$$= 244.5971 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 527180288 / (0.2 \cdot (247 - 8)^2)$$

$$= 275.8157 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + Cats + Catc$$

$$= 247 + 3 + 0$$

$$= 250.4657 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(250 ; 96)$$

$$= 250.4657 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |0 - 4950|$$

$$= 4950.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 264 / 1779 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 222)$$

$$= 13.108 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot |0 - 4950| + 3$$

$$= 96.0290 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 21.000mm

Min. Shell len. adjacent to the tubesheet is: 350.543 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1806^2 / (4 \cdot 21 (1806 + 21)) \cdot 0$$

$$= 0.0000 \text{ N./mm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 672690 / 21^2 \cdot [0.0093 \cdot 0 + 6(1 - 0.46) / 29202954 \cdot$$

$$1779 / 264^3 \cdot (1 + 264 \cdot 0.0093 / 2) \cdot (312042944 + 1779^2 / 32 \cdot$$

$$(0 - 4950))]$$

$$= -172.792 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 172.792 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 41.000 mm



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Min. Channel len. adjacent to the tubesheet is: 489.805mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1806^2 / (4 \cdot 41 (1806 + 41) \cdot \cos(0)) \cdot 4950$$

$$= 53.303 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 3523957 / 41^2 \cdot [(0.090274557 \cdot 1 \cdot 0.0066 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.46) / 29202954) \cdot 1779 / 264^3 \cdot (1 + 0.0066 \cdot 264 / 2 \cdot 1) \cdot (312042944 + 1779^2 / 32 \cdot (0 - 4950))]$$

$$= 364.963 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 418.267 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 187982736 \cdot \sqrt{(1.5 \cdot 148 / 418)}$$

$$= 136950544.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 552257536 / (0.2 (264 - 8)^2)$$

$$= 252.8 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 252 + 3 + 0 = 255 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) = \max(255; 96) = 255 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kg}$$



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Weld Strength Factor [fw]:

= S / Sw where Sw = min(S and St)

= 100/100

= 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot af \cdot (do + 0.67 \cdot af) Sw$ (but not > Ft)

= $0.55 \cdot \pi \cdot 1.65 \cdot (25.4 + 0.67 \cdot 1.65) 100$

= 773.7455 Kgf

Minimum Length of Weld [ar]:

= $2 \cdot (\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd}) - 0.75 \cdot do$

= $2 \cdot (\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1}) -$

$0.75 \cdot 25.4$)

= 2.141 mm

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is >= tube strength 1260 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	252.8 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	38.9 <=	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	172.8 <=	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	418.3 <=	498.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	255.038	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	246.847	...	238	276	38	110	Ps+Pt D1	Ok
D2uc	120.564	...	65	276	11	110	Ps+Pt D2	Ok
D3uc	208.879	...	175	276	29	110	Ps+Pt D3	Ok
D4uc	41.098	...	5	276	1	110	Ps+Pt D4	Ok
D1c	255.038	...	253	276	39	110	Ps+Pt-c D1	Ok
D2c	124.738	...	69	276	11	110	Ps+Pt-c D2	Ok
D3c	216.398	...	185	276	29	110	Ps+Pt-c D3	Ok
D4c	44.055	...	5	276	1	110	Ps+Pt-c D4	Ok
Max:	255.0381	... mm	0.916	0.352	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:



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Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Reqd. Thk.			Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	157	207	382	498	Ok
D2uc	...	...	142	207	59	222	Ok
D3uc	...	...	64	207	332	498	Ok
D4uc	...	...	3	207	8	222	Ok
D1c	...	...	173	207	418	498	Ok
D2c	...	...	161	207	65	222	Ok
D3c	...	...	69	207	362	498	Ok
D4c	...	...	4	207	9	222	Ok
Max	...	...	0.835		0.840		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	255.038	267.000	Ok
Tube thickness :	0.753	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.651	Ok
Tube-tubesheet groove weld leg :	1.070	1.651	Ok

Given Shell Thickness: 24.0000 mm

Given Channel Thickness: 44.0000 mm

Min Shell length of thk, (24.000) adj. to tubesheet: 374.123 mm

Min Channel length of thk, (44.000) adj. to tubesheet: 506.565 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5368.38	0.997	5379.21	1.000
Tubesheet Shear Stress	14047.22	1.000	14047.22	1.000
Shell Stress (Axial, Junction)	5368.38	0.997	3478.19	1.000
Tubesheet-Channel Junction	5368.38	0.997	3478.19	0.654
Tube Pressure Stress	13751.57	1.000	3908.61	1.000
Minimum MAWP	5368.38		3478.19	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
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Tubesheet Bending Stress	5690.86	1.000	5690.86	1.000
Tubesheet Shear Stress	14206.85	1.000	14206.85	1.000
Shell Stress (Axial, Junction)	14206.85	0.854	4699.55	1.000
Tubesheet-Channel Junction	5690.86	1.000	4699.55	0.838
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	5690.86		4699.55	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(267/4, \min(267, 24)) \\
 &= 66.750 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(1250/4700, 1) \\
 &= 0.266
 \end{aligned}$$

Thickness Ratio = 0.27, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(267/4, \min(267, 44)) \\
 &= 66.750 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(4950/5691, 1) \\
 &= 0.870
 \end{aligned}$$

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	35 °C

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Input Echo, Tubesheet Number 2, Description:

Shell Data:

Main Shell Description: Port shell-1

Shell Maximum Design Pressure	Psd,max	1250.00 KPa.
Shell Maximum Operating Pressure	Psox,max	510.00 KPa.
Shell Minimum Operating Pressure	Psox,min	15.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1800.000 mm
Shell Temperature for Internal Pressure	Ts	235.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4316.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	44.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1800.000 mm
Channel Design Temperature	TEMPC	275.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2660
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	3033.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		100.35 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	111.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	876.669 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc:

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	23585.6 cm ²
Perimeter of Tube Layout	Cp	5684.2 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

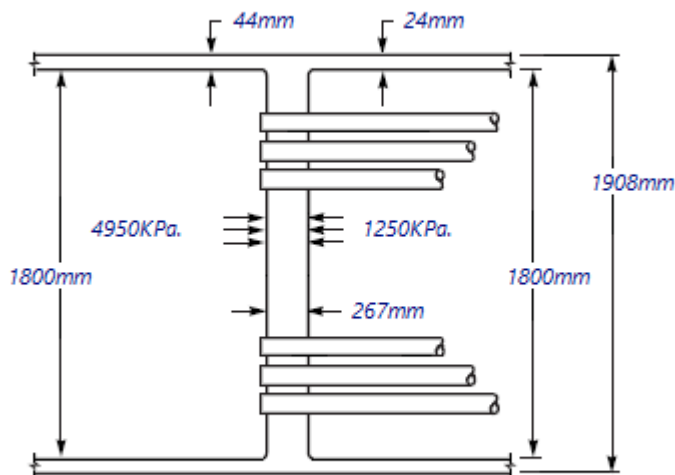
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	267.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1908.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 103.0000 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc:

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Elastic Mod. at Design Temperature	235.0 °C	0.19009E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.17745E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(5053 \cdot (25.4/2 - 0)) / (100 \cdot 1 + 0.4 \cdot 5053)$

= $0.6269 + 0.0000 = 0.6269$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	275.00 °C
Elastic Modulus for Material			172442880.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	3909.67	15.39	50.0000	0.0046419	45.13

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 45.1) / (3 \cdot 15.4) = 3909$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7098	25.40	3909.67	35.79	50.0000	0.0008589	33.56

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 33.6) / (3 \cdot 35.8) = 1250$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7098 mm
Allowable Internal Pressure at Corroded thickness	13751.58 KPa.
Required Internal Design Pressure	5053.00 KPa.
Allowable External Pressure at Corroded thickness	3908.62 KPa.
Required External Design Pressure	1250.00 KPa.
Required Thickness due to Shell Side pressure	0.7098 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 877 + 25.4$

= 1778.738 mm



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ASME TS Calc:

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Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 267$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944 / 187982736) \cdot (100 / 138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.8 / \sqrt{ 1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 8.409449$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (8.409) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1800 + 24) 24 }$$

$$= 0.0087 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0087 \cdot 190085616 \cdot 24^3 / (6 (1 - 0.3^2))$$

$$= 4181460480.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1800 + 24) \cdot 4.18E+09 / 267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2)$$

$$= 14450448.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 1800^2 / (4 \cdot 190085616 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2 (1800 + 24)))$$



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ASME TS Calc:

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$$= 0.15126E+00 \text{ mm/N./mm}^2$$

Parameter [ws]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.01 \cdot 4.18E+09 \cdot 0.0087 \cdot 0.15126048 (1 + 267 \cdot 0.0087) \\ &= 18462.3730 \text{ mm}^2 \end{aligned}$$

Channel coefficient [βc]:

$$\begin{aligned} &= \left(\left(12(1 - \nu_c^2) \right)^{1/4} \right) / \sqrt{\left((D_c + t_c) t_c \right)} \\ &= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \sqrt{\left((1800 + 44) 44 \right)} \\ &= 0.0064 \text{ 1/mm} \end{aligned}$$

Channel coefficient [kc]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\ &= 0.0064 \cdot 187982736 \cdot 44^3 / (6(1 - 0.3^2)) \\ &= 18716825600.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$\begin{aligned} &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ &= 6(1800 + 44 / \cos(0)) \cdot 18.72E+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1) \\ &= 45212188.0 \text{ KPa} \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned} &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ &= 1800^2 / (4 \cdot 187982736 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44))) \\ &= 0.83586E-01 \text{ mm/N./mm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ &= 1.01 \cdot 18.72E+09 [0.083585896 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)] \\ &= 27321.2422 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1908 / 1779 \\ &= 1.0727 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 29202954 \cdot (14450448 + 45212188 + 187982736 \cdot \ln(1.07)) \\ &= 1.3507 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ &= 1779^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot -103 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 4950] \\ &= -24174880.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$



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ASME TS Calc:

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$$= -24174880 + 27321 \cdot 4950 - 18462 \cdot -103$$

$$= 112966888.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((112966888) - 1779^2/32 \cdot 1.35 (-103 - 4950)) / (1 + 1.35)$$

$$= 335121440.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 335121440 + 1779^2/64 \cdot (3 + 0.46) \cdot (-103 - 4950)$$

$$= -528831648.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 335121440 | ; | -528831648 |)$$

$$= 528831648.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 528831648 / (0.2 (267 - 8)^2)$$

$$= 236.5181 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 519488768 / (0.2 (246 - 8)^2)$$

$$= 275.7398 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 246 + 0 + 0$$

$$= 245.7451 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB} ; H_{reqS})$$

$$= \max(246 ; 95)$$

$$= 245.7451 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | -103 - 4950 |$$

$$= 5053.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 267 / 1779 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 222)$$

$$= 13.257 \text{ N./mm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot | -103 - 4950 | + 0$$



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ASME TS Calc:

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= 94.9647 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 24.000mm

Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1800^2 / (4 \cdot 24 (1800 + 24)) \cdot -103$$

$$= -1.9059 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 940051 / 24^2 \cdot [0.0087 \cdot -0.016 + 6(1 - 0.46) / 29202954 \cdot$$

$$1779 / 267^3 \cdot (1 + 267 \cdot 0.0087 / 2) \cdot (335121440 + 1779^2 / 32 \cdot$$

$$(-103 - 4950))]$$

$$= -166.765 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s, m} | + | \sigma_{s, b} | \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 168.671 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 44.000 mm

Min. Channel len. adjacent to the tubesheet is: 506.565mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1800^2 / (4 \cdot 44 (1800 + 44) \cdot \cos(0)) \cdot 4950$$

$$= 49.420 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4207806 / 44^2 \cdot [(0.083585896 \cdot 1 \cdot 0.0064 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.46) / 29202954) \cdot 1779 / 267^3 \cdot (1 +$$

$$0.0064 \cdot 267 / 2 \cdot 1) \cdot (335121440 + 1779^2 / 32 \cdot (-103 - 4950))]$$

$$= 336.879 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c, m} | + | \sigma_{c, b} | \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 386.299 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 187982736 \cdot \sqrt{(1.5 \cdot 148 / 386)}$$

$$= 142504576.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu^* (h - h'g)^2)$$

$$= 6 \cdot 544361344 / (0.2 (267 - 8)^2)$$

$$= 243.5 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.



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ASME TS Calc:

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Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + \text{Cats} + \text{Catc} = 250 + 0 + 0 = 250 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(\text{HreqB}, \text{HreqS}) = \max(250; 95) = 250 \text{ mm}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 1.65) 1.65 \\ &= 1.231 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 1.65 (25.4 - 1.65) 100 \\ &= 1259.763 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100 \\ &= 1195.7887 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 100/100 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100 \\ &= 773.7455 \text{ Kg} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right) \\ &= 2.141 \text{ mm} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is $\geq$ tube strength 1260 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	243.5 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	39.3 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	168.7 $\leq$	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	386.3 $\leq$	498.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
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ASME TS Calc:

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Tubesheet Thickness :	249.754	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D2 un-corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 877 + 25.4$$

$$= 1778.738 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1800 / 1779$$

$$= 1.0120$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 267$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t T / E) \cdot (St T / S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944 / 187982736) \cdot (100 / 138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 8.409449$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $v^* = 0.458572$

Note: As h/p (8.409) is > 2 , data values for $h/p = 2$ were used.



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ASME TS Calc:

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ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\frac{12(1 - \nu_s^2)}{E_s} \right)^{1/4} / \sqrt{\left(\frac{D_s + t_s}{t_s} \right)}$$

$$= \left(\frac{12(1 - 0.3^2)}{190085616} \right)^{1/4} / \sqrt{\left(\frac{1800 + 24}{24} \right)}$$

$$= 0.0087 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0087 \cdot 190085616 \cdot 24^3 / (6(1 - 0.3^2))$$

$$= 4181460480.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1800 + 24) \cdot 4.18E+09 / 267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2)$$

$$= 14450448.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1800^2 / (4 \cdot 190085616 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 24)))$$

$$= 0.15126E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 4.18E+09 \cdot 0.0087 \cdot 0.15126048 (1 + 267 \cdot 0.0087)$$

$$= 18462.3730 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\frac{12(1 - \nu_c^2)}{E_c} \right)^{1/4} / \sqrt{\left(\frac{D_c + t_c}{t_c} \right)}$$

$$= \left(\frac{12(1 - 0.3^2)}{187982736} \right)^{1/4} / \sqrt{\left(\frac{1800 + 44}{44} \right)}$$

$$= 0.0064 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0064 \cdot 187982736 \cdot 44^3 / (6(1 - 0.3^2))$$

$$= 18716825600.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1800 + 44 / \cos(0)) \cdot 18.72E+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1)$$

$$= 45212188.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1800^2 / (4 \cdot 187982736 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44)))$$

$$= 0.83586E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 18.72E+09 [0.083585896 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)]$$

$$= 27321.2422 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:



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ASME TS Calc:

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$$= A / Do$$

$$= 1908/1779$$

$$= 1.0727$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 29202954 \cdot (14450448 + 45212188 + 187982736 \cdot \ln(1.07))$$

$$= 1.3507$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 1779^2/16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 1250 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 0]$$

$$= 5980328.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= 5980328 + 27321 \cdot 0 - 18462 \cdot 1250$$

$$= -17097640.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - Do^2/32 \cdot F (Ps - Pt) / (1 + F)$$

$$= (-17097640) - 1779^2/32 \cdot 1.35 (1250 - 0) / (1 + 1.35)$$

$$= -78286776.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= -78286776 + 1779^2/64 \cdot (3 + 0.46) \cdot (1250 - 0)$$

$$= 135436048.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|-78286776|; |135436048|)$$

$$= 135436048.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 135436048 / (0.2 (267 - 8)^2)$$

$$= 60.5733 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 108154152 / (0.2 (117 - 8)^2)$$

$$= 274.9643 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 117 + 0 + 0$$

$$= 116.6317 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(117; 23.5)$$



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ASME TS Calc:

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= 116.6317 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

= $|P_s - P_t|$

= $|1250 - 0|$

= 1250.000 KPa.

Shear Stress check [τ_{limit}]:

= $4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$

= $4 \cdot 0.2 \cdot 267 / 1779 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222)$

= 13.257 N./mm²

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

= $1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$

= $1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot |1250 - 0| + 0$

= 23.4922 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 24.000mm

Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

= $D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$

= $1800^2 / (4 \cdot 24 (1800 + 24)) \cdot 1250$

= 23.1305 N./mm²

Axial Shell Bending Stress [$\sigma_{s,b}$]:

= $6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot$

$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$

= $6 \cdot 940051 / 24^2 \cdot [0.0087 \cdot 0.19 + 6(1 - 0.46) / 29202954 \cdot$

$1779 / 267^3 \cdot (1 + 267 \cdot 0.0087 / 2) \cdot (-78286776 + 1779^2 / 32 \cdot$

$(1250 - 0))]$

= 115.870 N./mm²

Shell Membrane + Bending Stress [σ_s]:

= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq 1.5 \cdot S_s$)

= 139.001 N./mm² (Should be ≤ 207.00)

Note: For a given Channel thickness of: 44.000 mm

Min. Channel len. adjacent to the tubesheet is: 506.565mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$

= $1800^2 / (4 \cdot 44 (1800 + 44) \cdot \cos(0)) \cdot 0$

= 0.000 N./mm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:

= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$

$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$

$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$

= $6 \cdot 4207806 / 44^2 \cdot [(0.083585896 \cdot 1 \cdot 0.0064 - 0.000 \cdot 1) \cdot 0 -$

$6((1 - 0.46) / 29202954) \cdot 1779 / 267^3 \cdot (1 +$

$0.0064 \cdot 267 / 2 \cdot 1) \cdot (-78286776 + 1779^2 / 32 \cdot (1250 - 0))]$

= -50.596 N./mm²

Channel Membrane + Bending Stress [σ_c]:

= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq 1.5 \cdot S_c$)

= 50.596 N./mm² (Should be ≤ 222.00)



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ASME TS Calc:

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Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 100/100$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 773.7455 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is $\geq$ tube strength 1260 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 un-corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	60.6 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	9.7 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	139.0 $\leq$	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	50.6 $\leq$	222.0 N./mm ²	Ok

Thickness results for loadcase D2 un-corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	116.632	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)



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ASME TS Calc:

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Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 877 + 25.4 \\ &= 1778.738 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1800 / 1779 \\ &= 1.0120 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1800 / 1779 \\ &= 1.0120 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 267 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\ &= \max(25.4 - 2 \cdot 1.65 (177454944 / 187982736) \cdot \\ &\quad (100 / 138) \cdot (0); 25.4 - 2 \cdot 1.65) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) } \\ &= 31.7500 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.2000 \end{aligned}$$

$h/p = 8.409449$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (8.409) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1800 + 24) 24) } \\ &= 0.0087 \text{ 1/mm} \end{aligned}$$

Shell coefficient [β_k]:



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ASME TS Calc:

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$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\
 &= 0.0087 \cdot 190085616 \cdot 24^3 / (6(1 - 0.3^2)) \\
 &= 4181460480.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(1800 + 24) \cdot 4.18E+09 / 267^3 \cdot (1 + 267 \cdot 0.0087 + (267 \cdot 0.0087)^2 / 2) \\
 &= 14450448.0 \text{ KPa}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\
 &= 1800^2 / (4 \cdot 190085616 \cdot 24) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 24))) \\
 &= 0.15126E+00 \text{ mm/N./mm}^2
 \end{aligned}$$

Parameter [ω_s]:

$$\begin{aligned}
 &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\
 &= 1.01 \cdot 4.18E+09 \cdot 0.0087 \cdot 0.15126048 (1 + 267 \cdot 0.0087) \\
 &= 18462.3730 \text{ mm}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned}
 &= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c}) \\
 &= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1800 + 44)44}) \\
 &= 0.0064 \text{ 1/mm}
 \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\
 &= 0.0064 \cdot 187982736 \cdot 44^3 / (6(1 - 0.3^2)) \\
 &= 18716825600.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\
 &= 6(1800 + 44 / \cos(0)) \cdot 18.72E+09 / 267^3 \cdot (1 + 267 \cdot 0.0064 \cdot (1 + 1) / 2 + (267 \cdot 0.0064)^2 / 2 \cdot 1) \\
 &= 45212188.0 \text{ KPa}
 \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned}
 &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\
 &= 1800^2 / (4 \cdot 187982736 \cdot 44) \cdot (1 - 1800 \cdot 0.3 / (2(1800 + 44))) \\
 &= 0.83586E-01 \text{ mm/N./mm}^2
 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$\begin{aligned}
 &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\
 &= 1.01 \cdot 18.72E+09 [0.083585896 (1 + 0.0064 \cdot 1 \cdot 267) - 0.000 (1/267 + 1 \cdot 0.0064/2)] \\
 &= 27321.2422 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 1908 / 1779 \\
 &= 1.0727
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.46) / 29202954 \cdot (14450448 + 45212188 + 187982736 \cdot \ln(1.07))
 \end{aligned}$$



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ASME TS Calc:

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= 1.3507

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= \frac{D_o^2}{16} \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= \frac{1779^2}{16} \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 1250 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 4950]$$

$$= -17701772.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + w_c \cdot P_t - w_s \cdot P_s$$

$$= -17701772 + 27321 \cdot 4950 - 18462 \cdot 1250$$

$$= 94460408.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - \frac{D_o^2}{32} \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((94460408) - \frac{1779^2}{32} \cdot 1.35 (1250 - 4950)) / (1 + 1.35)$$

$$= 250383840.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + \frac{D_o^2}{64} \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 250383840 + \frac{1779^2}{64} \cdot (3 + 0.46) \cdot (1250 - 4950)$$

$$= -382235648.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 250383840 | ; | -382235648 |)$$

$$= 382235648.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 382235648 / (0.2 (267 - 8)^2)$$

$$= 170.9535 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 360561312 / (0.2 (206 - 8)^2)$$

$$= 275.0895 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 206 + 0 + 0$$

$$= 206.3014 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(206 ; 69.5)$$

$$= 206.3014 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 1250 - 4950 |$$

$$= 3700.000 \text{ KPa}$$



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ASME TS Calc:

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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 267 / 1779 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222)$$

$$= 13.257 \text{ N./mm}^2$$

| $P_s - P_t$ | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot |1250 - 4950| + 0$$

$$= 69.5368 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 24.000mm

Min. Shell len. adjacent to the tubesheet is: 374.123 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1800^2 / (4 \cdot 24 (1800 + 24)) \cdot 1250$$

$$= 23.1305 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 940051 / 24^2 \cdot [0.0087 \cdot 0.19 + 6(1 - 0.46) / 29202954 \cdot 1779 / 267^3 \cdot (1 + 267 \cdot 0.0087 / 2) \cdot (250383840 + 1779^2 / 32 \cdot (1250 - 4950))]$$

$$= -41.347 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 64.477 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 44.000 mm

Min. Channel len. adjacent to the tubesheet is: 506.565mm

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 1800^2 / (4 \cdot 44 (1800 + 44)) \cdot \cos(0) \cdot 4950$$

$$= 49.420 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 4207806 / 44^2 \cdot [(0.083585896 \cdot 1 \cdot 0.0064 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.46) / 29202954) \cdot 1779 / 267^3 \cdot (1 + 0.0064 \cdot 267 / 2 \cdot 1) \cdot (250383840 + 1779^2 / 32 \cdot (1250 - 4950))]$$

$$= 282.114 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 331.534 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$



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ASME TS Calc:

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$$= 187982736 \cdot \sqrt{(1.5 \cdot 148/332)}$$

$$= 153824832.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 390234336 / (0.2 \cdot (267 - 8)^2)$$

$$= 174.5 \text{ N./mm}^2 \text{ (Should be } \leq 276.0 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + \text{Cats} + \text{Catc} = 209 + 0 + 0 = 209 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS}) = \max(209; 69.5) = 209 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 100/100$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 773.7455 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is $\geq$ tube strength 1260 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):



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ASME TS Calc:

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Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	174.5 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	28.8 <=	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	64.5 <=	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	331.5 <=	498.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	208.879	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 877 + 25.4$$

$$= 1778.738 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1806 / 1779$$

$$= 1.0153$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1806 / 1779$$

$$= 1.0153$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 264$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944 / 187982736) \cdot (100 / 138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) }$$



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ASME TS Calc:

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= 31.7500 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$

= $(31.8 - 25.4) / 31.8$

= 0.2000

$h/p = 8.314960$; $\mu^* = 0.200000$; $E^* = 29202954$ KPa.

$E^*/E = 0.155349$; $v^* = 0.458572$

Note: As h/p (8.315) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

= $((12(1 - v_s^2))^{1/4}) / \sqrt{((D_s + t_s)t_s)}$

= $((12(1 - 0.3^2))^{1/4}) / \sqrt{((1806 + 21)21)}$

= 0.0093 1/mm

Shell coefficient [k_s]:

= $\beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$

= $0.0093 \cdot 190085616 \cdot 21^3 / (6(1 - 0.3^2))$

= 2992204288.0000 KPa·mm²

Shell coefficient [λ_s]:

= $6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$

= $6(1806 + 21) \cdot 2.99E+09 / 264^3 \cdot (1 + 264 \cdot 0.0093 + (264 \cdot 0.0093)^2 / 2)$

= 11500910.0 KPa.

Shell coefficient [δ_s]:

= $D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$

= $1806^2 / (4 \cdot 190085616 \cdot 21) \cdot (1 - 1806 \cdot 0.3 / (2(1806 + 21)))$

= 0.17397E+00 mm/N./mm²

Parameter [ω_s]:

= $\rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$

= $1.02 \cdot 2.99E+09 \cdot 0.0093 \cdot 0.17397252 (1 + 264 \cdot 0.0093)$

= 16924.2324 mm²

Channel coefficient [β_c]:

= $((12(1 - v_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$

= $((12(1 - 0.3^2))^{1/4}) / \sqrt{((1806 + 41)41)}$

= 0.0066 1/mm

Channel coefficient [k_c]:

= $\beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$

= $0.0066 \cdot 187982736 \cdot 41^3 / (6(1 - 0.3^2))$

= 15674986496.0000 KPa·mm²

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

CqDelta = CqTheta = CmTheta = CmDelta = 1

Channel coefficient [λ_c]:

= $6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$

= $6(1806 + 41 / \cos(0)) \cdot 15.67E+09 / 264^3 \cdot (1 + 264 \cdot 0.0066 \cdot (1 + 1) / 2 + (264 \cdot 0.0066)^2 / 2 \cdot 1)$

= 40261988.0 KPa.

Channel coefficient [δ_c]:

= $D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$

= $1806^2 / (4 \cdot 187982736 \cdot 41) \cdot (1 - 1806 \cdot 0.3 / (2(1806 + 41)))$

= 0.90275E-01 mm/N./mm²

Channel coefficient [zetac]:



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ASME TS Calc:

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$$\text{zetac} = 0$$

Parameter [wc]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)] \\ &= 1.02 \cdot 15.67E+09 [0.090274557 (1 + 0.0066 \cdot 1 \cdot 264) - \\ &\quad 0.000 (1/264 + 1 \cdot 0.0066/2)] \\ &= 26044.1445 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1908/1779 \\ &= 1.0727 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 29202954 \cdot (11500910 + 40261988 + \\ &\quad 187982736 \cdot \ln(1.07)) \\ &= 1.2042 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t] \\ &= 1779^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -103 - \\ &\quad (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950] \\ &= -31101566.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s \\ &= -31101566 + 26044 \cdot 4950 - 16924 \cdot -103 \\ &= 99560152.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= ((M^*) - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F) \\ &= ((99560152) - 1779^2 / 32 \cdot 1.2 \cdot (-103 - 4950)) / (1 + 1.2) \\ &= 318109824.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t) \\ &= 318109824 + 1779^2 / 64 \cdot (3 + 0.46) \cdot (-103 - 4950) \\ &= -545843264.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|318109824|; |-545843264|) \\ &= 545843264.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 545843264 / (0.2 (264 - 8)^2) \\ &= 249.8817 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:



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ASME TS Calc:

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$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 539886144 / (0.2 \cdot (250 - 8)^2) \\
 &= 275.8504 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + \text{Cats} + \text{Catc} \\
 &= 250 + 3 + 0 \\
 &= 253.3190 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(\text{HreqB}; \text{HreqS}) \\
 &= \max(253; 98) \\
 &= 253.3190 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= | \text{Ps} - \text{Pt} | \\
 &= | -103 - 4950 | \\
 &= 5053.000 \text{ KPa.}
 \end{aligned}$$

Shear Stress check [τlimit]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\
 &= 4 \cdot 0.2 \cdot 264 / 1779 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222) \\
 &= 13.108 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\
 &= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot | -103 - 4950 | + 3 \\
 &= 97.9647 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 21.000mm

Min. Shell len. adjacent to the tubesheet is: 350.543 mm

Axial Shell Membrane Stress [σs,m]:

$$\begin{aligned}
 &= \text{Ds}^2 / (4 \cdot \text{ts} \cdot (\text{Ds} + \text{ts})) \cdot \text{Ps} \\
 &= 1806^2 / (4 \cdot 21 \cdot (1806 + 21)) \cdot -103 \\
 &= -2.1892 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σs,b]:

$$\begin{aligned}
 &= 6 \cdot \text{ks} / \text{ts}^2 \cdot [\beta S \cdot \text{P}'s + 6(1 - \nu^*) / E^* \cdot \\
 &\quad \text{Do} / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\
 &= 6 \cdot 672690 / 21^2 \cdot [0.0093 \cdot -0.018 + 6(1 - 0.46) / 29202954 \cdot \\
 &\quad 1779 / 264^3 \cdot (1 + 264 \cdot 0.0093 / 2) \cdot (318109824 + 1779^2 / 32 \cdot \\
 &\quad (-103 - 4950))] \\
 &= -183.573 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σs]:

$$\begin{aligned}
 &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s) \\
 &= 185.763 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)
 \end{aligned}$$

Note: For a given Channel thickness of: 41.000 mm

Min. Channel len. adjacent to the tubesheet is: 489.805mm

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned}
 &= \text{Dc}^2 / (4 \cdot \text{tc} \cdot (\text{Dc} + \text{tc}) \cdot \cos(\theta)) \cdot \text{Pt} \\
 &= 1806^2 / (4 \cdot 41 \cdot (1806 + 41) \cdot \cos(0)) \cdot 4950 \\
 &= 53.303 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σc,b]:



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$$\begin{aligned}
 &= 6 \cdot kc/tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - \\
 &\quad 6((1 - \nu^*) / E^*) \cdot Do/h^3 \cdot (Cm\Theta + \\
 &\quad h \cdot \beta_c/2 \cdot Cm\Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 3523957/41^2 \cdot [(0.090274557 \cdot 1 \cdot 0.0066 - 0.000 \cdot 1) \cdot 4950 - \\
 &\quad 6((1 - 0.46)/29202954) \cdot 1779/264^3 \cdot (1 + \\
 &\quad 0.0066 \cdot 264/2 \cdot 1) \cdot (318109824 + 1779^2/32 \cdot (-103 - 4950))] \\
 &= 369.601 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot Sc)$$

$$= 422.904 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= Ec \cdot \sqrt{(1.5 \cdot Sc / \sigma_c)}$$

$$= 187982736 \cdot \sqrt{(1.5 \cdot 148/423)}$$

$$= 136197632.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 564517248 / (0.2 \cdot (264 - 8)^2)$$

$$= 258.4 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + Cats + Catc = 255 + 3 + 0 = 258 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) = \max(258; 98) = 258 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 100/100$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$



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ASME TS Calc:

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Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 773.7455 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is $\geq$ tube strength 1260 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	258.4 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	39.7 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	185.8 $\leq$	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	422.9 $\leq$	498.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	258.461	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 877 + 25.4$$

$$= 1778.738 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Ds / Do \text{ (Configurations a, b, c)}$$

$$= 1806 / 1779$$

$$= 1.0153$$

Ratio [pc]:



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ASME TS Calc:

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= Dc / Do (Configurations a, e, f)
= 1806/1779
= 1.0153

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

= ltx / h
= 0/264
= 0.000 (must be $0 \leq \rho \leq 1$)

Effective tube hole Diameter [d*]:

= max(dt - 2tt(EtT/E) • (StT/S) • (ρ); dt - 2tt)
= max(25.4 - 2 • 1.65 (177454944/187982736) •
(100/138) • (0); 25.4 - 2 • 1.65)
= 25.4000 mm

Effective tube pitch [p*]:

= $p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$
= 31.8 / $\sqrt{(1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2))}$
= 31.7500 mm

Effective ligament efficiency for bending [μ*]:

= (p* - d*) / p*
= (31.8 - 25.4) / 31.8
= 0.2000

h/p = 8.314960 ; μ* = 0.200000 ; E* = 29202954 KPa.

E*/E = 0.155349 ; ν* = 0.458572

Note: As h/p (8.315) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

= $((12(1 - \nu_s^2))^{1/4}) / \sqrt{(Ds + ts)ts}$
= $((12(1 - 0.3^2))^{1/4}) / \sqrt{(1806 + 21)21}$
= 0.0093 1/mm

Shell coefficient [ks]:

= βs • Es • ts³ / (6(1 - νs²))
= 0.0093 • 190085616 • 21³ / (6(1 - 0.3²))
= 2992204288.0000 KPa. • mm²

Shell coefficient [λs]:

= 6(Ds + ts) • ks / h³ • (1 + h • βs + (h • βs)² / 2)
= 6(1806 + 21) • 2.99E+09 / 264³ • (1 + 264 • 0.0093 + (264 • 0.0093)² / 2)
= 11500910.0 KPa.

Shell coefficient [δs]:

= Ds² / (4 • Es • ts) • (1 - Ds • νs / (2(Ds + ts)))
= 1806² / (4 • 190085616 • 21) • (1 - 1806 • 0.3 / (2(1806 + 21)))
= 0.17397E+00 mm/N./mm²

Parameter [ws]:

= ps • ks • βs • δs (1 + h • βs)
= 1.02 • 2.99E+09 • 0.0093 • 0.17397252 (1 + 264 • 0.0093)
= 16924.2324 mm²

Channel coefficient [βc]:

= $((12(1 - \nu_c^2))^{1/4}) / \sqrt{(Dc + tc)tc}$
= $((12(1 - 0.3^2))^{1/4}) / \sqrt{(1806 + 41)41}$
= 0.0066 1/mm

Channel coefficient [kc]:



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ASME TS Calc:

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$$\begin{aligned}
 &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\
 &= 0.0066 \cdot 187982736 \cdot 41^3 / (6(1 - 0.3^2)) \\
 &= 15674986496.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\
 &\quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta) \\
 &= 6(1806 + 41 / \cos(0)) \cdot 15.67E+09 / 264^3 \cdot (1 + \\
 &\quad 264 \cdot 0.0066 \cdot (1 + 1) / 2 + (264 \cdot 0.0066)^2 / 2 \cdot 1) \\
 &= 40261988.0 \text{ KPa}
 \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned}
 &= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\
 &= 1806^2 / (4 \cdot 187982736 \cdot 41) \cdot (1 - 1806 \cdot 0.3 / (2(1806 + 41))) \\
 &= 0.90275E-01 \text{ mm/N./mm}^2
 \end{aligned}$$

Channel coefficient [zeta α]:

$$zeta\alpha = 0$$

Parameter [ω_c]:

$$\begin{aligned}
 &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \\
 &\quad \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\
 &= 1.02 \cdot 15.67E+09 [0.090274557 (1 + 0.0066 \cdot 1 \cdot 264) - \\
 &\quad 0.000 (1/264 + 1 \cdot 0.0066/2)] \\
 &= 26044.1445 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 1908 / 1779 \\
 &= 1.0727
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.46) / 29202954 \cdot (11500910 + 40261988 + \\
 &\quad 187982736 \cdot \ln(1.07)) \\
 &= 1.2042
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt] \\
 &= 1779^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 1250 - \\
 &\quad (1.02 - 1) \cdot (1.02^2 + 1) \cdot 0] \\
 &= 7693837.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\
 &= 7693837 + 26044 \cdot 0 - 16924 \cdot 1250 \\
 &= -13461455.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= (M^*) - D_o^2 / 32 \cdot F(Ps - Pt) / (1 + F) \\
 &= (-13461455) - 1779^2 / 32 \cdot 1.2 (1250 - 0) / (1 + 1.2) \\
 &= -73626792.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\
 &= -73626792 + 1779^2 / 64 \cdot (3 + 0.46) \cdot (1250 - 0)
 \end{aligned}$$



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ASME TS Calc:

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$$= 140096032.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|-73626792|; |140096032|)$$

$$= 140096032.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 140096032 / (0.2 \cdot (264 - 8)^2)$$

$$= 64.1346 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 108472992 / (0.2 \cdot (117 - 8)^2)$$

$$= 275.6872 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 117 + 3 + 0$$

$$= 119.6490 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(120; 26.5)$$

$$= 119.6490 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| $P_s - P_t$ |]:

$$= |P_s - P_t|$$

$$= |1250 - 0|$$

$$= 1250.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 264 / 1779 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222)$$

$$= 13.108 \text{ N./mm}^2$$

| $P_s - P_t$ | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot |1250 - 0| + 3$$

$$= 26.4922 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 21.000mm

Min. Shell len. adjacent to the tubesheet is: 350.543 mm

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1806^2 / (4 \cdot 21 (1806 + 21)) \cdot 1250$$

$$= 26.5676 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:



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ASME TS Calc:

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$$\begin{aligned}
 &= 6 \cdot ks/ts^2 \cdot [\beta S \cdot P's + 6(1 - v^*)/E^* \cdot \\
 &\quad Do/h^3 \cdot (1 + h \cdot \beta S/2) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 672690/21^2 \cdot [0.0093 \cdot 0.22 + 6(1 - 0.46)/29202954 \cdot \\
 &\quad 1779/264^3 \cdot (1 + 264 \cdot 0.0093/2) \cdot (-73626792 + 1779^2/32 \cdot \\
 &\quad (1250 - 0))] \\
 &= 130.844 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 157.412 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 41.000 mm

Min. Channel len. adjacent to the tubesheet is: 489.805mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= Dc^2 / (4 \cdot tc (Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 1806^2 / (4 \cdot 41 (1806 + 41) \cdot \cos(0)) \cdot 0$$

$$= 0.000 \text{ N./mm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot kc/tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt -$$

$$6((1 - v^*)/E^*) \cdot Do/h^3 \cdot (Cm\theta +$$

$$h \cdot \beta_c/2 \cdot Cm\Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 3523957/41^2 \cdot [(0.090274557 \cdot 1 \cdot 0.0066 - 0.000 \cdot 1) \cdot 0 -$$

$$6((1 - 0.46)/29202954) \cdot 1779/264^3 \cdot (1 +$$

$$0.0066 \cdot 264/2 \cdot 1) \cdot (-73626792 + 1779^2/32 \cdot (1250 - 0))]$$

$$= -56.277 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 56.277 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 100/100$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 773.7455 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd) - 0.75 \cdot do}$$

$$= 2 \sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1) -$$



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ASME TS Calc:

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$$0.75 \cdot 25.4) \\ = 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is $\geq$ tube strength 1260 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	64.1 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	9.8 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	157.4 $\leq$	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	56.3 $\leq$	222.0 N./mm ²	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	119.649	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt \\ = 2 \cdot 877 + 25.4 \\ = 1778.738 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p \\ = (31.8 - 25.4) / 31.8 \\ = 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)} \\ = 1806 / 1779 \\ = 1.0153$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)} \\ = 1806 / 1779 \\ = 1.0153$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$



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ASME TS Calc:

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$$= 0/264$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944/187982736) \cdot$$

$$(100/138) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1779 \cdot 31.8) / (\pi \cdot 1779^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 8.314960$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (8.315) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1806 + 21) 21 }$$

$$= 0.0093 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0093 \cdot 190085616 \cdot 21^3 / (6(1 - 0.3^2))$$

$$= 2992204288.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1806 + 21) \cdot 2.99E+09 / 264^3 \cdot (1 + 264 \cdot 0.0093 + (264 \cdot 0.0093)^2 / 2)$$

$$= 11500910.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1806^2 / (4 \cdot 190085616 \cdot 21) \cdot (1 - 1806 \cdot 0.3 / (2(1806 + 21)))$$

$$= 0.17397E+00 \text{ mm/N} \cdot \text{mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 2.99E+09 \cdot 0.0093 \cdot 0.17397252 (1 + 264 \cdot 0.0093)$$

$$= 16924.2324 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1806 + 41) 41 }$$

$$= 0.0066 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0066 \cdot 187982736 \cdot 41^3 / (6(1 - 0.3^2))$$

$$= 15674986496.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:



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ASME TS Calc:

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$$= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta c)^2/2 \cdot Cq\Delta)$$

$$= 6(1806 + 41/\cos(0)) \cdot 15.67E+09/264^3 \cdot (1 + 264 \cdot 0.0066 \cdot (1 + 1)/2 + (264 \cdot 0.0066)^2/2 \cdot 1)$$

$$= 40261988.0 \text{ KPa.}$$

Channel coefficient [δc]:

$$= Dc^2/(4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc)))$$

$$= 1806^2/(4 \cdot 187982736 \cdot 41) \cdot (1 - 1806 \cdot 0.3/(2(1806 + 41)))$$

$$= 0.90275E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ωc]:

$$= \rho c \cdot kc [\delta c (Cm\theta + \beta c + Cq\Delta \cdot h \cdot \beta c^2) - \zeta c (Cm\theta + h \cdot Cq\theta + h \cdot \beta c / 2)]$$

$$= 1.02 \cdot 15.67E+09 [0.090274557 (1 + 0.0066 \cdot 1 \cdot 264) - 0.000 (1/264 + 1 \cdot 0.0066/2)]$$

$$= 26044.1445 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 1908/1779$$

$$= 1.0727$$

Coefficient [F]:

$$= (1 - v*)/E* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.46)/29202954 \cdot (11500910 + 40261988 + 187982736 \cdot \ln(1.07))$$

$$= 1.2042$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 1779^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 1250 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -22773758.0000 \text{ KPa.}\cdot\text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -22773758 + 26044 \cdot 4950 - 16924 \cdot 1250$$

$$= 84989472.0000 \text{ KPa.}\cdot\text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= (84989472) - 1779^2/32 \cdot 1.2 (1250 - 4950) / (1 + 1.2)$$

$$= 238416128.0000 \text{ KPa.}\cdot\text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$$

$$= 238416128 + 1779^2/64 \cdot (3 + 0.46) \cdot (1250 - 4950)$$

$$= -394203392.0000 \text{ KPa.}\cdot\text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp |; | Mo |)$$

$$= \max(| 238416128 |; | -394203392 |)$$

$$= 394203392.0000 \text{ KPa.}\cdot\text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:



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ASME TS Calc:

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Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 394203392 / (0.2 \cdot (264 - 8)^2)$$

$$= 180.4625 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 374831328 / (0.2 \cdot (210 - 8)^2)$$

$$= 275.2319 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + Cats + Catc$$

$$= 210 + 3 + 0$$

$$= 213.1351 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(213; 72.5)$$

$$= 213.1351 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |1250 - 4950|$$

$$= 3700.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 264 / 1779 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222)$$

$$= 13.108 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 23586 / 5684) \cdot |1250 - 4950| + 3$$

$$= 72.5368 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 21.000mm

Min. Shell len. adjacent to the tubesheet is: 350.543 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1806^2 / (4 \cdot 21 \cdot (1806 + 21)) \cdot 1250$$

$$= 26.5676 \text{ N./mm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 672690 / 21^2 \cdot [0.0093 \cdot 0.22 + 6(1 - 0.46) / 29202954 \cdot 1779 / 264^3 \cdot (1 + 264 \cdot 0.0093 / 2) \cdot (238416128 + 1779^2 / 32 \cdot (1250 - 4950))]$$

$$= -41.948 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:



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ASME TS Calc:

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$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 68.515 \text{ N./mm}^2 \text{ (Should be } \leq 207.00)$$

Note: For a given Channel thickness of: 41.000 mm
Min. Channel len. adjacent to the tubesheet is: 489.805mm

Axial Channel Membrane Stress $[\sigma_{c,m}]$:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1806^2 / (4 \cdot 41 (1806 + 41) \cdot \cos(0)) \cdot 4950$$

$$= 53.303 \text{ N./mm}^2$$

Axial Channel Bending Stress $[\sigma_{c,b}]$:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 3523957 / 41^2 \cdot [(0.090274557 \cdot 1 \cdot 0.0066 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.46) / 29202954) \cdot 1779 / 264^3 \cdot (1 + 0.0066 \cdot 264 / 2 \cdot 1) \cdot (238416128 + 1779^2 / 32 \cdot (1250 - 4950))]$$

$$= 308.686 \text{ N./mm}^2$$

Channel Membrane + Bending Stress $[\sigma_c]$:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 361.990 \text{ N./mm}^2 \text{ (Should be } \leq 222.00)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity $[E_c]$:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 187982736 \cdot \sqrt{(1.5 \cdot 148 / 362)}$$

$$= 147211744.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 404303616 / (0.2 (264 - 8)^2)$$

$$= 185.1 \text{ N./mm}^2 \text{ (Should be } \leq 276.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration $[H_{reqB}]$:

$$= h + C_{ats} + C_{atc} = 213 + 3 + 0 = 216 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) $[H_{req}]$:

$$= \max(H_{reqB}, H_{reqS}) = \max(216; 72.5) = 216 \text{ mm}$$

Tube Metal Area $[A_t]$:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength $[F_t]$:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 100$$

$$= 1259.763 \text{ Kg}$$

Groove Weld Strength $[F_g]$:



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ASME TS Calc:

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$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 1195.7887 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 100/100$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 100$$

$$= 773.7455 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1260 is >= tube strength 1260 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	185.1 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	29.1 <=	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	68.5 <=	207.0 N./mm ²	Ok
Stress in Channel at Tubesheet	362.0 <=	498.0 N./mm ²	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	216.398	267.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.651	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

	--Reqd. Thk. + CA		----	Tubesheet	Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	249.754	...	243	276	39	110	Ps+Pt D1	Ok
D2uc	116.632	...	61	276	10	110	Ps+Pt D2	Ok
D3uc	208.879	...	175	276	29	110	Ps+Pt D3	Ok
D4uc	41.086	...	5	276	1	110	Ps+Pt D4	Ok
D1c	258.461	...	258	276	40	110	Ps+Pt-c D1	Ok
D2c	119.649	...	64	276	10	110	Ps+Pt-c D2	Ok
D3c	216.398	...	185	276	29	110	Ps+Pt-c D3	Ok
D4c	44.041	...	5	276	1	110	Ps+Pt-c D4	Ok
Max:	258.4613	... mm		0.936		0.360	(Str. Ratio)	



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Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures

derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.							
Case#	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	169	207	386	498	Ok
D2uc	...	...	139	207	51	222	Ok
D3uc	...	...	64	207	332	498	Ok
D4uc	...	...	11	207	4	222	Ok
D1c	...	...	186	207	423	498	Ok
D2c	...	...	157	207	56	222	Ok
D3c	...	...	69	207	362	498	Ok
D4c	...	...	13	207	5	222	Ok
Max	...	...	0.897		0.849		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	258.461	267.000	Ok
Tube thickness :	0.710	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.651	Ok
Tube-tubesheet groove weld leg :	1.070	1.651	Ok

Given Shell Thickness: 24.0000 mm

Given Channel Thickness: 44.0000 mm

Min Shell length of thk, (24.000) adj. to tubesheet: 374.123 mm

Min Channel length of thk, (44.000) adj. to tubesheet: 506.565 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5368.38	0.997	5379.21	1.000
Tubesheet Shear Stress	14047.22	1.000	14047.22	1.000
Shell Stress (Axial, Junction)	5368.38	0.997	3478.19	1.000
Tubesheet-Channel Junction	5368.38	0.997	3478.19	0.654
Tube Pressure Stress	13751.57	1.000	3908.61	1.000
Minimum MAWP	5368.38		3478.19	



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ASME TS Calc:

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Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5690.86	1.000	5690.86	1.000
Tubesheet Shear Stress	14206.85	1.000	14206.85	1.000
Shell Stress (Axial, Junction)	14206.85	0.854	4699.55	1.000
Tubesheet-Channel Junction	5690.86	1.000	4699.55	0.838
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	5690.86		4699.55	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Input Echo, Tubesheet Number 3, Description:

Shell Data:

Main Shell Description: Port shell-1

Shell Maximum Design Pressure	Psd,max	1250.00 KPa.
Shell Maximum Operating Pressure	Psox,max	510.00 KPa.
Shell Minimum Operating Pressure	Psox,min	15.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1800.000 mm
Shell Temperature for Internal Pressure	Ts	235.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4316.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	44.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1800.000 mm
Channel Design Temperature	TEMPC	275.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2660
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	3033.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		100.35 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	111.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	876.669 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	23585.6 cm ²
Perimeter of Tube Layout	Cp	5684.2 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

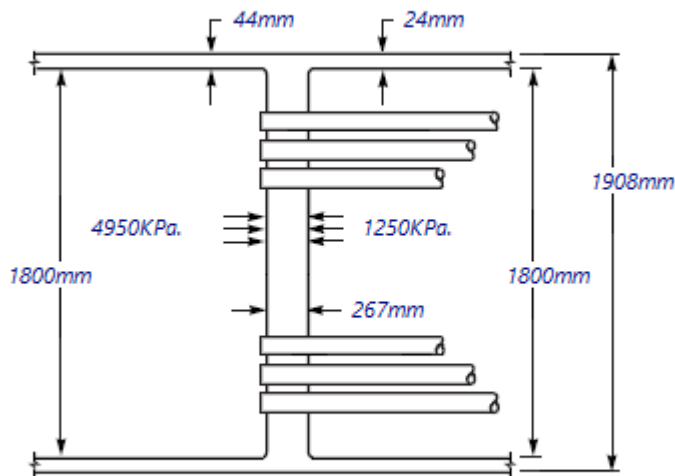
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	267.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1908.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc:

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Elastic Mod. at Design Temperature	235.0 °C	0.19009E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.17745E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4950 \cdot (25.4/2 - 0)) / (100 \cdot 1 + 0.4 \cdot 4950)$$

$$= 0.6144 + 0.0000 = 0.6144 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 275.00 °C
Elastic Modulus for Material 172442880.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	3909.67	15.39	50.0000	0.0046419	45.13

$$EAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 45.1) / (3 \cdot 15.4) = 3909 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7098	25.40	3909.67	35.79	50.0000	0.0008589	33.56

$$EAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 33.6) / (3 \cdot 35.8) = 1250 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7098 mm
Allowable Internal Pressure at Corroded thickness	13751.58 KPa.
Required Internal Design Pressure	4950.00 KPa.
Allowable External Pressure at Corroded thickness	3908.62 KPa.
Required External Design Pressure	1250.00 KPa.
Required Thickness due to Shell Side pressure	0.7098 mm

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	246.847	...	238	276	38 110	Ps+Pt D1 Ok
D2uc	116.632	...	61	276	10 110	Ps+Pt D2 Ok
D3uc	208.879	...	175	276	29 110	Ps+Pt D3 Ok
D1c	255.038	...	253	276	39 110	Ps+Pt-c D1 Ok
D2c	119.649	...	64	276	10 110	Ps+Pt-c D2 Ok
D3c	216.398	...	185	276	29 110	Ps+Pt-c D3 Ok

Max: 255.0381 ... mm 0.916 0.352 (Str. Ratio)

Load Case Definitions:



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ASME TS Calc:

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[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	157	207	382	498	Ok
D2uc	...	...	139	207	51	222	Ok
D3uc	...	...	64	207	332	498	Ok
D1c	...	...	173	207	418	498	Ok
D2c	...	...	157	207	56	222	Ok
D3c	...	...	69	207	362	498	Ok
Max	...	...	0.835		0.840		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	255.038	267.000	Ok
Tube thickness :	0.710	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.651	Ok
Tube-tubesheet groove weld leg :	1.070	1.651	Ok

Given Shell Thickness: 24.0000 mm

Given Channel Thickness: 44.0000 mm

Min Shell length of thk, (24.000) adj. to tubesheet: 374.123 mm

Min Channel length of thk, (44.000) adj. to tubesheet: 506.565 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	5368.38	0.997	5379.21	1.000
Tubesheet Shear Stress	14047.22	1.000	14047.22	1.000
Shell Stress (Axial, Junction)	5368.38	0.997	3478.19	1.000
Tubesheet-Channel Junction	5368.38	0.997	3478.19	0.654
Tube Pressure Stress	13751.57	1.000	3908.61	1.000
Minimum MAWP	5368.38		3478.19	



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ASME TS Calc:

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Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5690.86	1.000	5690.86	1.000
Tubesheet Shear Stress	14206.85	1.000	14206.85	1.000
Shell Stress (Axial, Junction)	14206.85	0.854	4699.55	1.000
Tubesheet-Channel Junction	5690.86	1.000	4699.55	0.838
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	5690.86		4699.55	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Problems/Failures Summary:

Step: 41

9:33am

Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Shell/Head in this vessel!**

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: